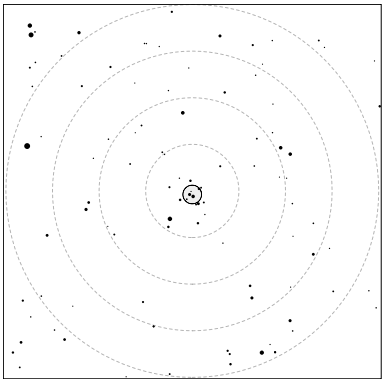
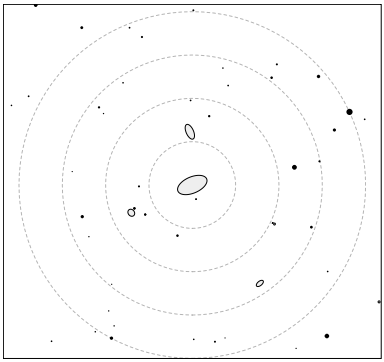
Pocket Finder-Chart Atlas





Correct-Image Version

Alan Watson Forster

Pocket Finder-Chart Atlas

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Correct-image version of 2022-03-26

Contents

The Messier Objects **1**

The Caldwell Objects **43**

The Urban Observing Program Objects **85**

The Messier Objects

The Messier objects are probably the most famous deep-sky objects. I find O'Meara's *Deep-Sky Companion: The Messier Objects* to be excellent on the origin of the catalog, the appearance of the objects, and their nature.

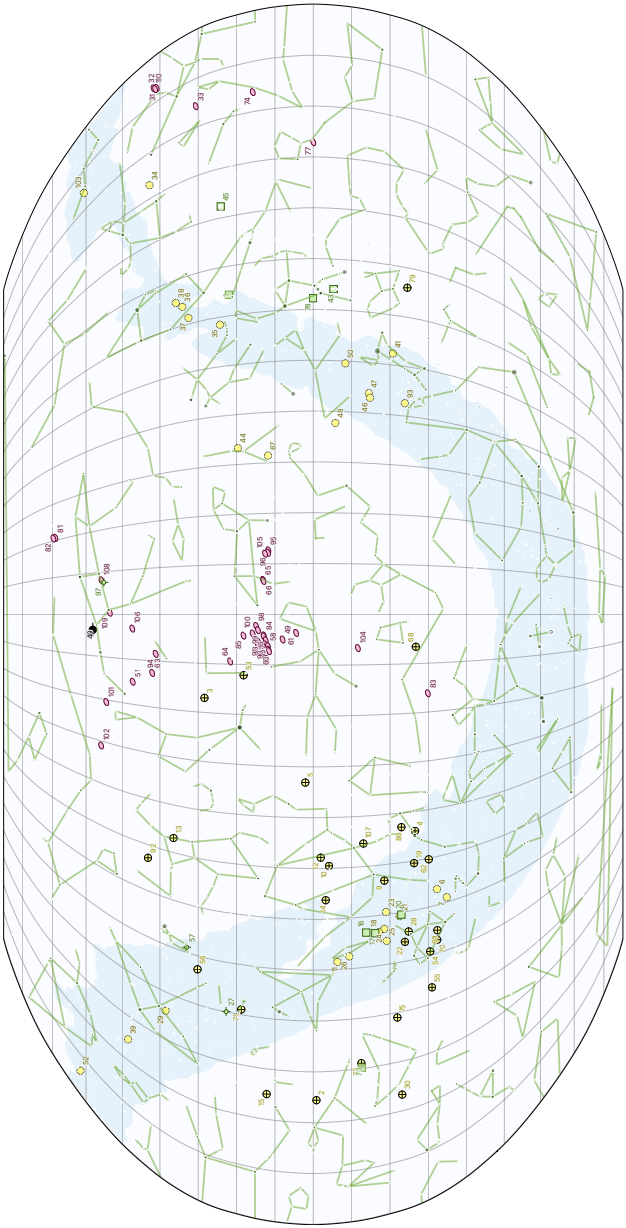
The following table lists the objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, and other names. I follow O'Meara in identifying M102 as NGC 5866.

For completeness, I include finder charts for all Messier objects, even bright ones like M45.

Name	Position	Type	Other Names
M1	05.6 +22	BN	BSC 119 = NGC 1952 = Crab Nebula
M2	21.6 −01	GC	U81 = BSC 56 = NGC 7089
M3	13.7 +28	GC	U51 = BSC 53 = NGC 5272
M4	16.4 −27	GC	U53 = BSC 43 = NGC 6121
M5	15.3 +02	GC	U52 = BSC 66 = NGC 5904
M6	17.7 −32	OC	U60 = BSC 18 = NGC 6405 = Butterfly Nebula
M7	17.9 −35	OC	U62 = BSC 10 = NGC 6475
M8	18.1 −24	BN	U64 = BSC 45 = NGC 6523/6530 = Lagoon Nebula
M9	17.3 −19	GC	BSC 123 = NGC 6333
M10	17.0 −04	GC	U57 = BSC 61 = NGC 6254
M11	18.9 −06	OC	U69 = NGC 6705 = Mel 213 = Cr 391 = Wild Duck Cluster
M12	16.8 −02	GC	U56 = BSC 94 = NGC 6218
M13	16.7 +36	GC	U54 = BSC 40 = NGC 6205 = Hercules Cluster
M14	17.6 −03	GC	BSC 116 = NGC 6402
M15	21.5 +12	GC	U80 = BSC 54 = NGC 7078
M16	18.3 −14	BN	BSC 46 = NGC 6611 = IC 4703 = Eagle Nebula
M17	18.3 −16	BN	U65 = BSC 47 = NGC 6618 = Omega Nebula
M18	18.3 −17	OC	BSC 87 = NGC 6613
M19	17.0 −26	GC	BSC 86 = NGC 6273
M20	18.0 −23	BN	BSC 57 = NGC 6514 = Cr 360 = Trifid Nebula
M21	18.1 −23	OC	BSC 65 = NGC 6531
M22	18.6 −24	GC	U67 = BSC 28 = NGC 6656
M23	17.9 −19	OC	BSC 69 = NGC 6494
M24	18.3 −19	SC	BSC 23 = IC 4715
M25	18.5 −19	OC	BSC 24 = IC 4725
M26	18.8 −09	OC	BSC 104 = NGC 6694
M27	20.0 +23	PN	U75 = BSC 88 = NGC 6853 = Dumbell Nebula
M28	18.4 −25	GC	BSC 93 = NGC 6626
M29	20.4 +38	OC	BSC 75 = NGC 6913
M30	21.7 −23	GC	BSC 95 = NGC 7099
M31	00.7 +41	Gal	U3 = BSC 11 = NGC 224 = Andromeda Galaxy
M32	00.7 +41	Gal	U2 = BSC 110 = NGC 221
M33	01.6 +31	Gal	BSC 39 = NGC 598
M34	02.7 +43	OC	BSC 35 = NGC 1039
M35	06.2 +24	OC	U27 = BSC 31 = NGC 2168
M36	05.6 +34	OC	U23 = BSC 58 = NGC 1960
M37	05.9 +33	OC	U26 = BSC 55 = NGC 2099
M38	05.5 +36	OC	U22 = BSC 82 = NGC 1912
M39	21.5 +48	OC	U82 = BSC 36 = NGC 7092
M40	12.4 +58	DS	
M41	06.8 −21	OC	U33 = BSC 21 = NGC 2287
M42	05.6 −05	BN	U24 = BSC 15 = NGC 1976 = Orion Nebula
M43	05.6 −05	BN	BSC 136 = NGC 1982
M44	08.7 +20	OC	U39 = BSC 12 = NGC 2632 = Mel 88 = Beehive Cluster
M45	03.8 +24	OC	U17 = BSC 4 = Mel 22 = Pleiades
M46	07.7 −15	OC	BSC 50 = NGC 2437
M47	07.6 −14	OC	BSC 19 = NGC 2422
M48	08.2 −06	OC	U38 = BSC 37 = NGC 2548
M49	12.5 +08	Gal	BSC 150 = NGC 4472
M50	07.0 −08	OC	U35 = BSC 44 = NGC 2323

Name	Position	Type	Other Names
M51	13.5 +47	Gal	BSC 120 = NGC 5194 = Whirlpool Galaxy
M52	23.4 +62	OC	BSC 27 = NGC 7654
M53	13.2 +18	GC	BSC 117 = NGC 5024
M54	18.9 −30	GC	BSC 118 = NGC 6715
M55	19.7 −31	GC	BSC 85 = NGC 6809
M56	19.3 +30	GC	BSC 113 = NGC 6779
M57	18.9 +33	PN	U71 = BSC 127 = NGC 6720 = Ring Nebula
M58	12.6 +12	Gal	NGC 4579
M59	12.7 +12	Gal	NGC 4621
M60	12.7 +12	Gal	NGC 4649
M61	12.4 +04	Gal	NGC 4303
M62	17.0 −30	GC	U58 = BSC 81 = NGC 6266
M63	13.3 +42	Gal	BSC 145 = NGC 5055 = Sunflower Galaxy
M64	12.9 +22	Gal	U50 = BSC 149 = NGC 4826 = Black-Eye Galaxy
M65	11.3 +13	Gal	NGC 3623
M66	11.3 +13	Gal	BSC 130 = NGC 3627
M67	08.9 +12	OC	U40 = BSC 51 = NGC 2682
M68	12.7 −27	GC	NGC 4590
M69	18.5 −32	GC	NGC 6637
M70	18.7 −32	GC	BSC 105 = NGC 6681
M71	19.9 +19	GC	BSC 52 = NGC 6838
M72	20.9 −13	GC	BSC 148 = NGC 6981
M73	21.0 −13	AST	NGC 6994 = Cr 426
M74	01.6 +16	Gal	NGC 628
M75	20.1 −22	GC	BSC 144 = NGC 6864
M76	01.7 +52	PN	NGC 651/650 = Little Dumbell
M77	02.7 −00	Gal	U12 = NGC 1068
M78	05.8 +00	BN	BSC 114 = NGC 2068
M79	05.4 −25	GC	BSC 126 = NGC 1904
M80	16.3 −23	GC	BSC 101 = NGC 6093
M81	09.9 +69	Gal	U41 = BSC 72 = NGC 3031
M82	09.9 +70	Gal	U42 = BSC 122 = NGC 3034
M83	13.6 −30	Gal	BSC 89 = NGC 5236
M84	12.4 +13	Gal	U45 = NGC 4374
M85	12.4 +18	Gal	NGC 4382
M86	12.4 +13	Gal	U46 = NGC 4406
M87	12.5 +12	Gal	U47 = NGC 4486
M88	12.5 +14	Gal	NGC 4501
M89	12.6 +13	Gal	NGC 4552
M90	12.6 +13	Gal	NGC 4569
M91	12.6 +14	Gal	NGC 4548
M92	17.3 +43	GC	U59 = BSC 59 = NGC 6341
M93	07.7 −24	OC	BSC 48 = NGC 2447 = Cr 160
M94	12.8 +41	Gal	U49 = BSC 134 = NGC 4736
M95	10.7 +12	Gal	NGC 3351
M96	10.8 +12	Gal	NGC 3368
M97	11.2 +55	PN	NGC 3587
M98	12.2 +15	Gal	NGC 4192
M99	12.3 +14	Gal	NGC 4254
M100	12.4 +16	Gal	NGC 4321

Name	Position	Type	Other Names
M101	14.1 +54	Gal	BSC 100 = NGC 5457
M102	15.1 +56	Gal	NGC 5866
M103	01.6 +61	OC	BSC 83 = NGC 581
M104	12.7 −12	Gal	U48 = BSC 133 = NGC 4594 = Sombrero Galaxy
M105	10.8 +13	Gal	NGC 3379
M106	12.3 +47	Gal	BSC 141 = NGC 4258
M107	16.5 −13	GC	BSC 129 = NGC 6171
M108	11.2 +56	Gal	NGC 3556
M109	12.0 +53	Gal	NGC 3992
M110	00.7 +42	Gal	BSC 132 = NGC 205



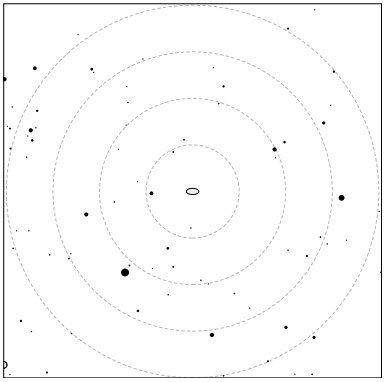
The distribution of Messier objects over the whole sky

M1

BSC 119 = NGC 1952 = Crab Nebula

RA : 05.6^h	PSA: 14	Type: BN
Dec: $+22°$	OSA: E3	Mag : 8.0
Con: Tau	CSA: 3/9	Size : $6.0' \times 4.0'$

See: OSA DSC:TMO

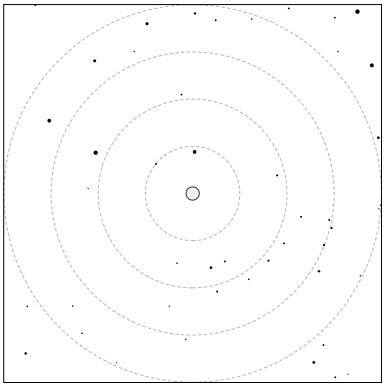


M2

U81 = BSC 56 = NGC 7089

RA : 21.6^h	PSA: 75/77	Type: GC
Dec: $-01°$	OSA: E24	Mag : 6.3
Con: Aqr	CSA: 13	Size : $16.0'$

See: OSA DSC:TMO

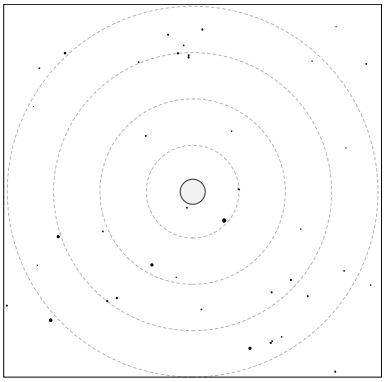


M3

U51 = BSC 53 = NGC 5272

RA : 13.7^h	PSA: 43/44	Type: GC
Dec: $+28°$	OSA: E15	Mag : 5.9
Con: CVn	CSA: 5	Size : $19.0'$

See: OSA DSC:TMO

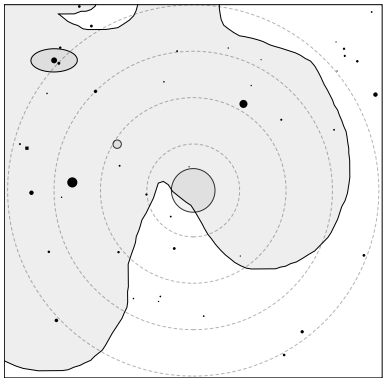


M4

U53 = BSC 43 = NGC 6121

RA : 16.4^h	PSA: 56/58	Type: GC
Dec: $-27°$	OSA: E18	Mag : 5.4
Con: Sco	CSA: 11/12/17/18	Size : 36.0′

See: OSA DSC:TMO

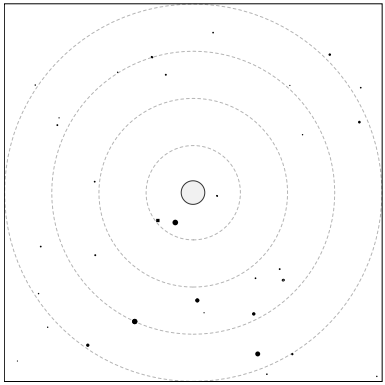


M5

U52 = BSC 66 = NGC 5904

RA : 15.3^h	PSA: 55/57	Type: GC
Dec: $+02°$	OSA: E15	Mag : 5.7
Con: Ser	CSA: 11/12	Size : 23.0′

See: OSA DSC:TMO



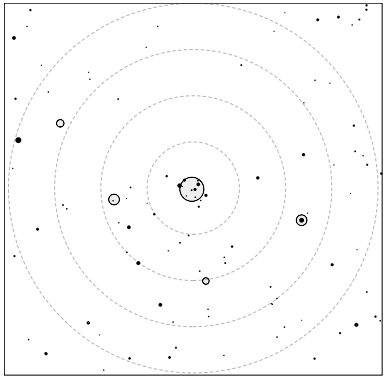
M6

U60 = BSC 18 = NGC 6405 = Butterfly Nebula

RA : 17.7^h	PSA: 58/69	Type: OC
Dec: $-32°$	OSA: E18	Mag : 4.2
Con: Sco	CSA: 18	Size : 33.0′

See: OSA DSC:TMO

λ Sco Shaula (mag 1.6)
→ 5° N
→ 1° E to M6/U60



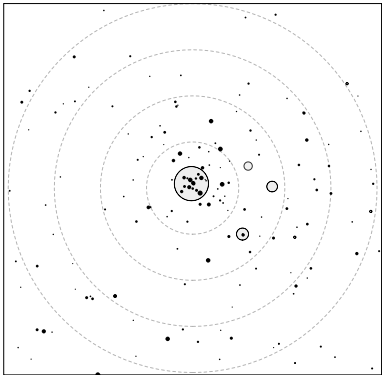
M7

U62 = BSC 10 = NGC 6475

RA : 17.9^h	PSA: 58/69	Type: OC
Dec: $-35°$	OSA: E18	Mag : 2.8
Con: Sco	CSA: 18	Size : 75.0′

See: OSA DSC:TMO

ε Sgr Kaus Australis (mag 1.8)
→ 6° W to M7/U62



M8

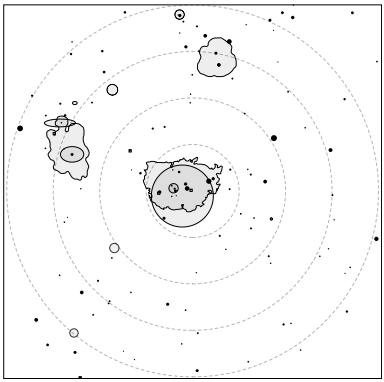
U64 = BSC 45 = NGC 6523/6530 = Lagoon Nebula

RA : 18.1^h	PSA: 67	Type: BN
Dec: $-24°$	OSA: E20	Mag : 3.0
Con: Sgr	CSA: 12/18	Size : 45.0′ × 30.0′

See: OSA DSC:TMO

Notes: M20 is 1.5° N. M21 is 2.0° N.

γ Sgr Alnasl (mag 3.0)
→ 6° N to M8/U64

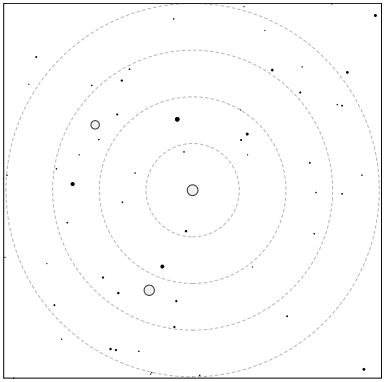


M9

BSC 123 = NGC 6333

RA : 17.3^h	PSA: 56	Type: GC
Dec: $-19°$	OSA: E17	Mag : 7.8
Con: Oph	CSA: 12/18	Size : 12.0′

See: OSA DSC:TMO

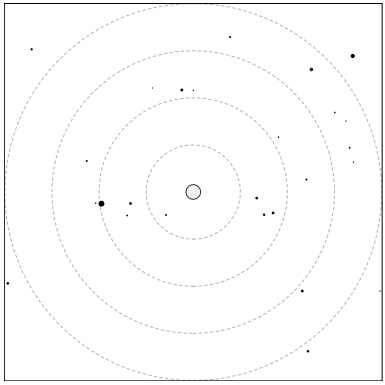


M10

U57 = BSC 61 = NGC 6254

RA : 17.0^h	PSA: 54/56	Type: GC
Dec: $-04°$	OSA: E17	Mag : 6.6
Con: Oph	CSA: 12	Size : 20.0′

See: OSA DSC:TMO

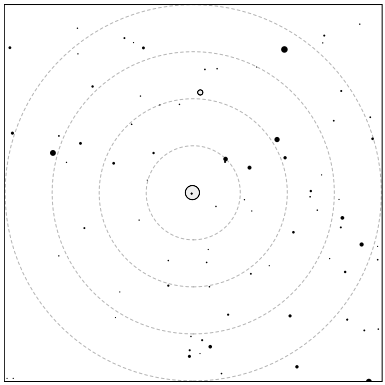


M11

U69 = NGC 6705 = Mel 213 = Cr 391 = Wild Duck Cluster

RA : 18.9^h	PSA: 65/67	Type: OC
Dec: $-06°$	OSA: E19	Mag : 5.3
Con: Sct	CSA: 12	Size : 13.0′

See: OSA DSC:TMO

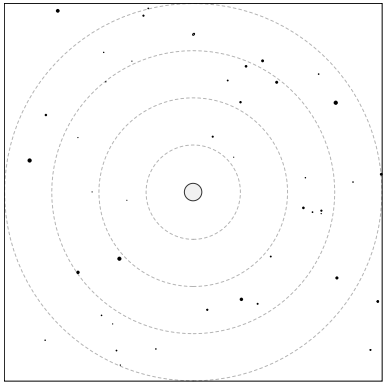


M12

U56 = BSC 94 = NGC 6218

RA : 16.8^h	PSA: 54/56	Type: GC
Dec: $-02°$	OSA: E17	Mag : 6.1
Con: Oph	CSA: 12	Size : 16.0′

See: OSA DSC:TMO



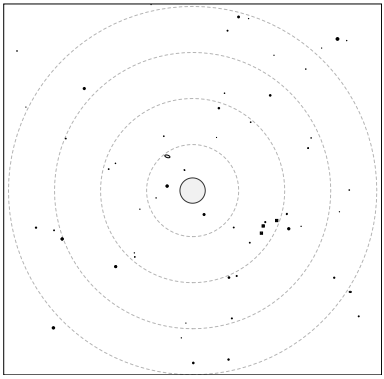
M13

U54 = BSC 40 = NGC 6205 = Hercules Cluster

RA : 16.7^h	PSA: 52	Type: GC
Dec: $+36°$	OSA: N14	Mag : 5.3
Con: Her	CSA: 5/6	Size : 20.0′

See: OSA DSC:TMO

ζ Her (mag 2.8)
→ 5° N (2/3 distance to η Her) to M13/U54

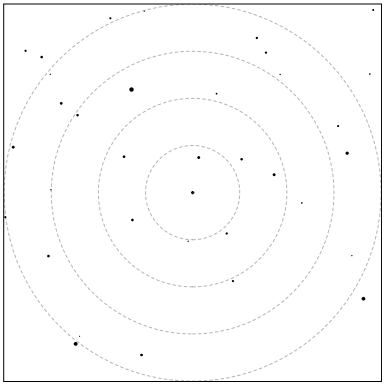


M14

BSC 116 = NGC 6402

RA : 17.6^h	PSA: 54	Type: GC
Dec: $-03°$	OSA: E17	Mag : 7.6
Con: Oph	CSA: 12	Size : 11.0′

See: OSA DSC:TMO

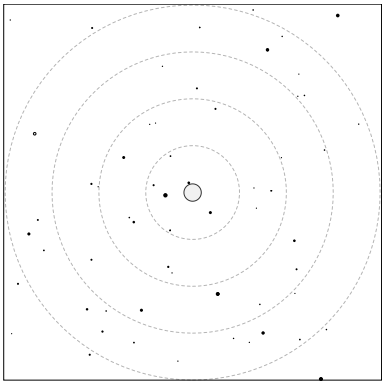


M15

U80 = BSC 54 = NGC 7078

RA : 21.5^h	PSA: 75	Type: GC
Dec: $+12°$	OSA: E23	Mag : 6.0
Con: Peg	CSA: 13	Size : 18.0′

See: OSA DSC:TMO



M16

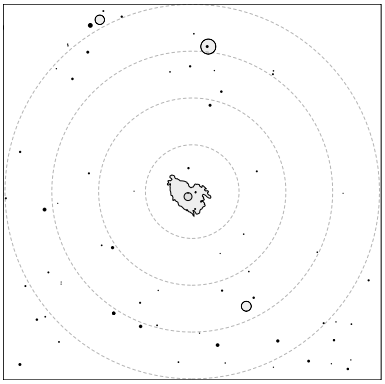
BSC 46 = NGC 6611 = IC 4703 = Eagle Nebula

RA : 18.3^h PSA: 67 Type: BN
Dec: $-14°$ OSA: E20 Mag : 6.0
Con: Ser CSA: 12 Size : 35.0′ × 28.0′

See: OSA DSC:TMO

ν Oph (mag 3.3)
→ 4° E
→ 4° S to M16

μ Sgr (mag 3.8)
→ 7° N
→ 1° E to M16



M17

U65 = BSC 47 = NGC 6618 = Omega Nebula

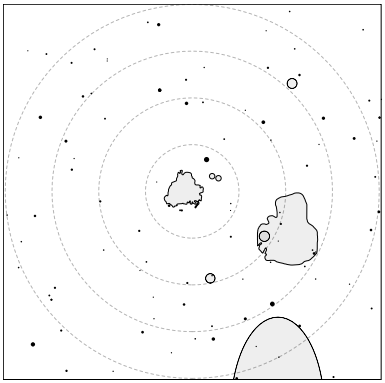
RA : 18.3^h PSA: 67 Type: BN
Dec: $-16°$ OSA: E20 Mag : 6.0
Con: Sgr CSA: 12 Size : 20.0′ × 15.0′

See: OSA DSC:TMO

Notes: M18 is 1° S. M24 is 3° SW.

ν Oph (mag 3.3)
→ 5° E
→ 6° S to M17/U65

μ Sgr (mag 3.8)
→ 5° N passing M24 and M18
→ 2° E to M17



M18

BSC 87 = NGC 6613

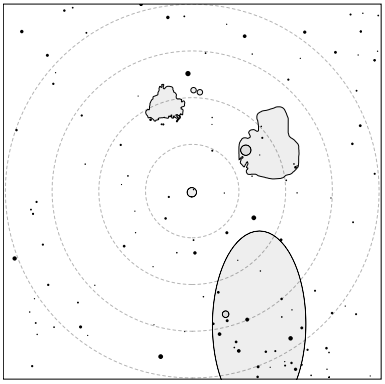
RA : 18.3^h PSA: 67 Type: OC
Dec: $-17°$ OSA: E20 Mag : 6.9
Con: Sgr CSA: 12/18 Size : 7.0′

See: OSA DSC:TMO

Notes: M17/U65 is 1° N. M24 is 2° SW.

ν Oph (mag 3.3)
→ 5° E
→ 7° S passing M17/U65 to M18

μ Sgr (mag 3.8)
→ 4° N passing M24
→ 2° E to M18

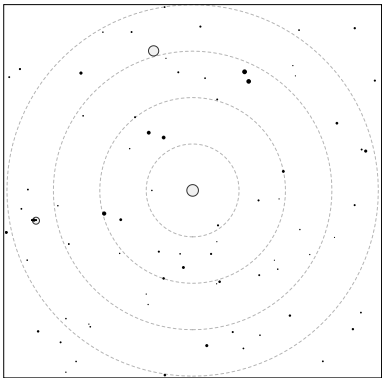


M19

BSC 86 = NGC 6273

RA : 17.0^h	PSA: 56	Type: GC
Dec: $-26°$	OSA: E18	Mag : 6.8
Con: Oph	CSA: 12/18	Size : 17.0′

See: OSA DSC:TMO



M20

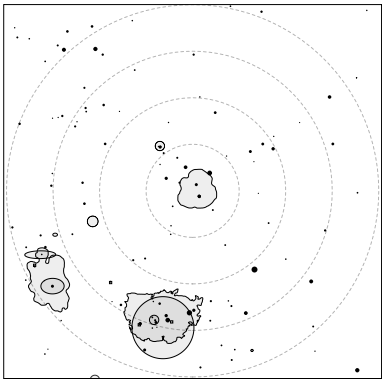
BSC 57 = NGC 6514 = Cr 360 = Trifid Nebula

RA : 18.0^h	PSA: 67	Type: BN
Dec: $-23°$	OSA: E20	Mag : 6.3
Con: Sgr	CSA: 12/18	Size : 20.0′

See: OSA DSC:TMO

Notes: M8/U64 is 1.5° S. M21 is 0.5° NE.

γ Sgr Alnasl (mag 3.0)
→ 7.5° N passing M8/U64 to M20



M21

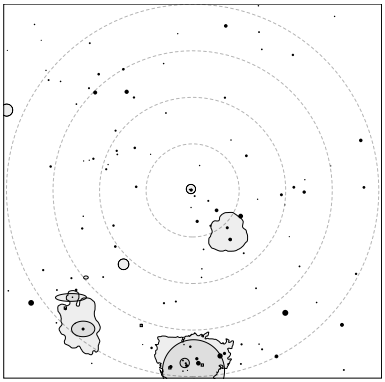
BSC 65 = NGC 6531

RA : 18.1^h	PSA: 67	Type: OC
Dec: $-23°$	OSA: E20	Mag : 5.9
Con: Sgr	CSA: 12/18	Size : 16.0′

See: OSA DSC:TMO

Notes: M8/U64 is 2.0° S. M20 is 0.5° SW.

γ Sgr Alnasl (mag 3.0)
→ 8.0° N passing M8/U64 and M20 to M21

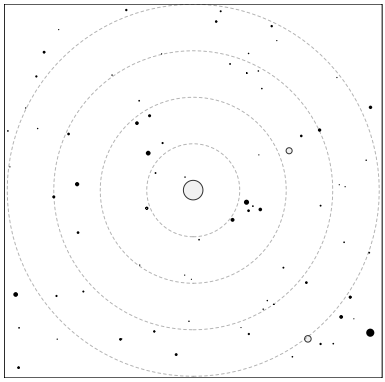


M22

U67 = BSC 28 = NGC 6656

RA : 18.6^h	PSA: 67	Type: GC
Dec: $-24°$	OSA: E20	Mag : 5.2
Con: Sgr	CSA: 12/18	Size : 32.0′

See: OSA DSC:TMO

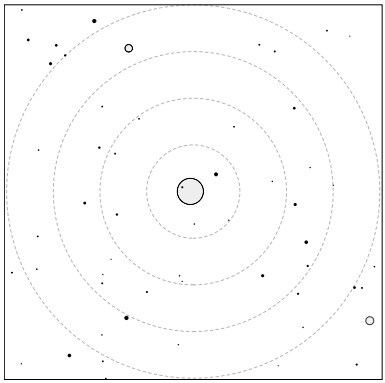


M23

BSC 69 = NGC 6494

RA : 17.9^h	PSA: 67	Type: OC
Dec: $-19°$	OSA: E20	Mag : 5.5
Con: Sgr	CSA: 12/18	Size : 25.0′

See: OSA DSC:TMO



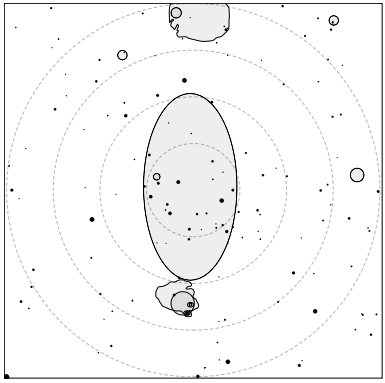
M24

BSC 23 = IC 4715

RA : 18.3^h	PSA: 67	Type: SC
Dec: $-19°$	OSA: E20	Mag : 2.5
Con: Sgr	CSA: 12/18	Size : 120.0′ × 60.0′

See: OSA DSC:TMO

Notes: M18 is 1.5° NE. M17/U65 is 3° NE.

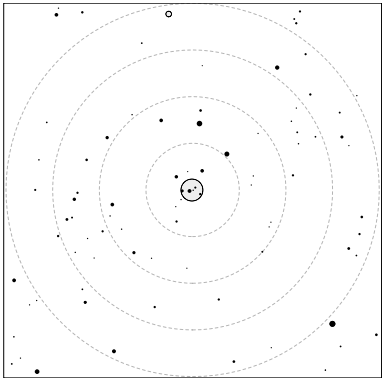


M25

BSC 24 = IC 4725

RA : 18.5^h	PSA: 67	Type: OC
Dec: $-19°$	OSA: E20	Mag : 4.6
Con: Sgr	CSA: 12/18	Size : 26.0′

See: OSA DSC:TMO

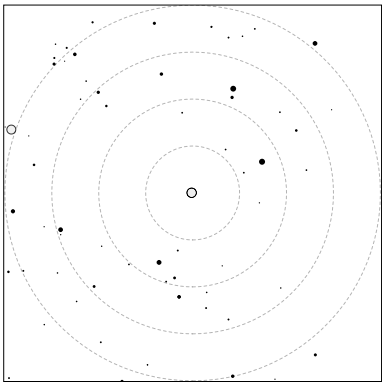


M26

BSC 104 = NGC 6694

RA : 18.8^h	PSA: 67	Type: OC
Dec: $-09°$	OSA: E19	Mag : 8.0
Con: Sct	CSA: 12	Size : 10.0′

See: OSA DSC:TMO

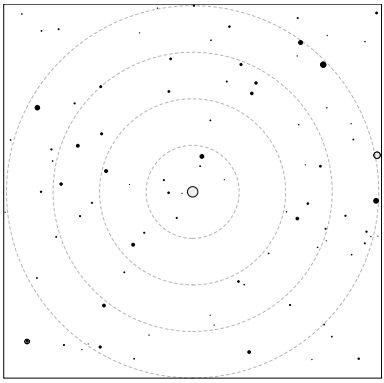


M27

U75 = BSC 88 = NGC 6853 = Dumbell Nebula

RA : 20.0^h	PSA: 64	Type: PN
Dec: $+23°$	OSA: E21	Mag : 7.3
Con: Vul	CSA: 6/7/12/13	Size : 8.0′×6.0′

See: OSA DSC:TMO

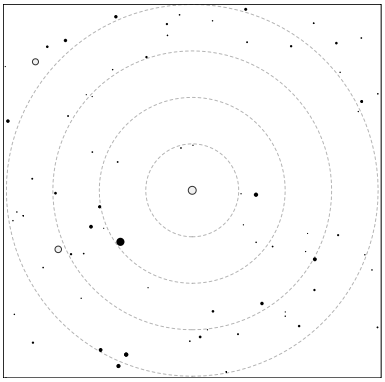


M28

BSC 93 = NGC 6626

RA : 18.4^h	PSA: 67/I	Type: GC
Dec: $-25°$	OSA: E20	Mag : 6.9
Con: Sgr	CSA: 12/18	Size : 13.8′

See: OSA DSC:TMO



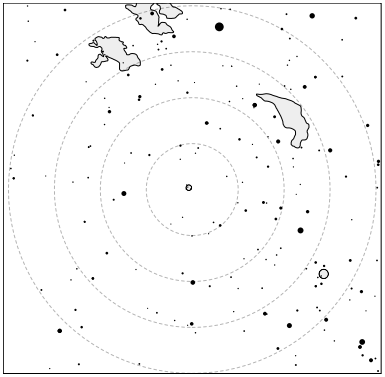
M29

BSC 75 = NGC 6913

RA : 20.4^h	PSA: 62	Type: OC
Dec: $+38°$	OSA: N20	Mag : 6.6
Con: Cyg	CSA: 6/7	Size : 10.0′

See: OSA DSC:TMO

γ Cyg Sadr (mag 2.2)
$\rightarrow 1.8°$ S to M29

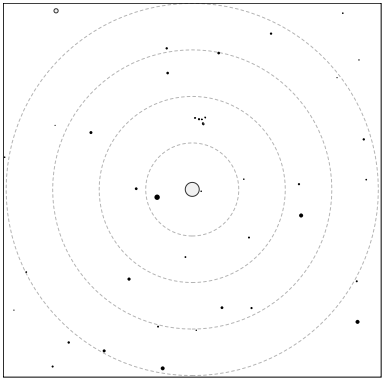


M30

BSC 95 = NGC 7099

RA : 21.7^h	PSA: 77	Type: GC
Dec: $-23°$	OSA: E22	Mag : 6.9
Con: Cap	CSA: 13/19	Size : 12.0′

See: OSA DSC:TMO



M31

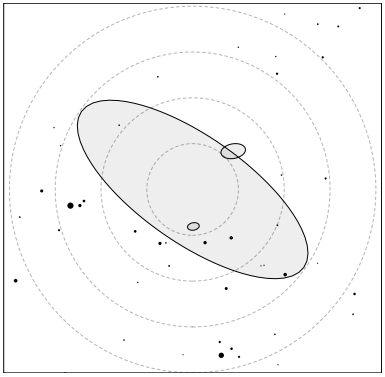
U3 = BSC 11 = NGC 224 = Andromeda Galaxy

RA : 00.7^h	PSA: 3	Type: Gal
Dec: +41°	OSA: N0	Mag : 3.4
Con: And	CSA: 2/7	Size : 3.0° × 1.0°

See: OSA DSC:TMO

Notes: M32/U2 is 0.5° S. M100 is 0.5° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 3.0° N to M31/U3



M32

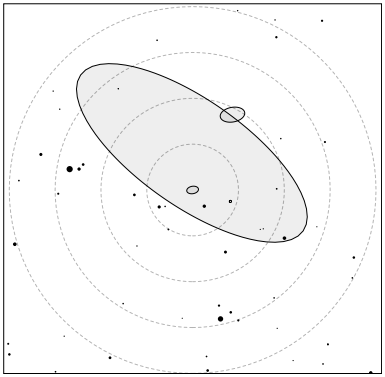
U2 = BSC 110 = NGC 221

RA : 00.7^h	PSA: 3	Type: Gal
Dec: +41°	OSA: N0	Mag : 8.2
Con: And	CSA: 2/7	Size : 11.0′ × 7.3′

See: OSA DSC:TMO

Notes: M31/U3 is 0.5° N. M110 is 1° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 2.5° N to M32/U2

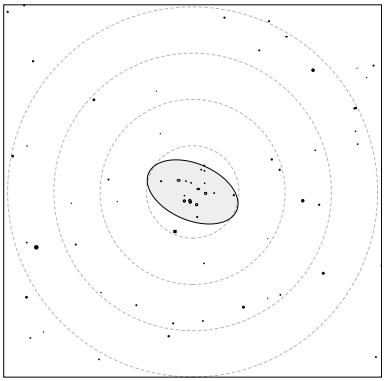


M33

BSC 39 = NGC 598

RA : 01.6^h	PSA: 3	Type: Gal
Dec: +31°	OSA: N0	Mag : 5.7
Con: Tri	CSA: 2	Size : 67.0′ × 41.5′

See: OSA DSC:TMO



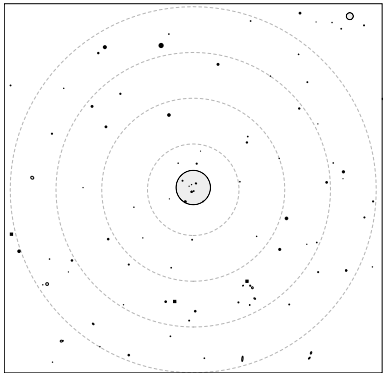
M34

BSC 35 = NGC 1039

RA : 02.7^h	PSA: 2	Type: OC
Dec: $+43°$	OSA: N4	Mag : 5.2
Con: Per	CSA: 2/3	Size : 25.0′

See: OSA DSC:TMO

β Per Algol (mag 2.1–3.4)
→ 2.0° N (1/2 distance to κ Per)
→ 4.5° W (2/5 distance to γ And) to M34

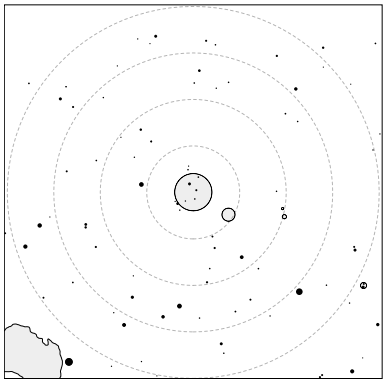


M35

U27 = BSC 31 = NGC 2168

RA : 06.2^h	PSA: 14	Type: OC
Dec: $+24°$	OSA: E7	Mag : 5.1
Con: Gem	CSA: 3/9	Size : 25.0′

See: OSA DSC:TMO

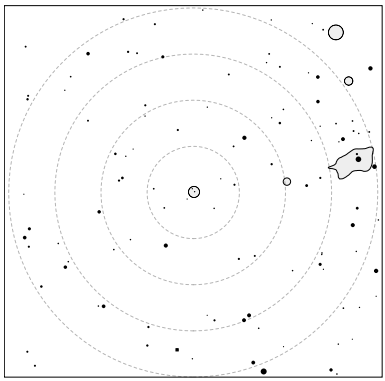


M36

U23 = BSC 58 = NGC 1960

RA : 05.6^h	PSA: 12	Type: OC
Dec: $+34°$	OSA: N6	Mag : 6.0
Con: Aur	CSA: 3	Size : 10.0′

See: OSA DSC:TMO



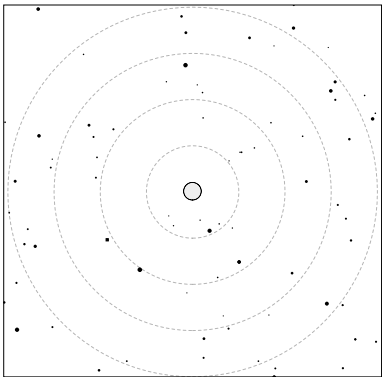
M37

U26 = BSC 55 = NGC 2099

RA : 05.9^h	PSA: 12	Type: OC
Dec: +33°	OSA: N6	Mag : 5.6
Con: Aur	CSA: 3	Size : 15.0′

See: OSA DSC:TMO

β Tau Elnath (mag 1.7)
→ 6° W to point 1° N of 136 Tau
→ 4° N to M37/U26



M38

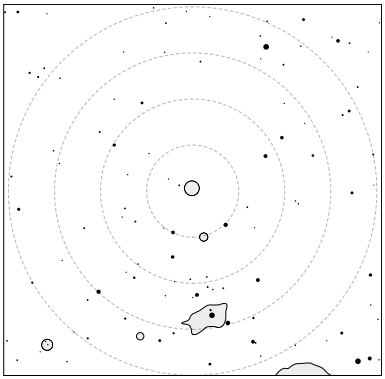
U22 = BSC 82 = NGC 1912

RA : 05.5^h	PSA: 12	Type: OC
Dec: +36°	OSA: N6	Mag : 6.4
Con: Aur	CSA: 3	Size : 15.0′

See: OSA DSC:TMO

Notes: M36/U23 is 2.5° SE.

β Tau Elnath (mag 1.7)
→ 7° N to M38/U23

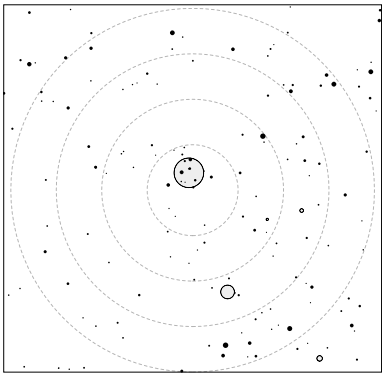


M39

U82 = BSC 36 = NGC 7092

RA : 21.5^h	PSA: 73	Type: OC
Dec: +48°	OSA: N24	Mag : 4.6
Con: Cyg	CSA: 1/6/7	Size : 31.0′

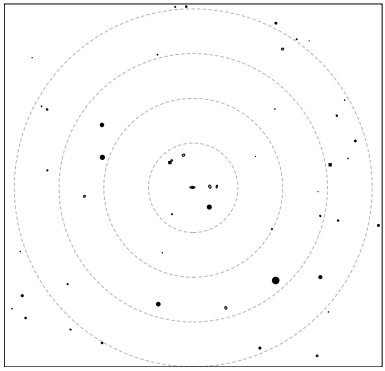
See: OSA DSC:TMO



M40

RA : 12.4^h PSA: 32 Type: DS
Dec: +58° OSA: N10 Mag : 9.0/9.6
Con: UMa CSA: 1/4/5 Size : 0.9′

See: OSA DSC:TMO



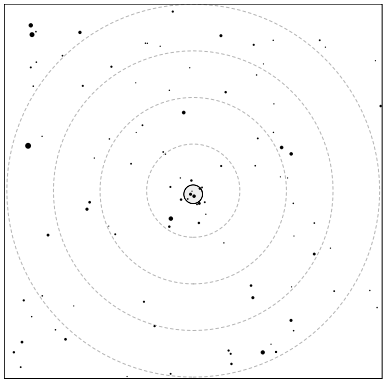
M41

U33 = BSC 21 = NGC 2287

RA : 06.8^h PSA: 27 Type: OC
Dec: −21° OSA: E6 Mag : 4.5
Con: CMa CSA: 9/15 Size : 39.0′

See: OSA DSC:TMO

α CMa Sirius (mag −1.5)
→ 4° S to M41/U33



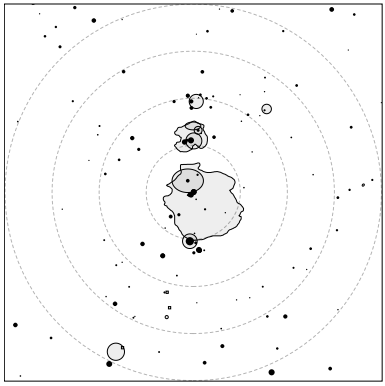
M42

U24 = BSC 15 = NGC 1976 = Orion Nebula

RA : 05.6^h PSA: 16/B Type: BN
Dec: −05° OSA: E4 Mag : 3.7
Con: Ori CSA: 9 Size : 90.0′ × 60.0′

See: OSA DSC:TMO

Notes: M42/U24 surrounds M43. U25 is 1° N.



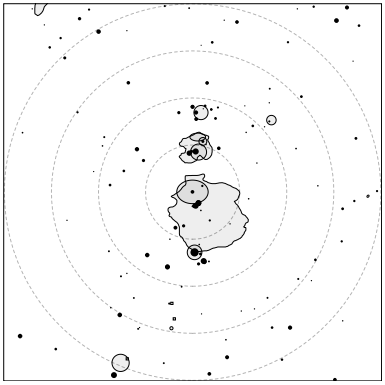
M43

BSC 136 = NGC 1982

RA : 05.6^h	PSA: 16/B	Type: BN
Dec: $-05°$	OSA: E4	Mag : 6.8
Con: Ori	CSA: 9	Size : 20.0′ × 15.0′

See: OSA DSC:TMO

Notes: M42/U24 surrounds M43. U25 is 1° N.

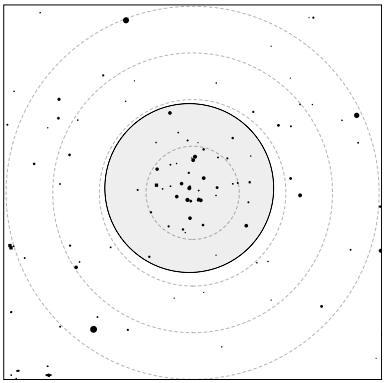


M44

U39 = BSC 12 = NGC 2632 = Mel 88 = Beehive Cluster

RA : 08.7^h	PSA: 24	Type: OC
Dec: $+20°$	OSA: E9	Mag : 3.1
Con: Cnc	CSA: 3/4/9/10	Size : 70.0′

See: OSA DSC:TMO

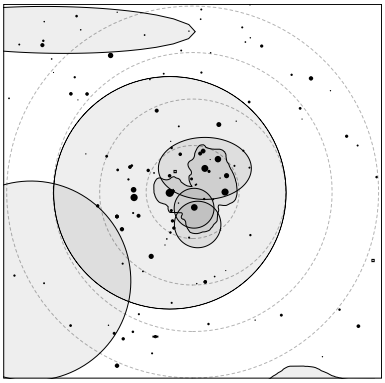


M45

U17 = BSC 4 = Mel 22 = Pleiades

RA : 03.8^h	PSA: 15/A	Type: OC
Dec: $+24°$	OSA: E3	Mag : 1.5
Con: Tau	CSA: 2/3/8/9	Size : 120.0′

See: OSA DSC:TMO



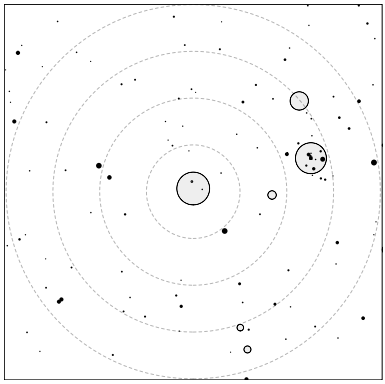
M46

BSC 50 = NGC 2437

RA : 07.7^h	PSA: 26	Type: OC
Dec: $-15°$	OSA: E8	Mag : 6.1
Con: Pup	CSA: 9/10/15/16	Size : 20.0′

See: OSA DSC:TMO

Notes: M47 is 1.3° W.



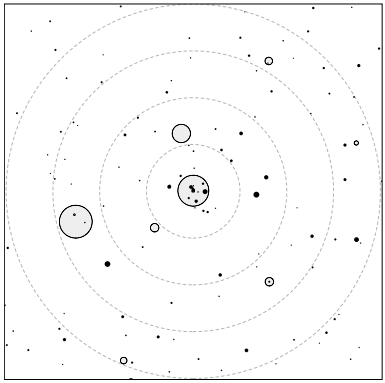
M47

BSC 19 = NGC 2422

RA : 07.6^h	PSA: 27	Type: OC
Dec: $-14°$	OSA: E8	Mag : 4.4
Con: Pup	CSA: 9/10/15/16	Size : 25.0′

See: OSA DSC:TMO

Notes: M46 is 1.3° E.

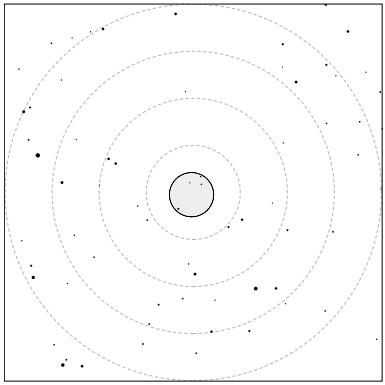


M48

U38 = BSC 37 = NGC 2548

RA : 08.2^h	PSA: 26	Type: OC
Dec: $-06°$	OSA: E10	Mag : 5.8
Con: Hya	CSA: 9/10	Size : 30.0′

See: OSA DSC:TMO



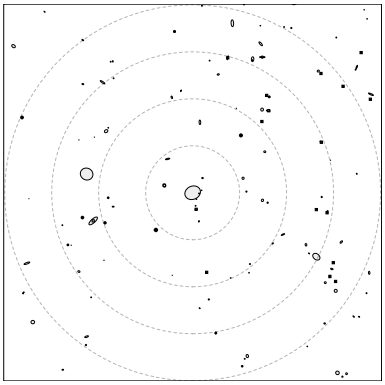
M49

BSC 150 = NGC 4472

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: $+08°$	OSA: E14	Mag : 8.4
Con: Vir	CSA: 10/11	Size : $8.1' \times 7.1'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



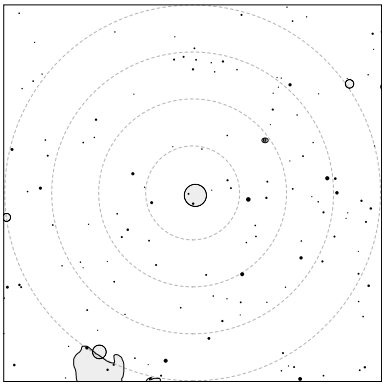
M50

U35 = BSC 44 = NGC 2323

RA : 07.0^h	PSA: 27	Type: OC
Dec: $-08°$	OSA: E8	Mag : 5.9
Con: Mon	CSA: 9	Size : 15.0′

See: OSA DSC:TMO

α CMa Sirius (mag −1.5)
→ 3° E to ι CMa
→ 5° N passing μ CMa to θ CMa
→ 4° N to asterism of three stars
→ 2° E to M50/U35

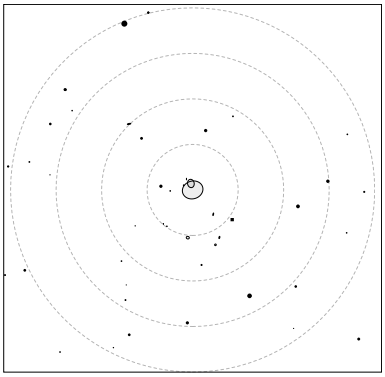


M51

BSC 120 = NGC 5194 = Whirlpool Galaxy

RA : 13.5^h	PSA: 43	Type: Gal
Dec: $+47°$	OSA: N12	Mag : 8.4
Con: CVn	CSA: 4/5	Size : $11.2' \times 6.9'$

See: OSA DSC:TMO



M52

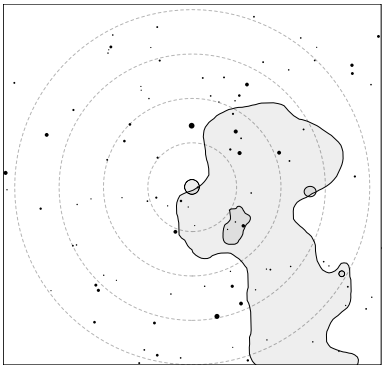
BSC 27 = NGC 7654

RA : 23.4^h	PSA: 72	Type: OC
Dec: $+62°$	OSA: N22	Mag : 6.9
Con: Cas	CSA: 2/7	Size : 16.0′

See: OSA DSC:TMO

Notes: C11 is 0.6° to SW.

β Cas Chaph (mag 2.3)
→ 2° N
→ 6° W to M52

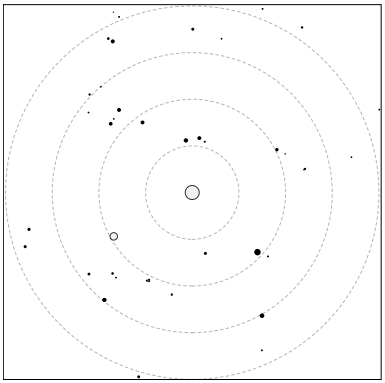


M53

BSC 117 = NGC 5024

RA : 13.2^h	PSA: 45	Type: GC
Dec: $+18°$	OSA: E13	Mag : 7.7
Con: Com	CSA: 5/11	Size : 13.0′

See: OSA DSC:TMO

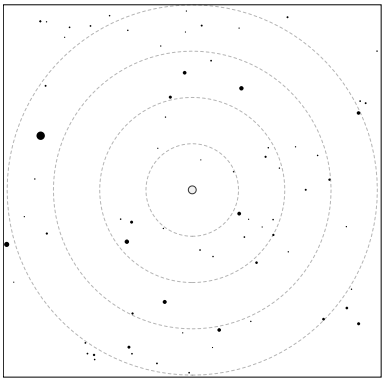


M54

BSC 118 = NGC 6715

RA : 18.9^h	PSA: 69	Type: GC
Dec: $-30°$	OSA: E20	Mag : 7.2
Con: Sgr	CSA: 18/19	Size : 12.0′

See: OSA DSC:TMO

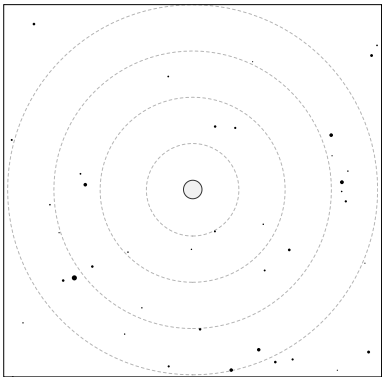


M55

BSC 85 = NGC 6809

RA : 19.7^h	PSA: 68	Type: GC
Dec: $-31°$	OSA: E22	Mag : 6.3
Con: Sgr	CSA: 18/19	Size : 19.0′

See: OSA DSC:TMO



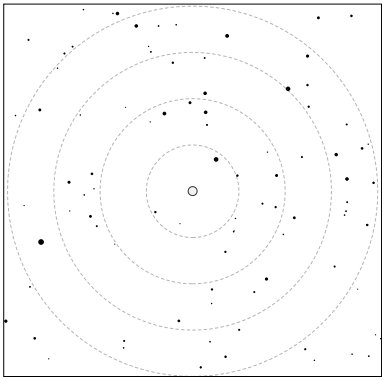
M56

BSC 113 = NGC 6779

RA : 19.3^h	PSA: 63	Type: GC
Dec: $+30°$	OSA: N18	Mag : 8.3
Con: Lyr	CSA: 6/7	Size : 7.0′

See: OSA DSC:TMO

γ Lyr Sulaphat (mag 3.2)
→ 3.5° E
→ 2.5° S to M56



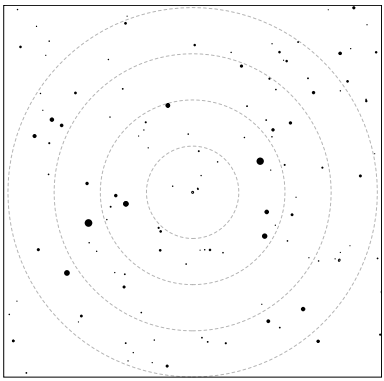
M57

U71 = BSC 127 = NGC 6720 = Ring Nebula

RA : 18.9^h	PSA: 63	Type: PN
Dec: $+33°$	OSA: N18	Mag : 8.8
Con: Lyr	CSA: 6/7	Size : 1.1′

See: OSA DSC:TMO

γ Lyr Sulaphat (mag 3.2)
→ 1.2° W
→ 0.4° N to M57/U71



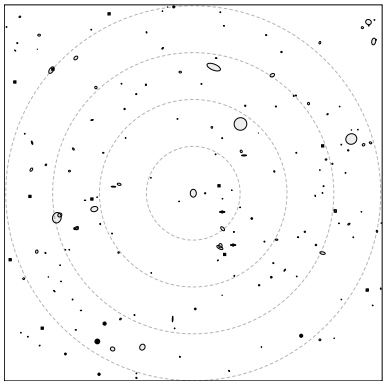
M58

NGC 4579

RA : 12.6^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 9.6
Con: Vir	CSA: 10/11	Size : 5.9′×4.7′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



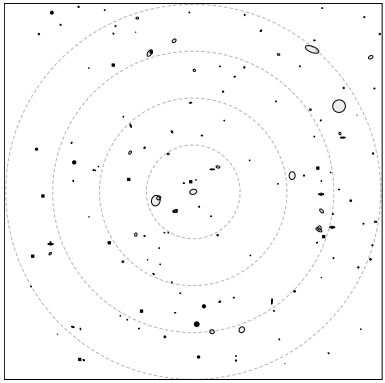
M59

NGC 4621

RA : 12.7^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 9.6
Con: Vir	CSA: 11	Size : 5.4′×3.7′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



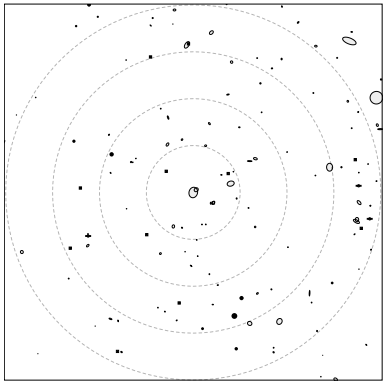
M60

NGC 4649

RA : 12.7^h	PSA: 45/C	Type: Gal
Dec: +12°	OSA: E14	Mag : 8.8
Con: Vir	CSA: 11	Size : 7.1′×6.1′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



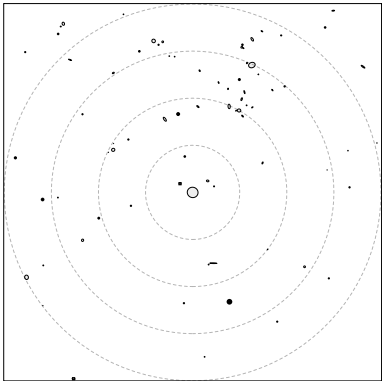
M61

NGC 4303

RA : 12.4^h	PSA: 45	Type: Gal
Dec: $+04°$	OSA: E14	Mag : 9.7
Con: Vir	CSA: 10/11	Size : $6.0' \times 5.9'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



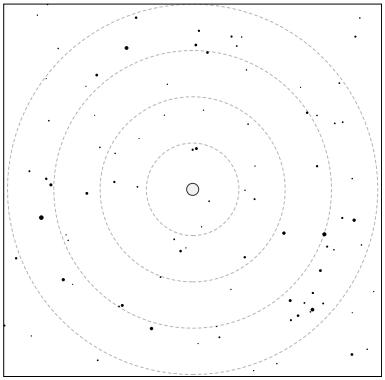
M62

U58 = BSC 81 = NGC 6266

RA : 17.0^h	PSA: 58	Type: GC
Dec: $-30°$	OSA: E18	Mag : 6.7
Con: Oph	CSA: 17/18	Size : $14.0'$

See: OSA DSC:TMO

ε Sco (mag 2.3)
$\rightarrow 2°$ E
$\rightarrow 4°$ N to M63

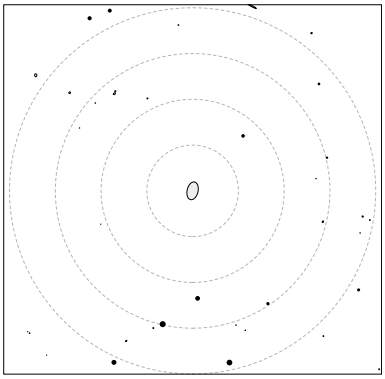


M63

BSC 145 = NGC 5055 = Sunflower Galaxy

RA : 13.3^h	PSA: 43	Type: Gal
Dec: $+42°$	OSA: N12	Mag : 8.6
Con: CVn	CSA: 4/5	Size : $13.5' \times 8.3'$

See: OSA DSC:TMO

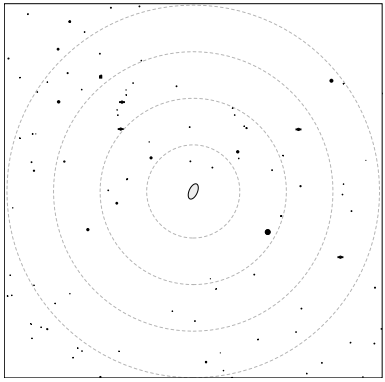


M64

U50 = BSC 149 = NGC 4826 = Black-Eye Galaxy

RA : 12.9^h	PSA: 45	Type: Gal
Dec: +22°	OSA: E13	Mag : 8.5
Con: Com	CSA: 5/11	Size : 9.2′×4.6′

See: OSA DSC:TMO



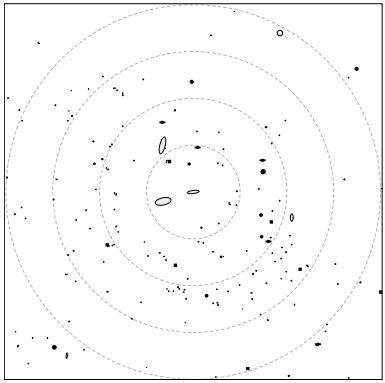
M65

NGC 3623

RA : 11.3^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 8.8
Con: Leo	CSA: 10/11	Size : 8.7′×2.2′

See: OSA DSC:TMO

Notes: M66 is 20′ E.



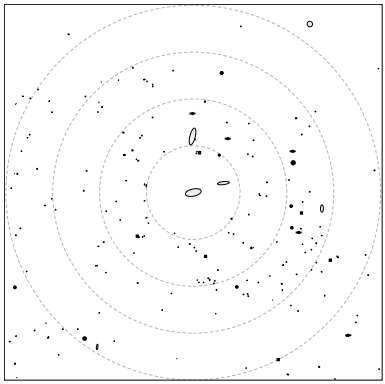
M66

BSC 130 = NGC 3627

RA : 11.3^h	PSA: 34	Type: Gal
Dec: +13°	OSA: E11	Mag : 8.9
Con: Leo	CSA: 10/11	Size : 8.2′×3.9′

See: OSA DSC:TMO

Notes: M65 is 20′ W.

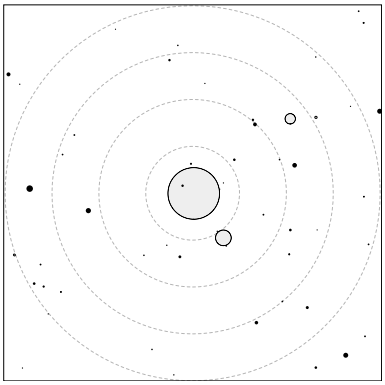


M67

U40 = BSC 51 = NGC 2682

RA : 08.9^h	PSA: 24	Type: OC
Dec: $+12°$	OSA: E9	Mag : 6.0
Con: Cnc	CSA: 10	Size : 25.0′

See: OSA DSC:TMO

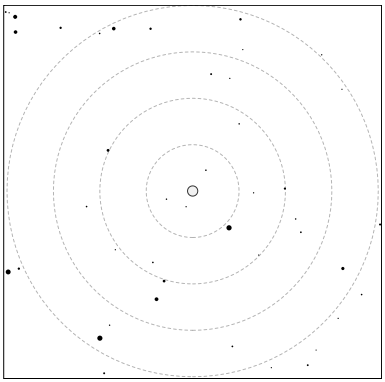


M68

NGC 4590

RA : 12.7^h	PSA: 47	Type: GC
Dec: $-27°$	OSA: E12	Mag : 7.6
Con: Hya	CSA: 10/11/16/17	Size : 11.0′

See: OSA DSC:TMO

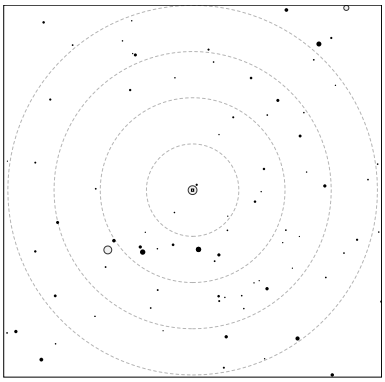


M69

NGC 6637

RA : 18.5^h	PSA: 67	Type: GC
Dec: $-32°$	OSA: E20	Mag : 7.4
Con: Sgr	CSA: 18	Size : 10.0′

See: OSA DSC:TMO

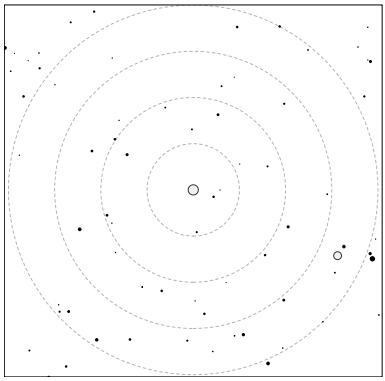


M70

BSC 105 = NGC 6681

RA : 18.7^h	PSA: 67	Type: GC
Dec: $-32°$	OSA: E20	Mag : 7.8
Con: Sgr	CSA: 18	Size : 8.0′

See: OSA DSC:TMO

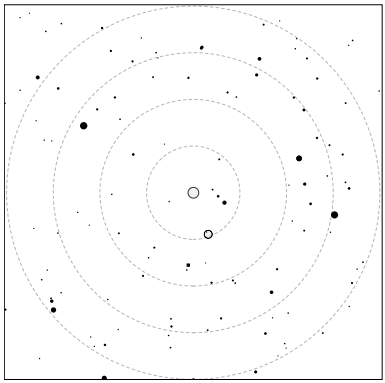


M71

BSC 52 = NGC 6838

RA : 19.9^h	PSA: 64	Type: GC
Dec: $+19°$	OSA: E21	Mag : 8.0
Con: Sge	CSA: 6/7/12/13	Size : 7.0′

See: OSA DSC:TMO



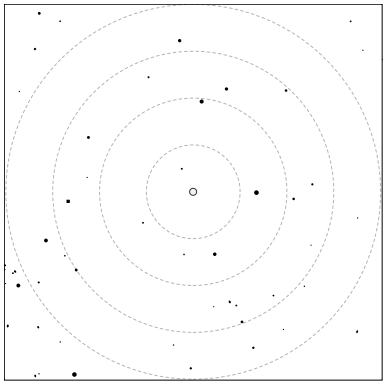
M72

BSC 148 = NGC 6981

RA : 20.9^h	PSA: 66/77	Type: GC
Dec: $-13°$	OSA: E24	Mag : 9.2
Con: Aqr	CSA: 13	Size : 7.0′

See: OSA DSC:TMO

Notes: M73 is 1.4° W.



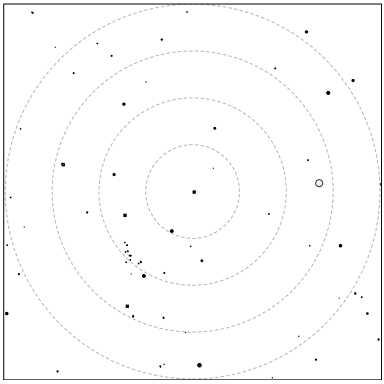
M73

NGC 6994 = Cr 426

RA : 21.0^h	PSA: 66/77	Type: AST
Dec: $-13°$	OSA: E24	Mag : 9.7
Con: Aqr	CSA: 13	Size : 1.4′

See: OSA DSC:TMO

Notes: Y-shaped asterism of four stars. M72 is 1.4° E.

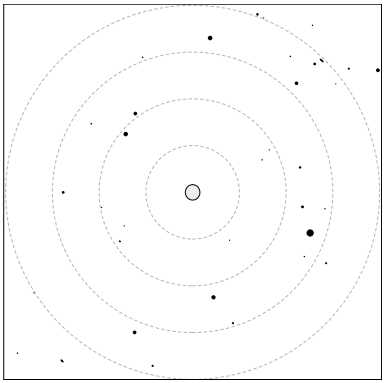


M74

NGC 628

RA : 01.6^h	PSA: 4/5	Type: Gal
Dec: $+16°$	OSA: E1	Mag : 8.5
Con: Psc	CSA: 8	Size : 11.0′

See: OSA DSC:TMO

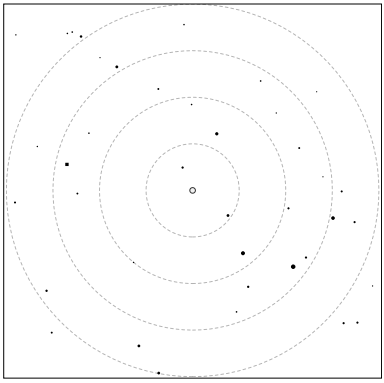


M75

BSC 144 = NGC 6864

RA : 20.1^h	PSA: 66	Type: GC
Dec: $-22°$	OSA: E22	Mag : 8.6
Con: Sgr	CSA: 12/13/18/19	Size : 7.0′

See: OSA DSC:TMO

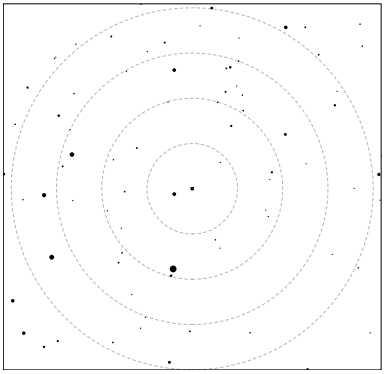


M76

NGC 651/650 = Little Dumbell

RA : 01.7^h	PSA: 2	Type: PN
Dec: $+52°$	OSA: N0	Mag : 10.4
Con: Per	CSA: 1/2/3/7	Size : $2.7' \times 1.8'$

See: OSA DSC:TMO

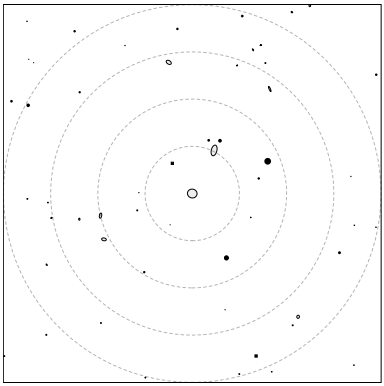


M77

U12 = NGC 1068

RA : 02.7^h	PSA: 4	Type: Gal
Dec: $-00°$	OSA: E0	Mag : 8.9
Con: Cet	CSA: 8	Size : $8.2' \times 7.3'$

See: OSA DSC:TMO

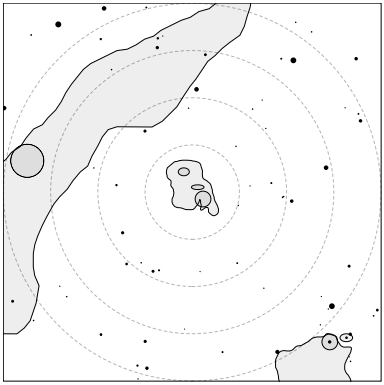


M78

BSC 114 = NGC 2068

RA : 05.8^h	PSA: 16	Type: BN
Dec: $+00°$	OSA: E5	Mag : 8.0
Con: Ori	CSA: 9	Size : $8.0' \times 6.0'$

See: OSA DSC:TMO

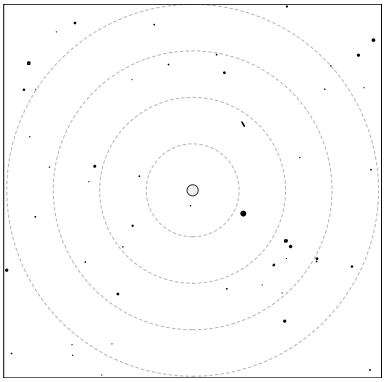


M79

BSC 126 = NGC 1904

RA : 05.4^h	PSA: 16	Type: GC
Dec: $-25°$	OSA: E4	Mag : 7.7
Con: Lep	CSA: 9/15	Size : 10.0′

See: OSA DSC:TMO



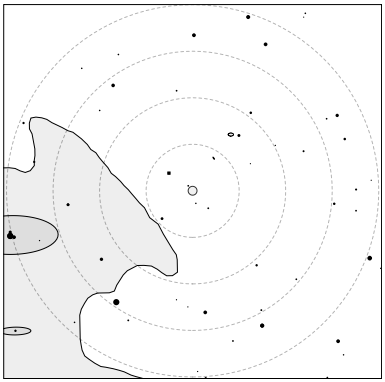
M80

BSC 101 = NGC 6093

RA : 16.3^h	PSA: 56	Type: GC
Dec: $-23°$	OSA: E18	Mag : 7.3
Con: Sco	CSA: 11/12/17/18	Size : 10.0′

See: OSA DSC:TMO

α Sco Antares (mag 0.9–1.1)
→ 3.0° W
→ 3.5° N to M80



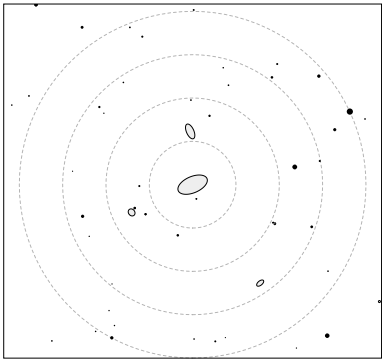
M81

U41 = BSC 72 = NGC 3031

RA : 09.9^h	PSA: 31	Type: Gal
Dec: $+69°$	OSA: N8	Mag : 6.9
Con: UMa	CSA: 1/4	Size : 24.0′ × 13.0′

See: OSA DSC:TMO

Notes: M82/U42 is 0.6° N.



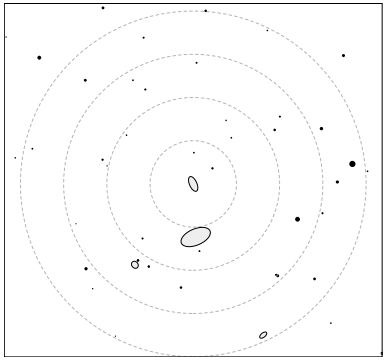
M82

U42 = BSC 122 = NGC 3034

RA : 09.9^h PSA: 31 Type: Gal
Dec: +70° OSA: N8 Mag : 8.4
Con: UMa CSA: 1/4 Size : 12.0′ × 5.6′

See: OSA DSC:TMO

Notes: M81/U41 is 0.6° S.

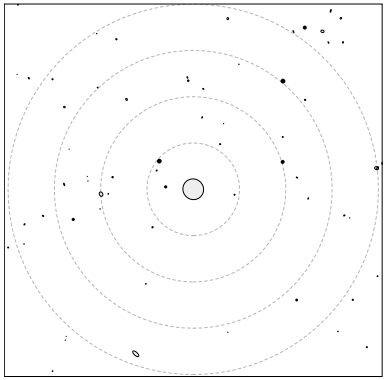


M83

BSC 89 = NGC 5236

RA : 13.6^h PSA: 47/48 Type: Gal
Dec: −30° OSA: E16 Mag : 7.6
Con: Hya CSA: 17 Size : 15.5′ × 13.0′

See: OSA DSC:TMO



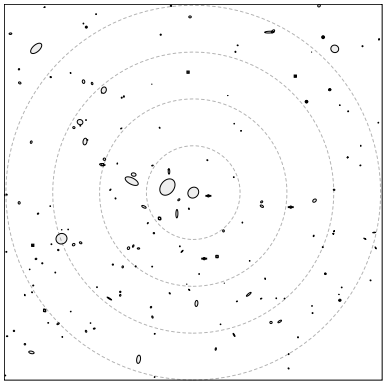
M84

U45 = NGC 4374

RA : 12.4^h PSA: 45/C Type: Gal
Dec: +13° OSA: E14 Mag : 9.1
Con: Vir CSA: 4/10/11 Size : 5.1′×4.1′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



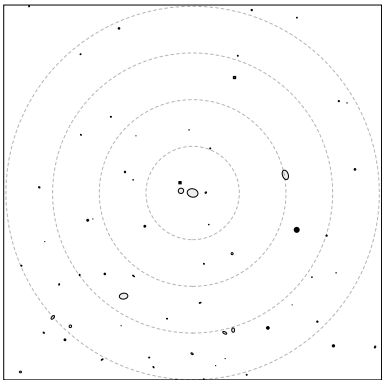
M85

NGC 4382

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: $+18°$	OSA: E14	Mag : 9.1
Con: Com	CSA: 4/5/10/11	Size : $7.5' \times 5.7'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



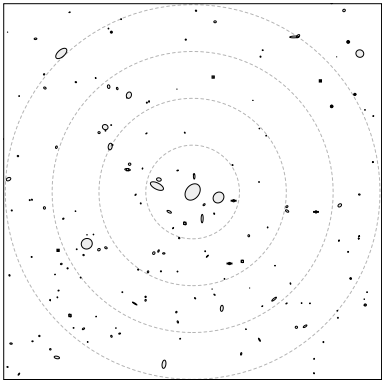
M86

U46 = NGC 4406

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: $+13°$	OSA: E14	Mag : 8.9
Con: Vir	CSA: 4/10/11	Size : $12.0' \times 9.3'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



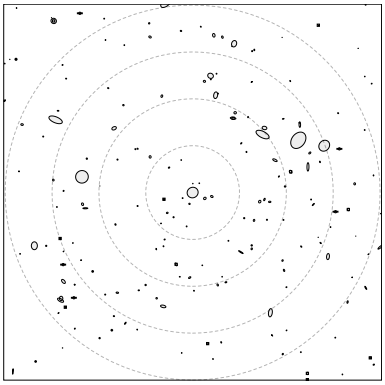
M87

U47 = NGC 4486

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: $+12°$	OSA: E14	Mag : 8.6
Con: Vir	CSA: 10/11	Size : $7.1'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



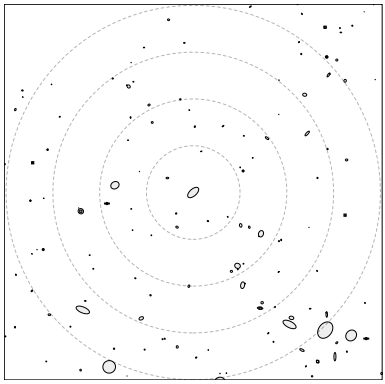
M88

NGC 4501

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: $+14°$	OSA: E14	Mag : 9.6
Con: Com	CSA: 5/10/11	Size : $6.1' \times 2.8'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



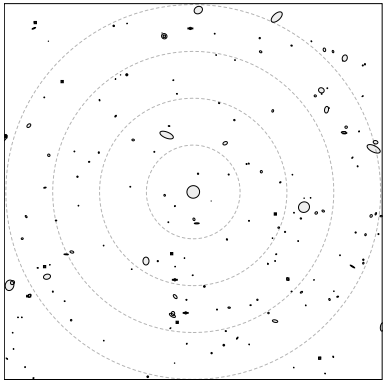
M89

NGC 4552

RA : 12.6^h	PSA: 45/C	Type: Gal
Dec: $+13°$	OSA: E14	Mag : 9.8
Con: Vir	CSA: 10/11	Size : $3.4'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



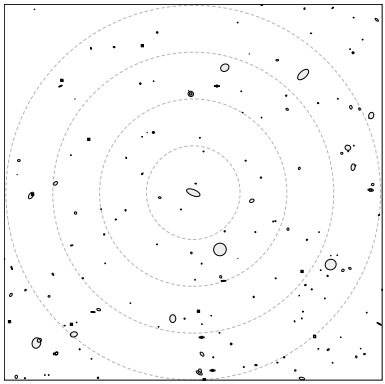
M90

NGC 4569

RA : 12.6^h	PSA: 45/C	Type: Gal
Dec: $+13°$	OSA: E14	Mag : 9.5
Con: Vir	CSA: 10/11	Size : $10.5' \times 4.4'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



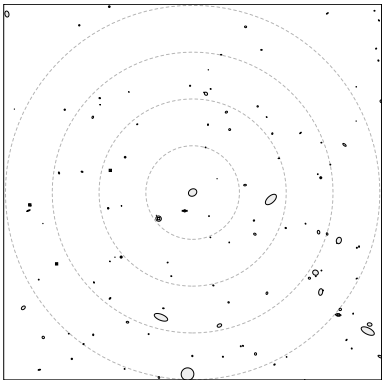
M91

NGC 4548

RA : 12.6^h	PSA: 45/C	Type: Gal
Dec: $+14°$	OSA: E14	Mag : 10.2
Con: Com	CSA: 5/10/11	Size : $5.0' \times 4.1'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



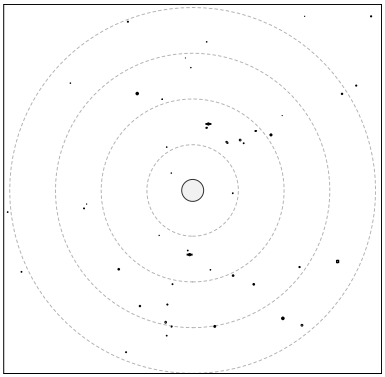
M92

U59 = BSC 59 = NGC 6341

RA : 17.3^h	PSA: 52	Type: GC
Dec: $+43°$	OSA: N14	Mag : 6.5
Con: Her	CSA: 5/6	Size : $14.0'$

See: OSA DSC:TMO

π Her (mag 3.2)
$\rightarrow$ 6.5° N to M92/U59

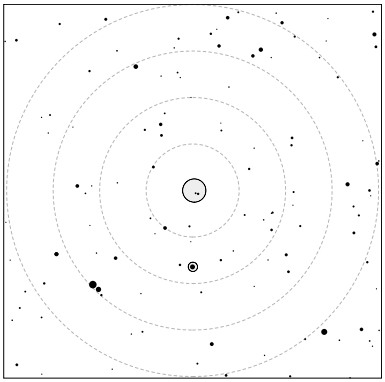


M93

BSC 48 = NGC 2447 = Cr 160

RA : 07.7^h	PSA: 26	Type: OC
Dec: $-24°$	OSA: E6	Mag : 6.2
Con: Pup	CSA: 9/10/15/16	Size : $10.0'$

See: OSA DSC:TMO

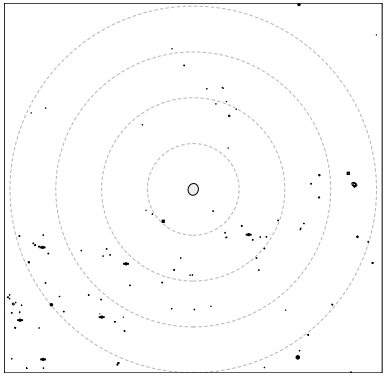


M94

U49 = BSC 134 = NGC 4736

RA : 12.8^h	PSA: 43	Type: Gal
Dec: +41°	OSA: N12	Mag : 8.2
Con: CVn	CSA: 4/5	Size : 11.5′ × 10.0′

See: OSA DSC:TMO



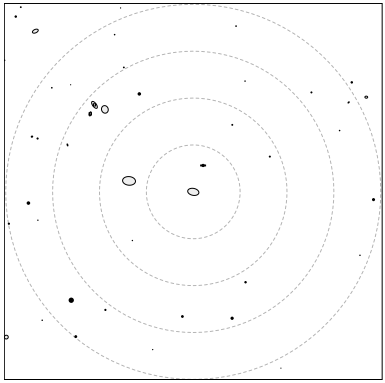
M95

NGC 3351

RA : 10.7^h	PSA: 34	Type: Gal
Dec: +12°	OSA: E11	Mag : 9.7
Con: Leo	CSA: 10	Size : 7.8′×4.5′

See: OSA DSC:TMO

Notes: M96 is 0.7° E. M105 is 1.3° NE.



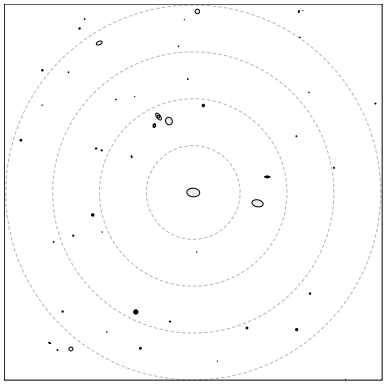
M96

NGC 3368

RA : 10.8^h	PSA: 34	Type: Gal
Dec: +12°	OSA: E11	Mag : 9.2
Con: Leo	CSA: 10	Size : 6.9′×4.6′

See: OSA DSC:TMO

Notes: M96 is 0.7° W. M105 is 0.8° NNE.



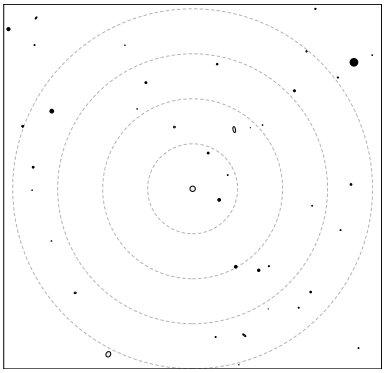
M97

NGC 3587

RA : 11.2^h	PSA: 32	Type: PN
Dec: +55°	OSA: N10	Mag : 9.8
Con: UMa	CSA: 1/4/5	Size : 3.4′

See: OSA DSC:TMO

Notes: M108 is 0.8° to NW.



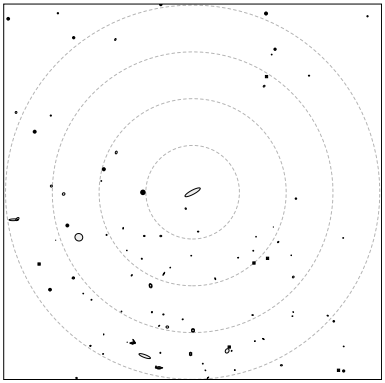
M98

NGC 4192

RA : 12.2^h	PSA: 45/C	Type: Gal
Dec: +15°	OSA: E14	Mag : 10.1
Con: Com	CSA: 4/5/10/11	Size : 9.1′×2.1′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



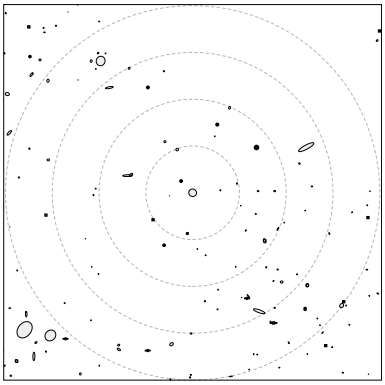
M99

NGC 4254

RA : 12.3^h	PSA: 45/C	Type: Gal
Dec: +14°	OSA: E14	Mag : 9.9
Con: Com	CSA: 4/5/10/11	Size : 4.6′×4.3′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



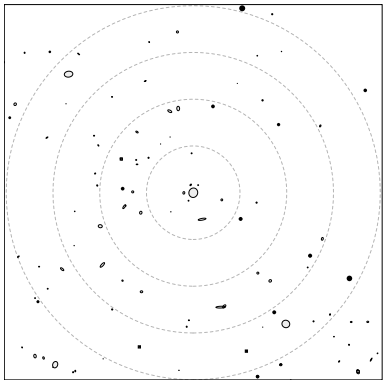
M100

NGC 4321

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: +16°	OSA: E14	Mag : 9.3
Con: Com	CSA: 4/5/10/11	Size : 6.2′×4.3′

See: OSA DSC:TMO

Notes: In the Virgo Cluster.

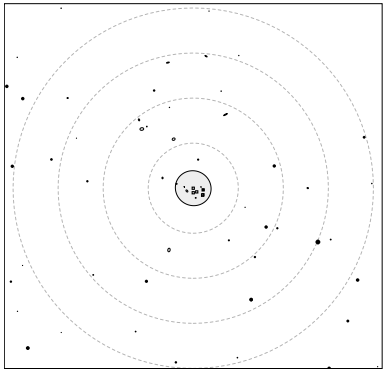


M101

BSC 100 = NGC 5457

RA : 14.1^h	PSA: 42	Type: Gal
Dec: +54°	OSA: N10	Mag : 7.9
Con: UMa	CSA: 1/4/5/6	Size : 26.0′

See: OSA DSC:TMO



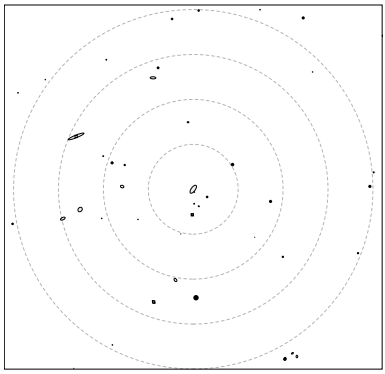
M102

NGC 5866

RA : 15.1^h	PSA: 42	Type: Gal
Dec: +56°	OSA: N16	Mag : 9.9
Con: Dra	CSA: 1/5/6	Size : 6.6′×3.2′

See: OSA DSC:TMO

Notes: Labelled 5866 in PSA



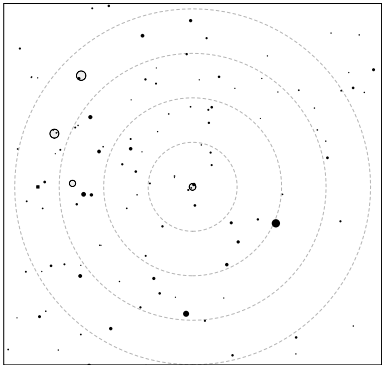
M103

BSC 83 = NGC 581

RA : 01.6^h	PSA: 3	Type: OC
Dec: $+61°$	OSA: N2	Mag : 7.4
Con: Cas	CSA: 1/2/3/7	Size : $6.0'$

See: OSA DSC:TMO

Notes: C10 is 1.7° ENE.

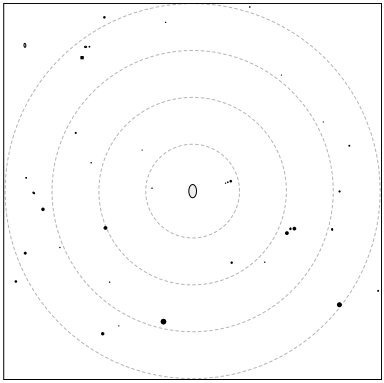


M104

U48 = BSC 133 = NGC 4594 = Sombrero Galaxy

RA : 12.7^h	PSA: 47	Type: Gal
Dec: $-12°$	OSA: E12	Mag : 8.0
Con: Vir	CSA: 10/11	Size : $7.1' \times 4.4'$

See: OSA DSC:TMO



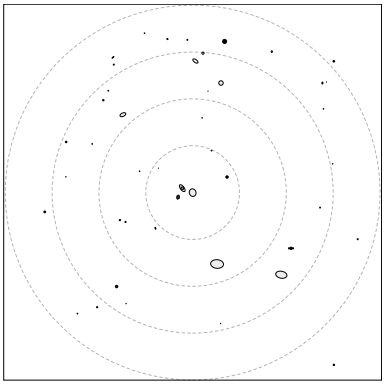
M105

NGC 3379

RA : 10.8^h	PSA: 34	Type: Gal
Dec: $+13°$	OSA: E11	Mag : 9.3
Con: Leo	CSA: 10	Size : $3.9'$

See: OSA DSC:TMO

Notes: M95 is 1.3° SW. M96 is 0.8° SSW.

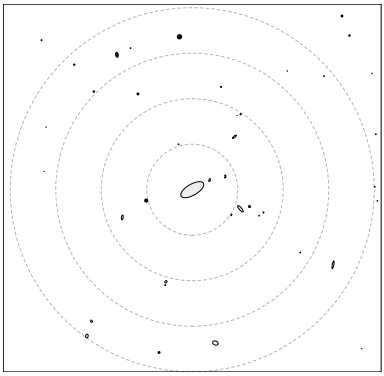


M106

BSC 141 = NGC 4258

RA : 12.3^h PSA: 43 Type: Gal
Dec: +47° OSA: N12 Mag : 8.4
Con: CVn CSA: 4/5 Size : 20.0′ × 8.4′

See: OSA DSC:TMO

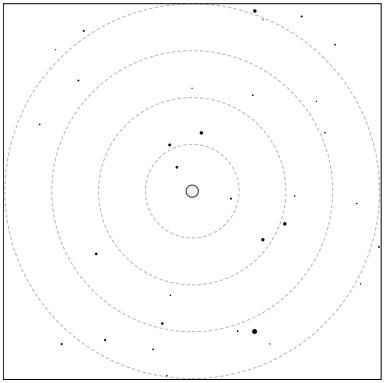


M107

BSC 129 = NGC 6171

RA : 16.5^h PSA: 56 Type: GC
Dec: −13° OSA: E17 Mag : 7.8
Con: Oph CSA: 11/12 Size : 13.0′

See: OSA DSC:TMO



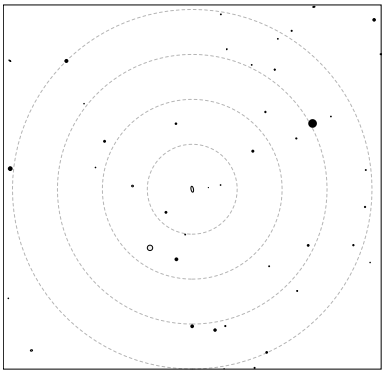
M108

NGC 3556

RA : 11.2^h PSA: 32/33 Type: Gal
Dec: +56° OSA: N10 Mag : 10.0
Con: UMa CSA: 1/4/5 Size : 8.2′×2.1′

See: OSA DSC:TMO

Notes: M97 is 0.8° to SE.

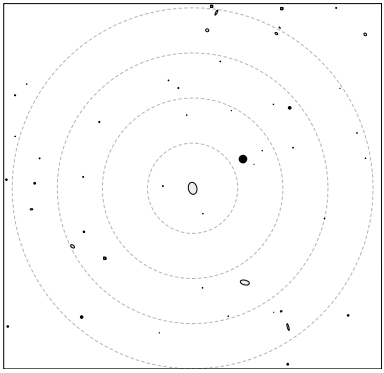


M109

NGC 3992

RA : 12.0^h	PSA: 32	Type: Gal
Dec: +53°	OSA: N10	Mag : 9.8
Con: UMa	CSA: 1/4/5	Size : 7.6′ ×4.3′

See: OSA DSC:TMO



M110

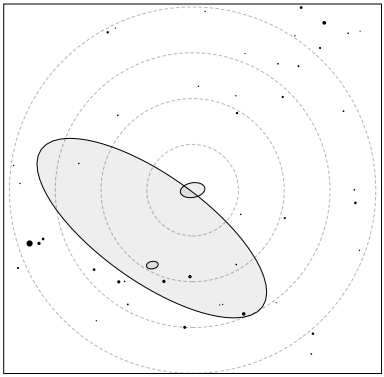
BSC 132 = NGC 205

RA : 00.7^h	PSA: 3	Type: Gal
Dec: +42°	OSA: N0	Mag : 8.1
Con: And	CSA: 2/7	Size : 19.5′ × 12.5′

See: OSA DSC:TMO

Notes: M31/U3 is 0.5° SE. M32/U2 is 1° SE.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 3.0° W
→ 3.5° N to M31/U3



The Caldwell Objects

The Caldwell objects were selected by Patrick Moore as a complement to the Messier objects. Again, I find O'Meara's *Deep-Sky Companion: The Caldwell Objects* to be excellent on the origin of the catalog, the appearance of the objects, and their nature.

I use C1 to C109 to label the Caldwell objects. This is common practice, but does not confirm to the IAU's recommendation on nomenclature, as O'Meara rightly notes (pp. 14–15).

The following table lists the objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, and other names. I follow the corrections given by O'Meara (p. 15) for C37, C49, C89, and C100.

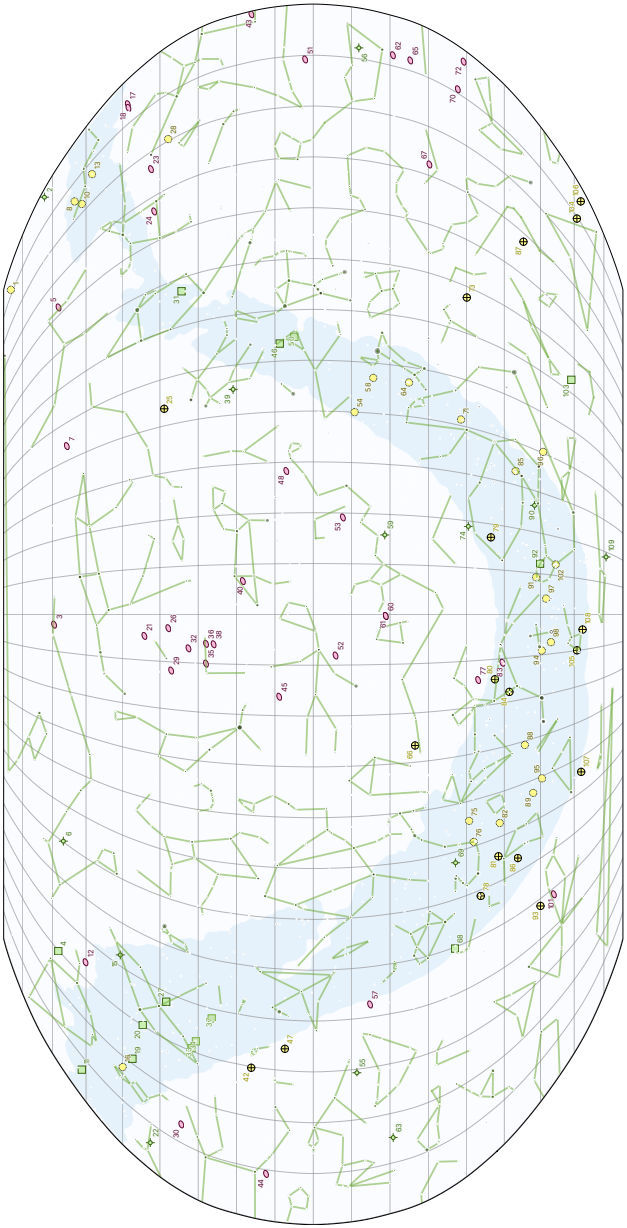
For completeness, I include finder charts for all the Caldwell objects, even bright ones like C41 (the Hyades).

I note that C41 (the Hyades) and C99 (the Coalsack Nebula) are much bigger than the finder charts. They are really binocular objects and are better located using a small-scale all-sky atlas.

Name	Position	Type	Other Names
C1	00.8 +85	OC	BSC 111 = NGC 188 = Mel 2
C2	00.2 +73	PN	NGC 40 = Bow-Tie Nebula
C3	12.3 +69	Gal	NGC 4236
C4	21.0 +68	BN	BSC 68 = NGC 7023 = Iris Nebula
C5	03.8 +68	Gal	BSC 142 = IC 342 = Hidden Galaxy
C6	18.0 +67	PN	NGC 6543 = Cat's Eye Nebula
C7	07.6 +66	Gal	BSC 131 = NGC 2403 = Fireworks Galaxy
C8	01.5 +63	OC	NGC 559 = Ghost's Goblet
C9	23.0 +63	BN	BSC 96 = Sh2-155 = Cave Nebula
C10	01.8 +61	OC	U5 = BSC 76 = NGC 663 = Horseshoe Cluster
C11	23.3 +61	BN	NGC 7635 = Bubble Nebula
C12	20.6 +60	Gal	NGC 6946 = Firecracker Galaxy
C13	01.3 +58	OC	U4 = BSC 62 = NGC 457 = E.T. Cluster = Owl Cluster
C14	02.3 +57	OC	U9/10 = NGC 869/884 = Double Cluster = h/χ Per Clusters
C15	19.7 +51	PN	BSC 128 = NGC 6826 = Blinking Planetary
C16	22.3 +50	OC	U85 = BSC 63 = NGC 7243
C17	00.6 +49	Gal	NGC 147
C18	00.6 +48	Gal	NGC 185
C19	21.9 +47	BN	BSC 78 = IC 5146 = Cr 470 = Cocoon Nebula
C20	21.0 +44	BN	BSC 16 = NGC 7000 = North America Nebula
C21	12.5 +44	Gal	NGC 4449 = Box Galaxy
C22	23.4 +43	PN	BSC 124 = NGC 7662 = Light Blue Snowball
C23	02.4 +42	Gal	NGC 891
C24	03.3 +42	Gal	NGC 1275 = Perseus A
C25	07.6 +39	GC	BSC 140 = NGC 2419
C26	12.3 +38	Gal	NGC 4244
C27	20.2 +38	BN	BSC 84 = NGC 6888 = Crescent Nebula
C28	01.9 +38	OC	U7 = NGC 752 = Mel 12
C29	13.2 +37	Gal	NGC 5005
C30	22.6 +34	Gal	NGC 7331
C31	05.3 +34	BN	BSC 49 = IC 405 = Flaming Star Nebula
C32	12.7 +33	Gal	NGC 4631 = Whale Galaxy
C33	20.9 +32	BN	BSC 73 = NGC 6992 = Eastern Veil Nebula
C34	20.8 +31	BN	BSC 74 = NGC 6960 = Western Veil Nebula
C35	13.0 +28	Gal	NGC 4889
C36	12.6 +28	Gal	NGC 4559
C37	20.2 +26	OC	NGC 6885
C38	12.6 +26	Gal	BSC 38 = NGC 4565 = Needle Galaxy
C39	07.5 +21	PN	U36 = NGC 2392 = Eskimo Nebula
C40	11.3 +18	Gal	NGC 3626
C41	04.4 +16	OC	U18 = BSC 1 = Mel 25 = Cr 50 = Hyades
C42	21.0 +16	GC	NGC 7006
C43	00.1 +16	Gal	NGC 7814
C44	23.1 +12	Gal	NGC 7479
C45	13.6 +09	Gal	NGC 5248
C46	06.7 +09	BN	BSC 137 = NGC 2261 = Hubble's Variable Nebula
C47	20.6 +07	GC	BSC 121 = NGC 6934
C48	09.2 +07	Gal	NGC 2775
C49	06.5 +05	BN	BSC 135 = NGC 2237/38/46 = Rosette Nebula
C50	06.5 +05	OC	U30 = BSC 25 = NGC 2244

Name	Position	Type	Other Names
C51	01.1 +02	Gal	IC 1613
C52	12.8 −06	Gal	NGC 4697
C53	10.1 −08	Gal	NGC 3115 = The Spindle
C54	08.0 −11	OC	BSC 90 = NGC 2506 = Mel 80 = Cr 170
C55	21.1 −11	PN	BSC 106 = NGC 7009 = Saturn Nebula
C56	00.8 −12	PN	BSC 107 = NGC 246
C57	19.7 −15	Gal	BSC 146 = NGC 6822 = IC 4895 = Barnard's Galaxy
C58	07.3 −16	OC	BSC 79 = NGC 2360 = Mel 64 = Caroline's Cluster
C59	10.4 −19	PN	U43 = BSC 125 = NGC 3242 = Ghost of Jupiter
C60	12.0 −19	Gal	NGC 4038 = NW Antennae Galaxy
C61	12.0 −19	Gal	NGC 4039 = SE Antennae Galaxy
C62	00.8 −21	Gal	NGC 247
C63	22.5 −21	PN	BSC 91 = NGC 7293 = Helix Nebula
C64	07.3 −25	OC	BSC 17 = NGC 2362 = τ CMa Cluster
C65	00.8 −25	Gal	BSC 108 = NGC 253 = Sculpter Galaxy
C66	14.7 −27	GC	NGC 5694
C67	02.8 −30	Gal	NGC 1097
C68	19.0 −37	BN	NGC 6729 = R CrA Nebula
C69	17.2 −37	PN	BSC 77 = NGC 6302 = Bug Nebula
C70	00.9 −38	Gal	BSC 138 = NGC 300
C71	07.9 −39	OC	BSC 41 = NGC 2477 = Mel 78
C72	00.2 −39	Gal	BSC 102 = NGC 55
C73	05.2 −40	GC	BSC 80 = NGC 1851
C74	10.1 −40	PN	NGC 3132
C75	16.4 −41	OC	BSC 42 = NGC 6124
C76	16.9 −42	OC	BSC 7 = NGC 6231 = Cr 315
C77	13.4 −43	Gal	NGC 5128 = Cen A
C78	18.1 −44	GC	BSC 60 = NGC 6541
C79	10.3 −46	GC	BSC 112 = NGC 3201
C80	13.4 −47	GC	BSC 14 = NGC 5139 = ω Cen
C81	17.4 −48	GC	BSC 98 = NGC 6352
C82	16.7 −49	OC	BSC 30 = NGC 6193
C83	13.1 −49	Gal	BSC 147 = NGC 4945
C84	13.8 −51	GC	BSC 92 = NGC 5286
C85	08.7 −53	OC	BSC 6 = IC 2391 = o Vel Cluster
C86	17.7 −54	GC	BSC 67 = NGC 6397
C87	03.2 −55	GC	BSC 115 = NGC 1261
C88	15.1 −56	OC	BSC 103 = NGC 5823 = Mel 131
C89	16.3 −58	OC	BSC 33 = NGC 6087 = S Nor Cluster
C90	09.4 −58	PN	NGC 2867
C91	11.1 −59	OC	BSC 9 = NGC 3532
C92	10.8 −60	BN	BSC 3 = NGC 3372 = η Car Nebula
C93	19.2 −60	GC	BSC 34 = NGC 6752
C94	12.9 −60	OC	BSC 20 = NGC 4755 = Mel 114 = Jewel Box
C95	16.1 −60	OC	BSC 29 = NGC 6025
C96	08.0 −61	OC	BSC 13 = NGC 2516 = Southern Beehive Cluster
C97	11.6 −62	OC	BSC 32 = NGC 3766 = Cr 248
C98	12.7 −63	OC	BSC 70 = NGC 4609 = Coalsack Cluster
C99	12.5 −64	DN	Coalsack
C100	11.6 −63	OC	BSC 22 = Cr 249 = λ Cen Cluster

Name	Position	Type	Other Names
C101	19.2 −64	Gal	BSC 143 = NGC 6744
C102	10.7 −64	OC	BSC 5 = IC 2602 = Mel 102 = Cr 229 = θ Car Cluster
C103	05.6 −69	BN	BSC 109 = NGC 2070 = Tarantula Nebula
C104	01.1 −71	GC	BSC 64 = NGC 362 = Mel 4
C105	13.0 −71	GC	BSC 97 = NGC 4833
C106	00.4 −72	GC	BSC 26 = NGC 104 = Mel 1 = 47 Tuc
C107	16.4 −72	GC	BSC 139 = NGC 6101
C108	12.4 −73	GC	BSC 99 = NGC 4372
C109	10.2 −81	PN	NGC 3195



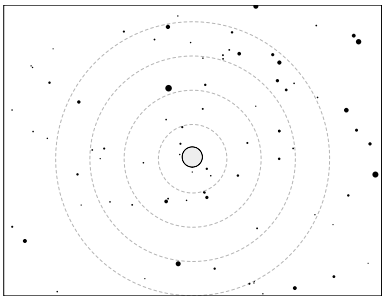
The distribution of Caldwell objects over the whole sky

C1

BSC 111 = NGC 188 = Mel 2

RA : 00.8^h	PSA: 1	Type: OC
Dec: +85°	OSA:	Mag : 7.1
Con: Cep	CSA: 1	Size : 15.0′

See: DSC:TCO



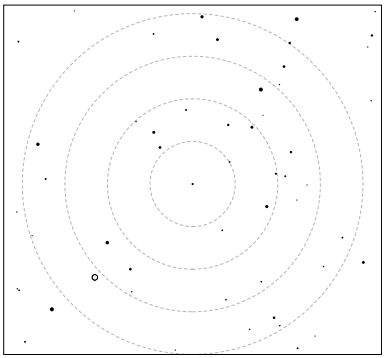
C2

NGC 40 = Bow-Tie Nebula

RA : 00.2^h	PSA: 1/71	Type: PN
Dec: +73°	OSA:	Mag : 12.3
Con: Cep	CSA: 1	Size : 0.6′

See: DSC:TCO

Notes: Central star is magnitude 11.6 (DSC:TCO).

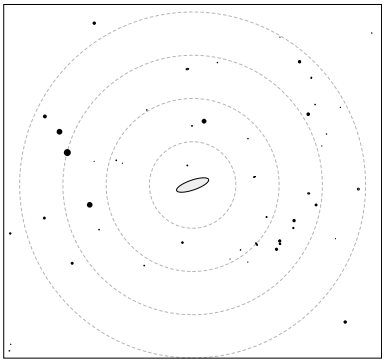


C3

NGC 4236

RA : 12.3^h	PSA: 31/41	Type: Gal
Dec: +69°	OSA:	Mag : 9.6
Con: Dra	CSA: 1/5	Size : 20.0′ × 8.0′

See: DSC:TCO



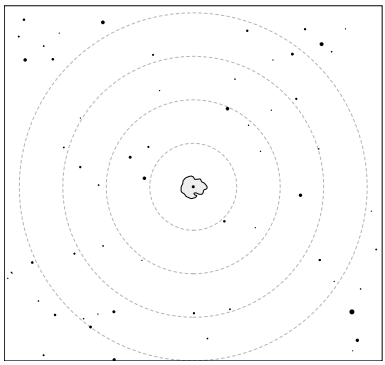
C4

BSC 68 = NGC 7023 = Iris Nebula

RA : 21.0^h	PSA: 61/71	Type: BN
Dec: +68°	OSA:	Mag : 7.7
Con: Cep	CSA: 1/6/7	Size : 10.0′ × 8.0′

See: DSC:TCO

Notes: Central star is magnitude 6.8 (DSC:TCO).

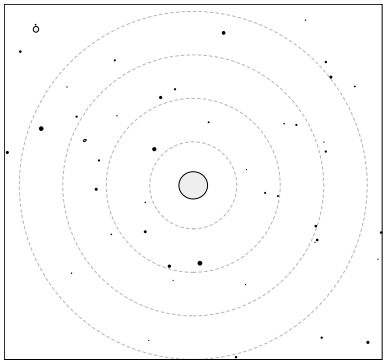


C5

BSC 142 = IC 342 = Hidden Galaxy

RA : 03.8^h	PSA: 1/11	Type: Gal
Dec: +68°	OSA:	Mag : 7.9
Con: Cam	CSA: 1/2/3	Size : 16.0′

See: DSC:TCO

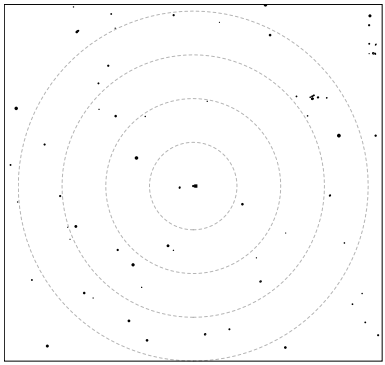


C6

NGC 6543 = Cat's Eye Nebula

RA : 18.0^h	PSA: 51/61	Type: PN
Dec: +67°	OSA: N16	Mag : 8.1
Con: Dra	CSA: 1/6	Size : 0.4′ × 0.3′

See: OSA DSC:TCO

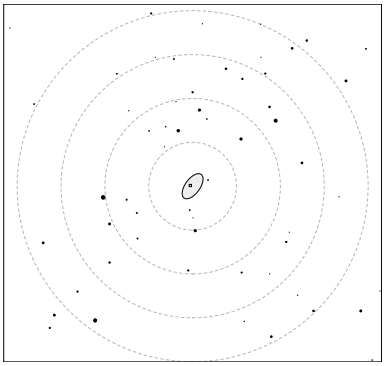


C7

BSC 131 = NGC 2403 = Fireworks Galaxy

RA : 07.6^h	PSA: 21	Type: Gal
Dec: +66°	OSA:	Mag : 7.3
Con: Cam	CSA: 1/3/4	Size : 24.0′ × 13.0′

See: OSA DSC:TCO



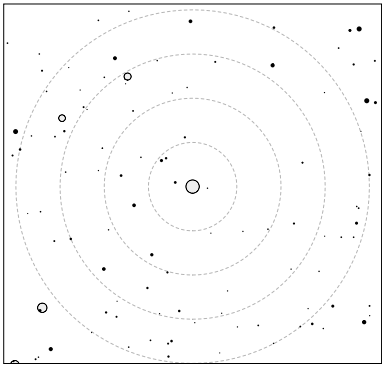
C8

NGC 559 = Ghost's Goblet

RA : 01.5^h	PSA: 1/3	Type: OC
Dec: +63°	OSA: N2	Mag : 9.5
Con: Cas	CSA: 1/2	Size : 7.0′

See: OSA DSC:TCO

Notes: C10 is 3° to SE.



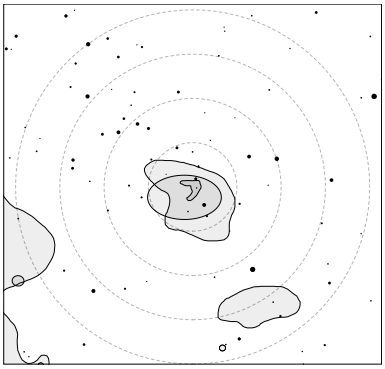
C9

BSC 96 = Sh2-155 = Cave Nebula

RA : 23.0^h	PSA: 71/72	Type: BN
Dec: +63°	OSA:	Mag : 10.0
Con: Cep	CSA: 1/7	Size : 50.0′ × 30.0′

See: DSC:TCO

Notes: Not shown in the CSA.



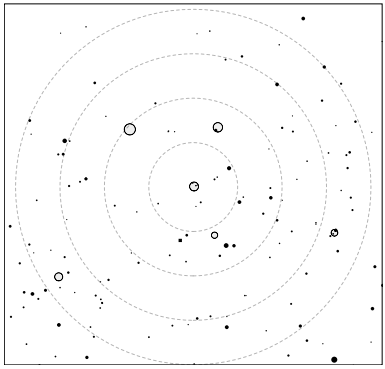
C10

U5 = BSC 76 = NGC 663 = Horseshoe Cluster

RA : 01.8^h	PSA: 1/2	Type: OC
Dec: +61°	OSA: N2	Mag : 6.7
Con: Cas	CSA: 1/2	Size : 15.0′

See: OSA DSC:TCO

Notes: M103 is 1.7° WSW. C8 is 3° NW.



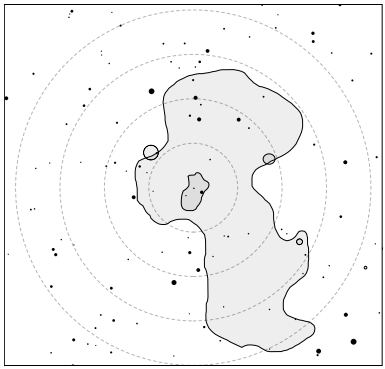
C11

NGC 7635 = Bubble Nebula

RA : 23.3^h	PSA: 71/72	Type: BN
Dec: +61°	OSA:	Mag : 10
Con: Cas	CSA: 1/2/7	Size : 15.0′ × 8.0′

See: DSC:TCO

Notes: M52 is 0.6° to NE.

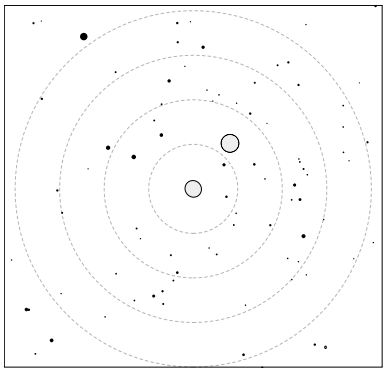


C12

NGC 6946 = Firecracker Galaxy

RA : 20.6^h	PSA: 61/62	Type: Gal
Dec: +60°	OSA: N22	Mag : 8.8
Con: Cep/Cyg	CSA: 1/6/7	Size : 11.0′ × 9.0′

See: OSA DSC:TCO

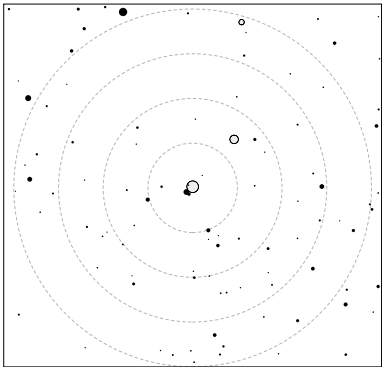


C13

U4 = BSC 62 = NGC 457 = E.T. Cluster = Owl Cluster

RA : 01.3^h	PSA: 1/3	Type: OC
Dec: $+58°$	OSA: N2	Mag : 6.4
Con:	CSA:	Size : $20.0'$

See: OSA DSC:TCO



C14

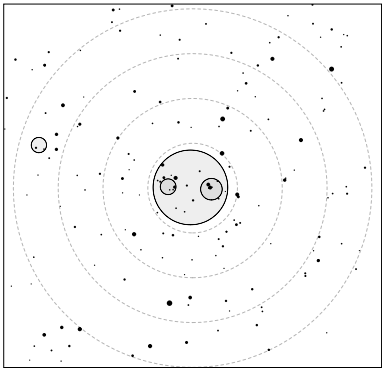
U9/10 = NGC 869/884 = Double Cluster = h/χ Per Clusters

RA : 02.3^h	PSA: 1/2	Type: OC
Dec: $+57°$	OSA: N2	Mag : 4.5/5.7
Con:	CSA:	Size : $18.0'$

See: OSA DSC:TCO

Notes: NGC 869 (h Per) is to W and brighter; NGC 884 (χ Per) is to E and fainter. Both are about $18'$ in diameter (DSC:TCO). U8 is $2°$ N. U11 is $2.5°$ SE.

α Per Mirfak (mag 1.8)
→ $9°$ E passing ι Per and θ Per to 64/65 And
→ $7°$ N to C14/U9/U10

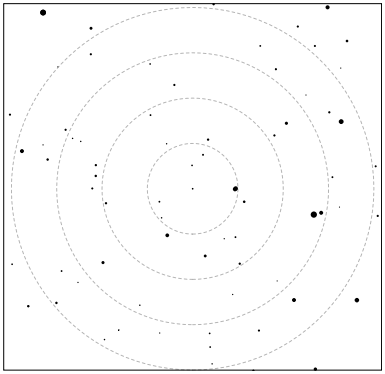


C15

BSC 128 = NGC 6826 = Blinking Planetary

RA : 19.7^h	PSA: 62	Type: PN
Dec: $+51°$	OSA: N18	Mag : 8.5
Con:	CSA:	Size : $0.5' \times 0.4'$

See: OSA DSC:TCO

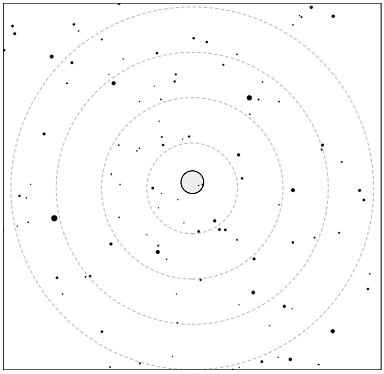


C16

U85 = BSC 63 = NGC 7243

RA : 22.3^h PSA: 73 Type: OC
Dec: +50° OSA: N24 Mag : 6.4
Con: CSA: Size : 30.0′

See: OSA DSC:TCO



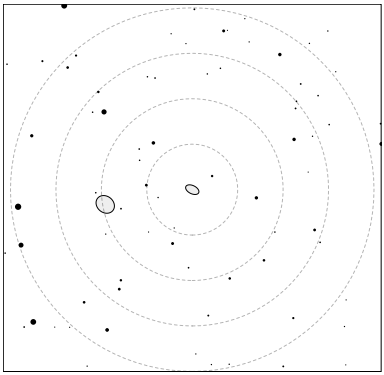
C17

NGC 147

RA : 00.6^h PSA: 3 Type: Gal
Dec: +49° OSA: Mag : 9.5
Con: CSA: Size : 18.0′ × 11.0′

See: DSC:TCO

Notes: C18 is 1° E.



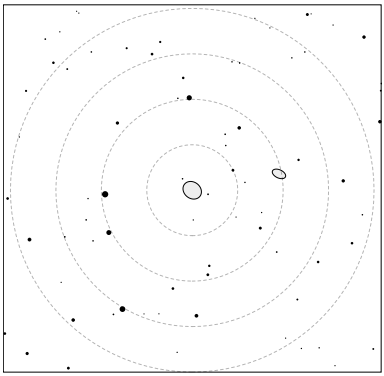
C18

NGC 185

RA : 00.6^h PSA: 3 Type: Gal
Dec: +48° OSA: Mag : 9.2
Con: CSA: Size : 17.0′ × 14.0′

See: DSC:TCO

Notes: C17 is 1° W.



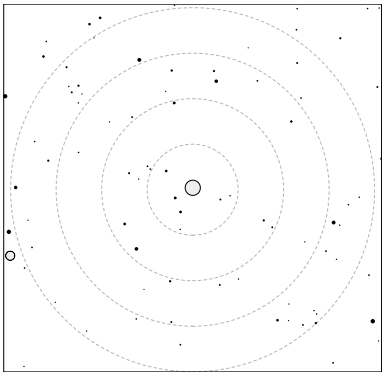
C19

BSC 78 = IC 5146 = Cr 470 = Cocoon Nebula

RA : 21.9^h	PSA: 73	Type: BN
Dec: +47°	OSA:	Mag : 9.3
Con:	CSA:	Size : 10.0′

See: DSC:TCO

Notes: Cluster is magitude 7.2 (DSC:TCO). U84 is 2° ESE.



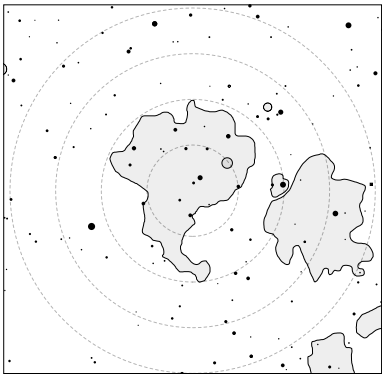
C20

BSC 16 = NGC 7000 = North America Nebula

RA : 21.0^h	PSA: 62	Type: BN
Dec: +44°	OSA: N20	Mag : 3.8
Con:	CSA:	Size : 100.0′ × 60.0′

See: OSA DSC:TCO

α Cyg Deneb (mag 1.3)
→ 1° S
→ 3° E to C20

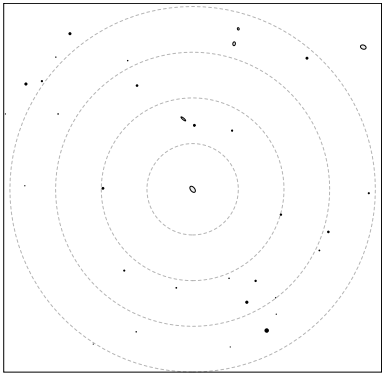


C21

NGC 4449 = Box Galaxy

RA : 12.5^h	PSA: 32/43	Type: Gal
Dec: +44°	OSA: N12	Mag : 9.6
Con:	CSA:	Size : 5.0′×4.0′

See: OSA DSC:TCO

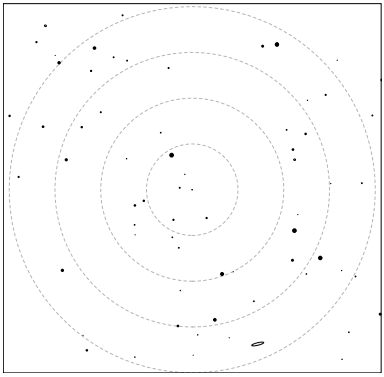


C22

BSC 124 = NGC 7662 = Light Blue Snowball

RA : 23.4^h	PSA: 72	Type: PN
Dec: +43°	OSA: N24	Mag : 8.3
Con:	CSA:	Size : 0.5′

See: OSA DSC:TCO

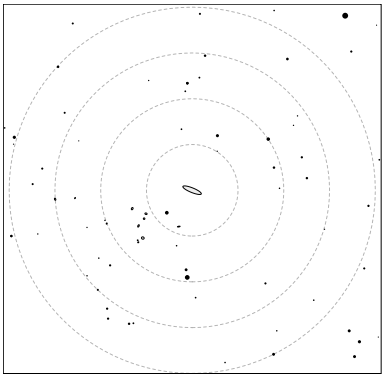


C23

NGC 891

RA : 02.4^h	PSA: 2	Type: Gal
Dec: +42°	OSA: N0	Mag : 9.9
Con:	CSA:	Size : 12.0′ × 3.0′

See: OSA DSC:TCO

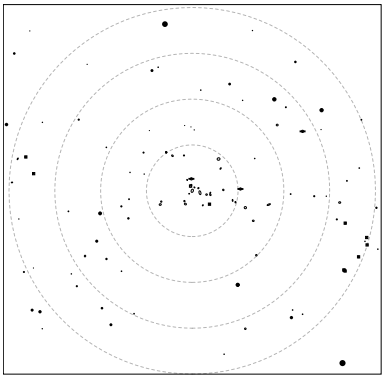


C24

NGC 1275 = Perseus A

RA : 03.3^h	PSA: 13	Type: Gal
Dec: +42°	OSA:	Mag : 11.9
Con:	CSA:	Size : 3.0′×2.0′

See: DSC:TCO

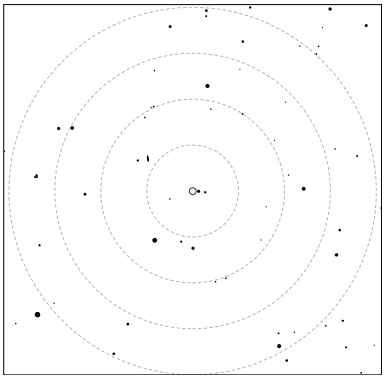


C25

BSC 140 = NGC 2419

RA : 07.6^h	PSA: 23	Type: GC
Dec: +39°	OSA:	Mag : 10.4
Con:	CSA:	Size : 4.6′

See: DSC:TCO

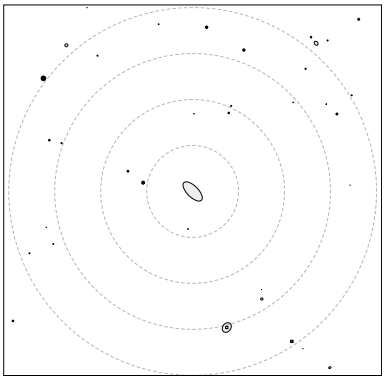


C26

NGC 4244

RA : 12.3^h	PSA: 32/43	Type: Gal
Dec: +38°	OSA: N12	Mag : 10.4
Con:	CSA:	Size : 16.0′ × 2.0′

See: OSA DSC:TCO

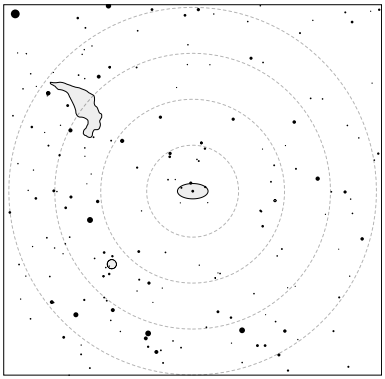


C27

BSC 84 = NGC 6888 = Crescent Nebula

RA : 20.2^h	PSA: 62	Type: BN
Dec: +38°	OSA:	Mag : 8.8
Con:	CSA:	Size : 18.0′ × 13.0′

See: DSC:TCO

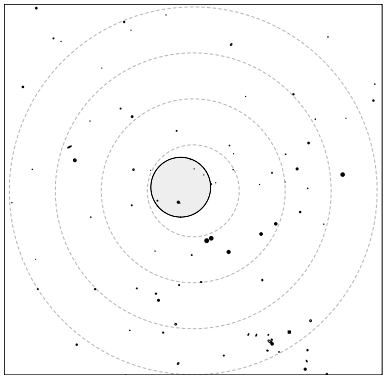


C28

U7 = NGC 752 = Mel 12

RA : 01.9^h	PSA: 2	Type: OC
Dec: $+38°$	OSA: N0	Mag : 5.7
Con:	CSA:	Size : 75.0′

See: OSA DSC:TCO

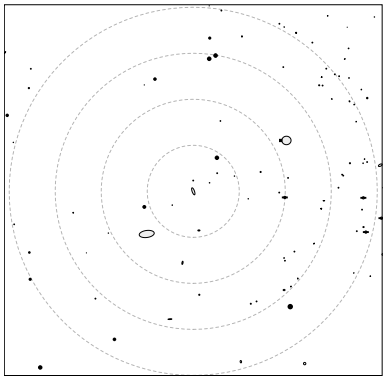


C29

NGC 5005

RA : 13.2^h	PSA: 43	Type: Gal
Dec: $+37°$	OSA:	Mag : 9.8
Con:	CSA:	Size : 6.0′×3.0′

See: DSC:TCO

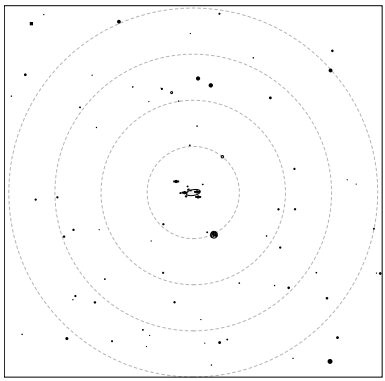


C30

NGC 7331

RA : 22.6^h	PSA: 72/74	Type: Gal
Dec: $+34°$	OSA: E23	Mag : 9.5
Con:	CSA:	Size : 9.7′×4.5′

See: OSA DSC:TCO



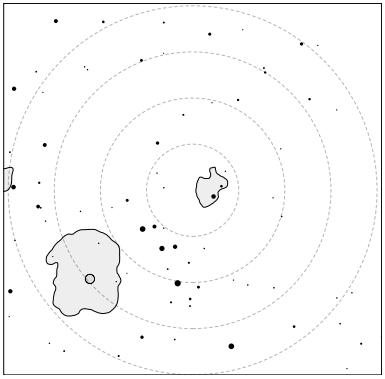
C31

BSC 49 = IC 405 = Flaming Star Nebula

RA : 05.3^h	PSA: 12	Type: BN	
Dec: $+34°$	OSA:	Mag : 9.2	
Con:	CSA:	Size : $30.0'$	×
		$20.0'$	

See: DSC:TCO

Notes: AE Aur (towards the SW edge) is magnitude 6 and variable (DSC:TCO).

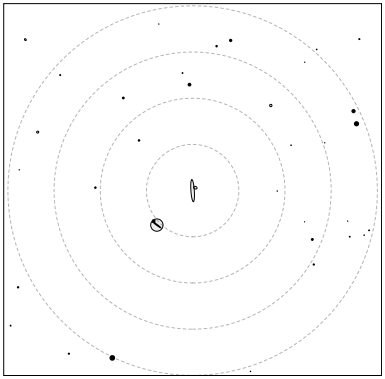


C32

NGC 4631 = Whale Galaxy

RA : 12.7^h	PSA: 43/45	Type: Gal	
Dec: $+33°$	OSA: E13	Mag : 9.2	
Con:	CSA:	Size : $15.0'$	×
		$4.0'$	

See: OSA DSC:TCO



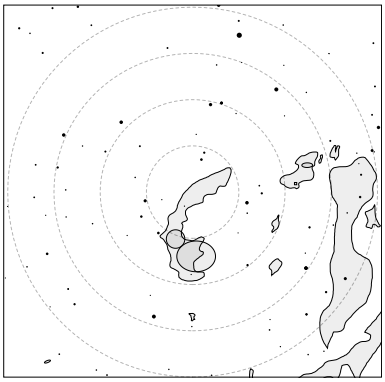
C33

BSC 73 = NGC 6992 = Eastern Veil Nebula

RA : 20.9^h	PSA: 62	Type: BN	
Dec: $+32°$	OSA: N20	Mag : 7.5	
Con:	CSA:	Size : $60.0'$	×
		$8.0'$	

See: OSA DSC:TCO

Notes: C34 is $2.5°$ SW.



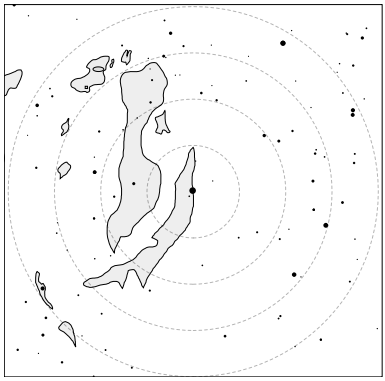
C34

BSC 74 = NGC 6960 = Western Veil Nebula

RA : 20.8^h	PSA: 62	Type: BN
Dec: +31°	OSA: N20	Mag : 7.9
Con:	CSA:	Size : 70.0′ × 6.0′

See: OSA DSC:TCO

Notes: C33 is 2.5° NE.

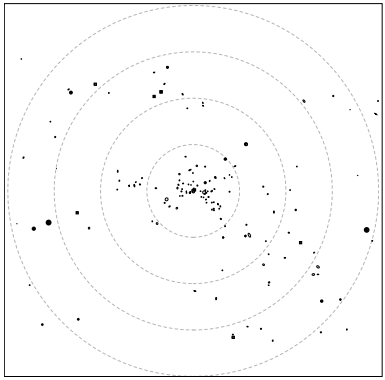


C35

NGC 4889

RA : 13.0^h	PSA: 43/45	Type: Gal
Dec: +28°	OSA:	Mag : 11.5
Con:	CSA:	Size : 3.0′×2.0′

See: DSC:TCO



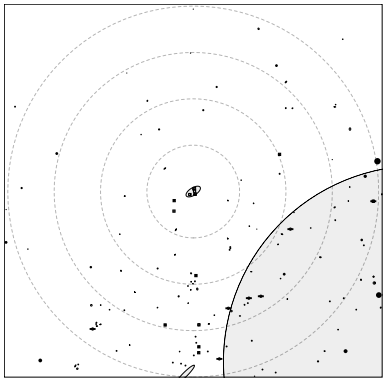
C36

NGC 4559

RA : 12.6^h	PSA: 43/45	Type: Gal
Dec: +28°	OSA: E13	Mag : 10.0
Con:	CSA:	Size : 11.0′ × 5.0′

See: OSA DSC:TCO

Notes: C38 is 2° S. U44 is 4° to SW.

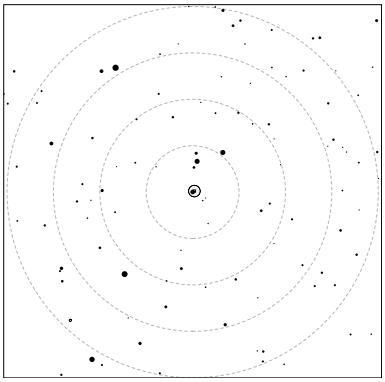


C37

NGC 6885

RA : 20.2^h	PSA: 62/64	Type: OC
Dec: +26°	OSA:	Mag : 8.1
Con:	CSA:	Size : 20.0′

See: DSC:TCO



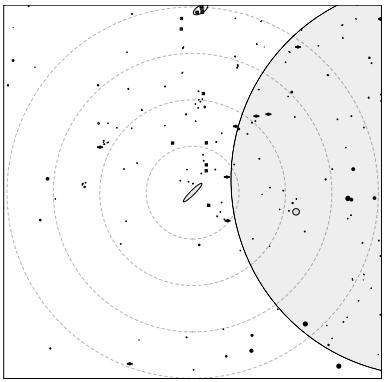
C38

BSC 38 = NGC 4565 = Needle Galaxy

RA : 12.6^h	PSA: 43/45	Type: Gal
Dec: +26°	OSA: E13	Mag : 9.6
Con:	CSA:	Size : 16.0′ × 2.0′

See: OSA DSC:TCO

Notes: C36 is 2° N. U44 is 3° to W.



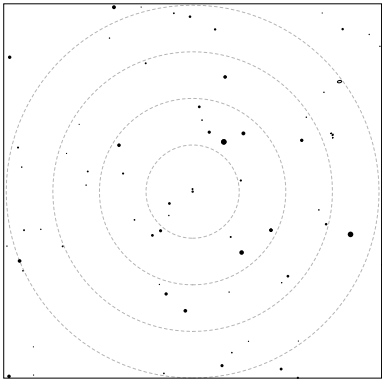
C39

U36 = NGC 2392 = Eskimo Nebula

RA : 07.5^h	PSA: 25	Type: PN
Dec: +21°	OSA: E7	Mag : 9.2
Con:	CSA:	Size : 0.8′×0.7′

See: OSA DSC:TMO DSC:TCO

Notes: The open cluster NGC 2420 is 2° N (see DSC:TCO).

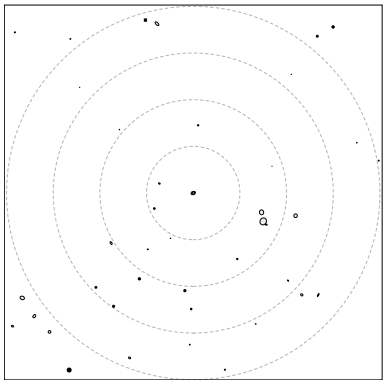


C40

NGC 3626

RA : 11.3^h	PSA: 34	Type: Gal
Dec: $+18°$	OSA:	Mag : 11.0
Con:	CSA:	Size : $3.0' \times 2.2'$

See: DSC:TCO



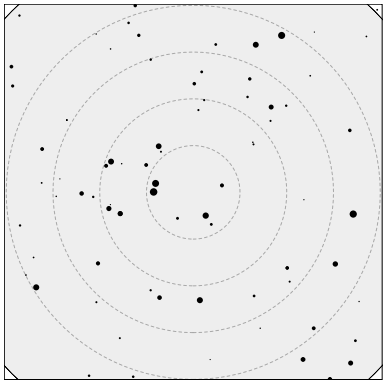
C41

U18 = BSC 1 = Mel 25 = Cr 50 = Hyades

RA : 04.4^h	PSA: 15	Type: OC
Dec: $+16°$	OSA: E3	Mag : 0.5
Con:	CSA:	Size : $5.5°$

See: OSA DSC:TCO

Notes: Much bigger than the finder chart.

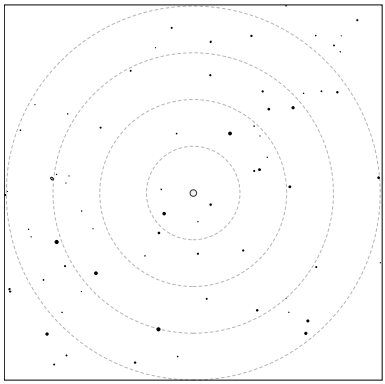


C42

NGC 7006

RA : 21.0^h	PSA: 64/75	Type: GC
Dec: $+16°$	OSA:	Mag : 10.6
Con:	CSA:	Size : $3.6'$

See: DSC:TCO

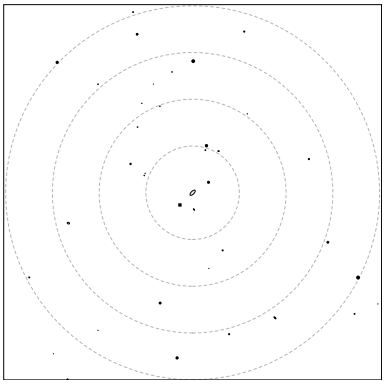


C43

NGC 7814

RA : 00.1^h	PSA: 5/74	Type: Gal
Dec: +16°	OSA:	Mag : 10.6
Con:	CSA:	Size : 6.0′×2.0′

See: DSC:TCO

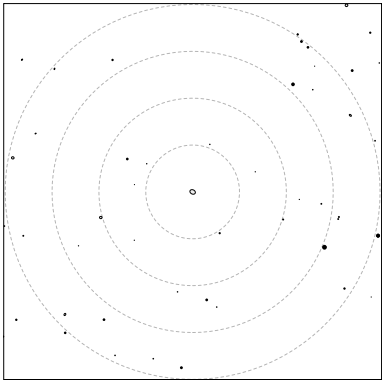


C44

NGC 7479

RA : 23.1^h	PSA: 74	Type: Gal
Dec: +12°	OSA:	Mag : 10.8
Con:	CSA:	Size : 4.0′×3.0′

See: DSC:TCO

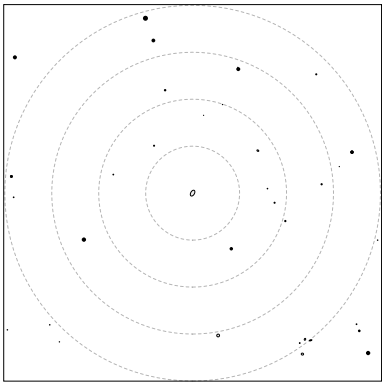


C45

NGC 5248

RA : 13.6^h	PSA: 44	Type: Gal
Dec: +09°	OSA:	Mag : 10.3
Con:	CSA:	Size : 6.0′×5.0′

See: DSC:TCO



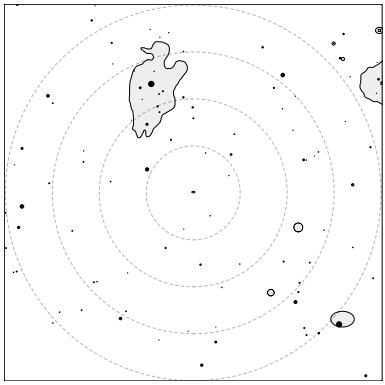
C46

BSC 137 = NGC 2261 = Hubble's Variable Nebula

RA : 06.7^h	PSA: 25/E	Type: BN
Dec: +09°	OSA: E7	Mag : 10
Con:	CSA:	Size : 4.0′×2.0′

See: OSA DSC:TCO

Notes: The magnitude of the nebula is variable (DSC:TCO). U31 is 1.25° NNE.

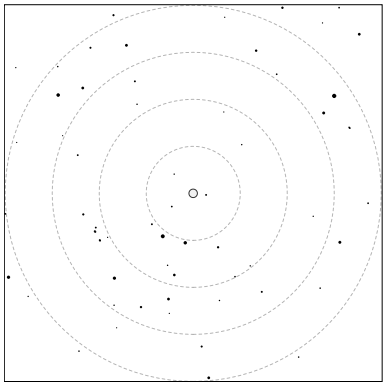


C47

BSC 121 = NGC 6934

RA : 20.6^h	PSA: 64	Type: GC
Dec: +07°	OSA:	Mag : 8.8
Con:	CSA:	Size : 7.1′

See: DSC:TCO

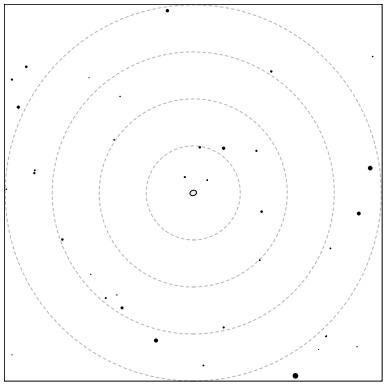


C48

NGC 2775

RA : 09.2^h	PSA: 24/35	Type: Gal
Dec: +07°	OSA:	Mag : 10.1
Con:	CSA:	Size : 5.0′×4.0′

See: DSC:TCO



C49

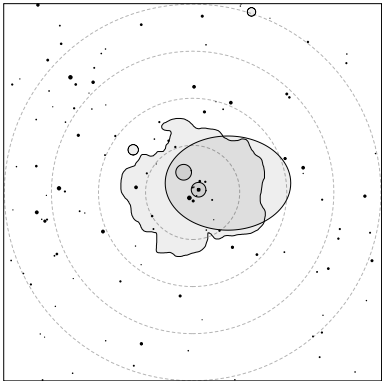
BSC 135 = NGC 2237/38/46 = Rosette Nebula

RA : 06.5^h	PSA: 25/E	Type: BN	
Dec: +05°	OSA: E9	Mag : 5	
Con:	CSA:	Size : 80.0′	×
		60.0′	

See: OSA DSC:TCO

Notes: Surrounding C50.

α Ori Betelgeuse (mag 0.3–0.9)
→ 3° S
→ 7° E to 8 Mon
→ 2° E to C49



C50

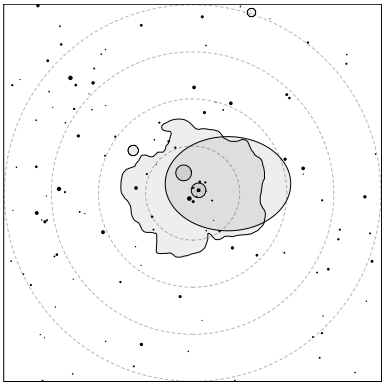
U30 = BSC 25 = NGC 2244

RA : 06.5^h	PSA: 25/E	Type: OC
Dec: +05°	OSA: E9	Mag : 4.4
Con:	CSA:	Size : 30.0′

See: OSA DSC:TCO

Notes: Surrounded by C49.

α Ori Betelgeuse (mag 0.3–0.9)
→ 3° S
→ 7° E to 8 Mon
→ 2° E to C50/U30

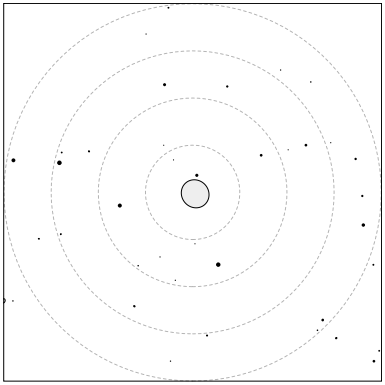


C51

IC 1613

RA : 01.1^h	PSA: 5/7	Type: Gal	
Dec: +02°	OSA:	Mag : 9.2	
Con:	CSA:	Size : 19.0′	×
		17.0′	

See: DSC:TCO

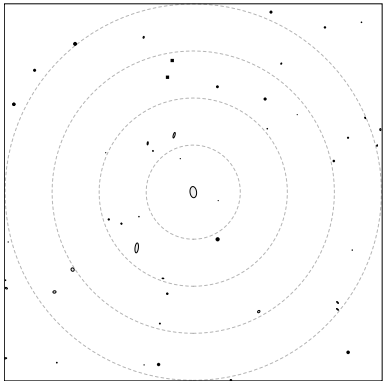


C52

NGC 4697

RA : 12.8^h	PSA: 47	Type: Gal
Dec: $-06°$	OSA: E12	Mag : 9.0
Con:	CSA:	Size : $7.0' \times 5.0'$

See: OSA DSC:TCO

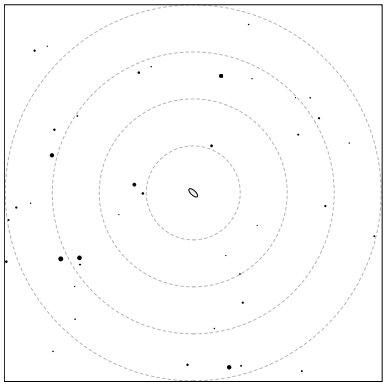


C53

NGC 3115 = The Spindle

RA : 10.1^h	PSA: 37	Type: Gal
Dec: $-08°$	OSA: E10	Mag : 8.9
Con:	CSA:	Size : $7.0' \times 3.0'$

See: OSA DSC:TCO

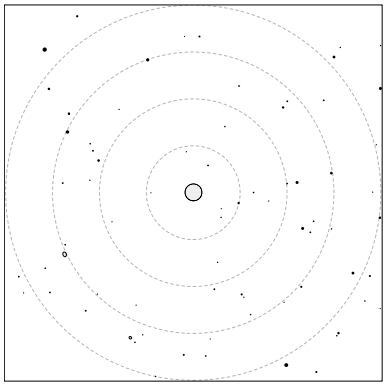


C54

BSC 90 = NGC 2506 = Mel 80 = Cr 170

RA : 08.0^h	PSA: 26	Type: OC
Dec: $-11°$	OSA:	Mag : 7.6
Con:	CSA:	Size : $12.0'$

See: DSC:TCO

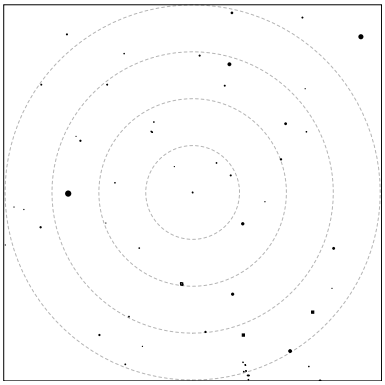


C55

BSC 106 = NGC 7009 = Saturn Nebula

RA : 21.1^h	PSA: 77	Type: PN
Dec: $-11°$	OSA: E24	Mag : 8.0
Con:	CSA:	Size : $0.7' \times 0.4'$

See: OSA DSC:TCO

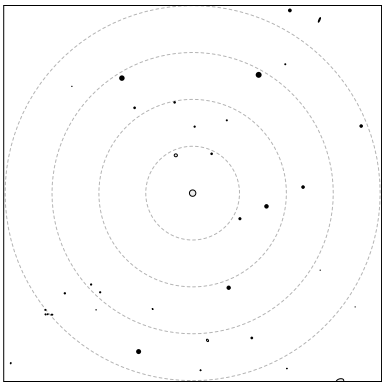


C56

BSC 107 = NGC 246

RA : 00.8^h	PSA: 7	Type: PN
Dec: $-12°$	OSA:	Mag : 10.9
Con:	CSA:	Size : $5.0' \times 4.0'$

See: DSC:TCO



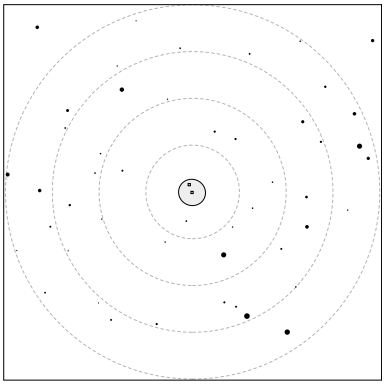
C57

BSC 146 = NGC 6822 = IC 4895 = Barnard's Galaxy

RA : 19.7^h	PSA: 66	Type: Gal
Dec: $-15°$	OSA: E22	Mag : 7.9
Con:	CSA:	Size : $15.0'$

See: OSA DSC:TCO

Notes: U73 is 0.7° NNW.

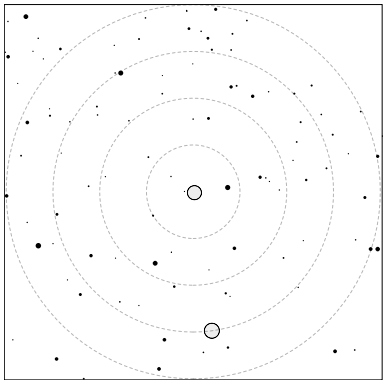


C58

BSC 79 = NGC 2360 = Mel 64 = Caroline's Cluster

RA : 07.3^h	PSA: 27	Type: OC
Dec: $-16°$	OSA: E8	Mag : 7.0
Con:	CSA:	Size : 14.0′

See: OSA DSC:TCO

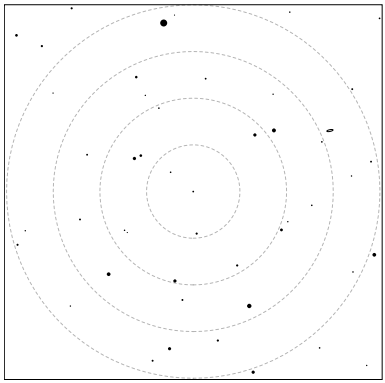


C59

U43 = BSC 125 = NGC 3242 = Ghost of Jupiter

RA : 10.4^h	PSA: 36/37	Type: PN
Dec: $-19°$	OSA: E10	Mag : 7.3
Con:	CSA:	Size : 0.8′×0.6′

See: OSA DSC:TMO DSC:TCO



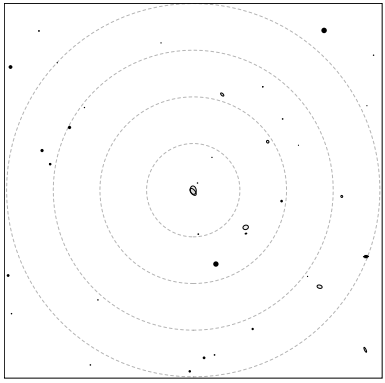
C60

NGC 4038 = NW Antennae Galaxy

RA : 12.0^h	PSA: 36/47	Type: Gal
Dec: $-19°$	OSA:	Mag : 10.5
Con:	CSA:	Size : 11.0′ × 6.0′

See: DSC:TCO

Notes: C61 is 1.2′ SW.



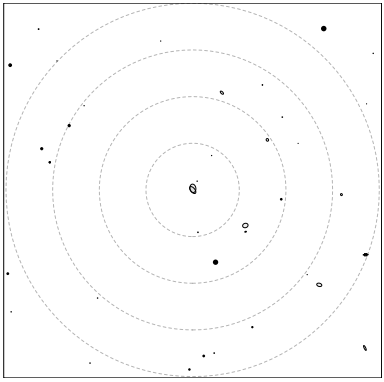
C61

NGC 4039 = SE Antennae Galaxy

RA : 12.0^h	PSA: 36/47	Type: Gal	
Dec: $-19°$	OSA:	Mag : 11.2	
Con:	CSA:	Size : 10.0′	×
		4.0′	

See: DSC:TCO

Notes: C60 is 1.2′ NW

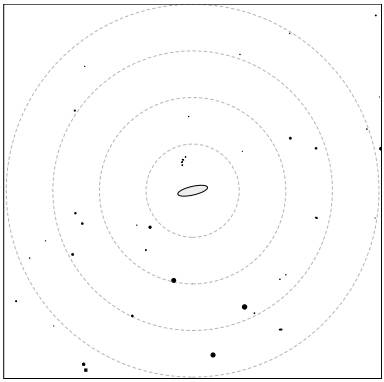


C62

NGC 247

RA : 00.8^h	PSA: 7	Type: Gal	
Dec: $-21°$	OSA: E0	Mag : 8.9	
Con:	CSA:	Size : 22.0′	×
		7.0′	

See: OSA DSC:TCO

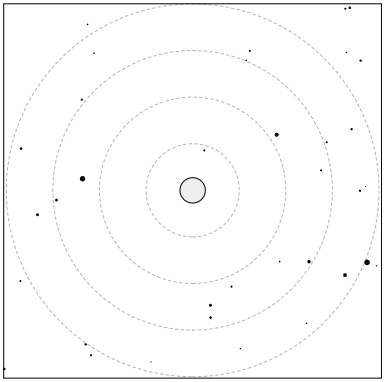


C63

BSC 91 = NGC 7293 = Helix Nebula

RA : 22.5^h	PSA: 76/77	Type: PN	
Dec: $-21°$	OSA: E24	Mag : 6.0	
Con:	CSA:	Size : 12.0′	×
		10.0′	

See: OSA DSC:TCO

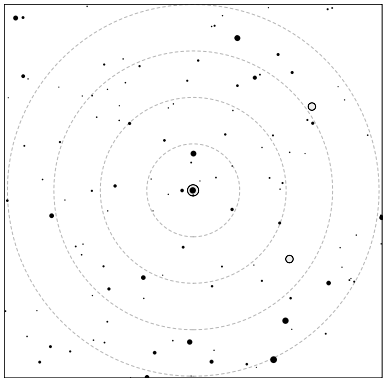


C64

BSC 17 = NGC 2362 = τ CMa Cluster

RA : 07.3^h	PSA: 27	Type: OC
Dec: $-25°$	OSA: E6	Mag : 4.1
Con:	CSA:	Size : 6.0′

See: OSA DSC:TCO



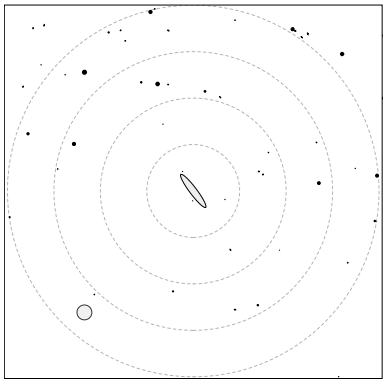
C65

BSC 108 = NGC 253 = Sculpter Galaxy

RA : 00.8^h	PSA: 7/9	Type: Gal
Dec: $-25°$	OSA: E0	Mag : 7.1
Con:	CSA:	Size : 26.0′ × 6.0′

See: OSA DSC:TCO

β Cet Deneb Kaitos (mag 2.0)
→ 7° S
→ 1° E to C65

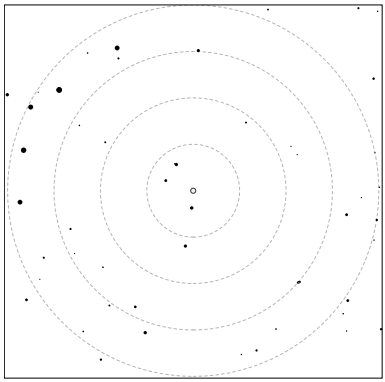


C66

NGC 5694

RA : 14.7^h	PSA: 46	Type: GC
Dec: $-27°$	OSA:	Mag : 10.2
Con:	CSA:	Size : 4.3′

See: DSC:TCO

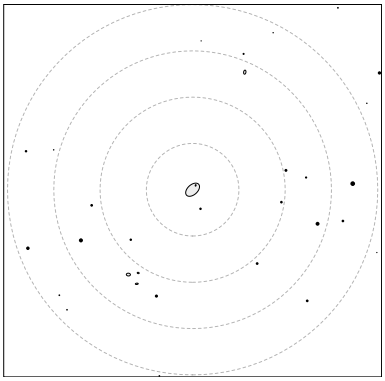


C67

NGC 1097

RA : 02.8^h	PSA: 6/8	Type: Gal
Dec: $-30°$	OSA:	Mag : 9.2
Con:	CSA:	Size : 10.0′ × 6.0′

See: DSC:TCO



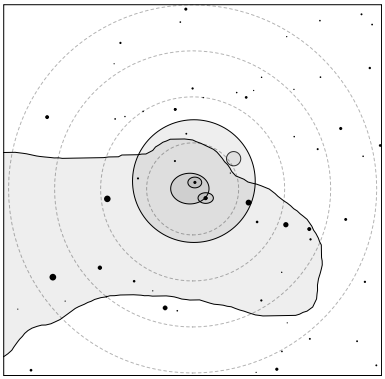
C68

NGC 6729 = R CrA Nebula

RA : 19.0^h	PSA: 69	Type: BN
Dec: $-37°$	OSA:	Mag : 9.5
Con:	CSA:	Size : 1.3′

See: DSC:TCO

Notes: The morphology and magnitude of the nebula are variable (DSC:TCO).

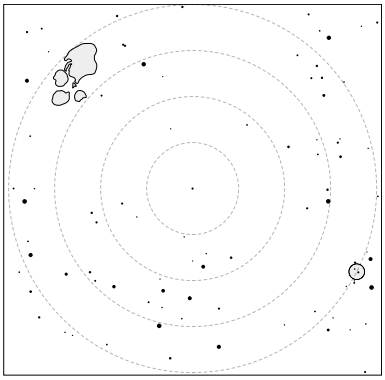


C69

BSC 77 = NGC 6302 = Bug Nebula

RA : 17.2^h	PSA: 58	Type: PN
Dec: $-37°$	OSA:	Mag : 9.6
Con:	CSA:	Size : 1.4′×0.4′

See: DSC:TCO

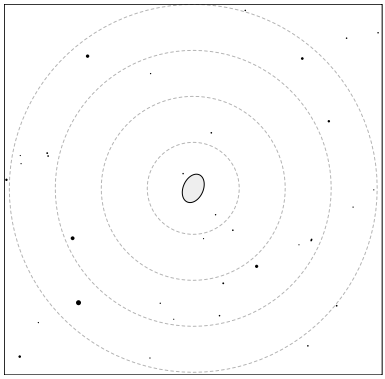


C70

BSC 138 = NGC 300

RA : 00.9^h PSA: 9 Type: Gal
Dec: −38° OSA: Mag : 7.2
Con: CSA: Size : 20.0′ × 13.0′

See: DSC:TCO

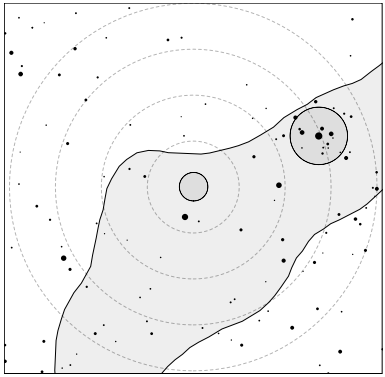


C71

BSC 41 = NGC 2477 = Mel 78

RA : 07.9^h PSA: 28 Type: OC
Dec: −39° OSA: S6 Mag : 5.8
Con: CSA: Size : 20.0′

See: OSA DSC:TCO

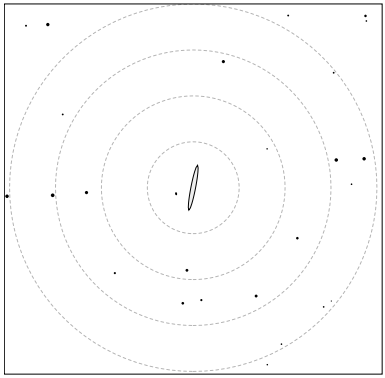


C72

BSC 102 = NGC 55

RA : 00.2^h PSA: 9 Type: Gal
Dec: −39° OSA: S0 Mag : 7.5
Con: CSA: Size : 34.0′ × 6.0′

See: OSA DSC:TCO

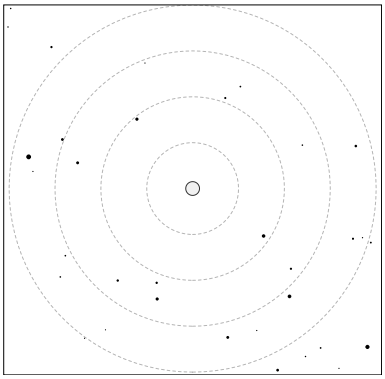


C73

BSC 80 = NGC 1851

RA : 05.2^h	PSA: 18	Type: GC
Dec: $-40°$	OSA: S0	Mag : 7.1
Con:	CSA:	Size : 12.0′

See: OSA DSC:TCO

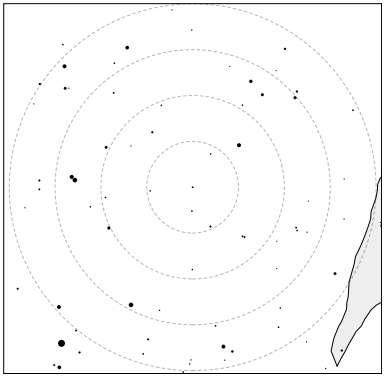


C74

NGC 3132

RA : 10.1^h	PSA: 39	Type: PN
Dec: $-40°$	OSA: S6	Mag : 9.2
Con:	CSA:	Size : 1.4′×0.9′

See: OSA DSC:TCO

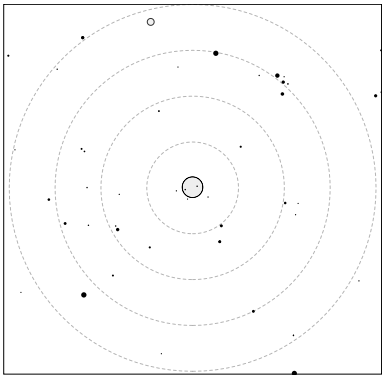


C75

BSC 42 = NGC 6124

RA : 16.4^h	PSA: 58	Type: OC
Dec: $-41°$	OSA: S21	Mag : 5.3
Con:	CSA:	Size : 40.0′

See: OSA DSC:TCO

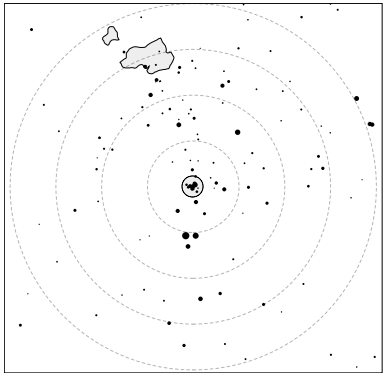


C76

BSC 7 = NGC 6231 = Cr 315

RA : 16.9^h	PSA: 58	Type: OC
Dec: $-42°$	OSA: S21	Mag : 2.6
Con:	CSA:	Size : 14.0′

See: OSA DSC:TCO

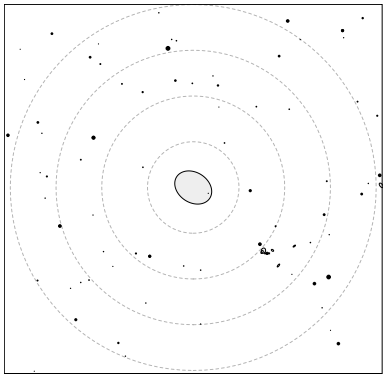


C77

NGC 5128 = Cen A

RA : 13.4^h	PSA: 48/49	Type: Gal
Dec: $-43°$	OSA: S15	Mag : 6.6
Con:	CSA:	Size : 17.0′ × 13.0′

See: OSA DSC:TCO

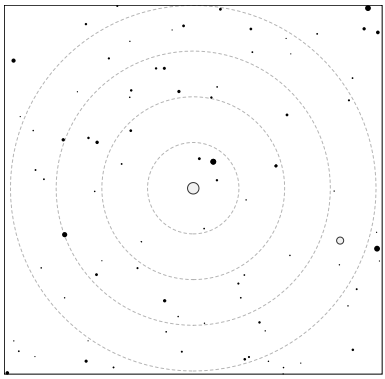


C78

BSC 60 = NGC 6541

RA : 18.1^h	PSA: 69	Type: GC
Dec: $-44°$	OSA: S21	Mag : 6.3
Con:	CSA:	Size : 15.0′

See: OSA DSC:TCO

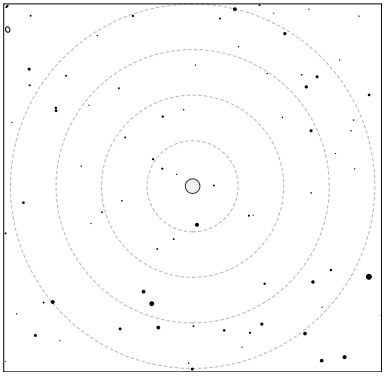


C79

BSC 112 = NGC 3201

RA : 10.3^h	PSA: 39	Type: GC
Dec: $-46°$	OSA: S6	Mag : 6.7
Con:	CSA:	Size : 20.0′

See: OSA DSC:TCO

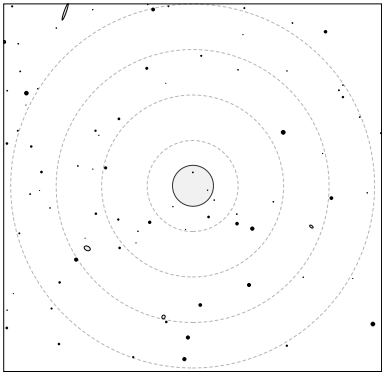


C80

BSC 14 = NGC 5139 = ω Cen

RA : 13.4^h	PSA: 48/49	Type: GC
Dec: $-47°$	OSA: S15	Mag : 3.9
Con:	CSA:	Size : 55.0′

See: OSA DSC:TCO

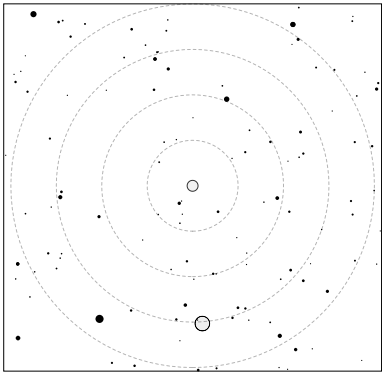


C81

BSC 98 = NGC 6352

RA : 17.4^h	PSA: 58	Type: GC
Dec: $-48°$	OSA:	Mag : 7.8
Con:	CSA:	Size : 9.0′

See: DSC:TCO

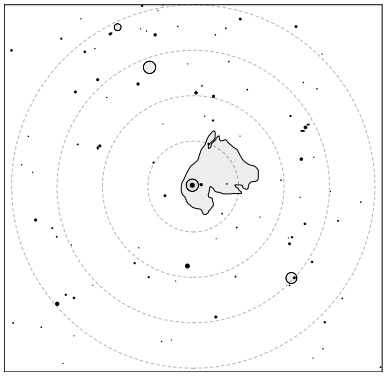


C82

BSC 30 = NGC 6193

RA : 16.7^h	PSA: 58	Type: OC
Dec: $-49°$	OSA:	Mag : 4.8
Con:	CSA:	Size : 14.0′

See: DSC:TCO

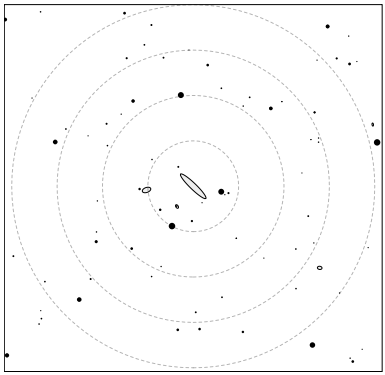


C83

BSC 147 = NGC 4945

RA : 13.1^h	PSA: 49	Type: Gal
Dec: $-49°$	OSA:	Mag : 8.8
Con:	CSA:	Size : 19.0′ × 4.0′

See: DSC:TCO

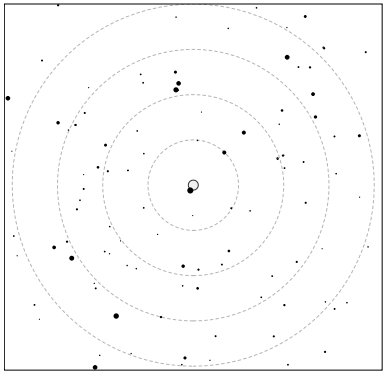


C84

BSC 92 = NGC 5286

RA : 13.8^h	PSA: 48	Type: GC
Dec: $-51°$	OSA:	Mag : 7.2
Con:	CSA:	Size : 11.0′

See: DSC:TCO

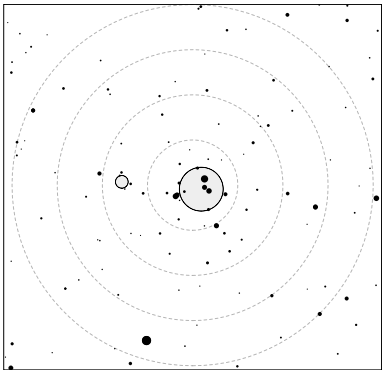


C85

BSC 6 = IC 2391 = *o* Vel Cluster

RA : 08.7^h	PSA: 28	Type: OC
Dec: $-53°$	OSA: S6	Mag : 2.8
Con:	CSA:	Size : 60.0′

See: OSA DSC:TCO

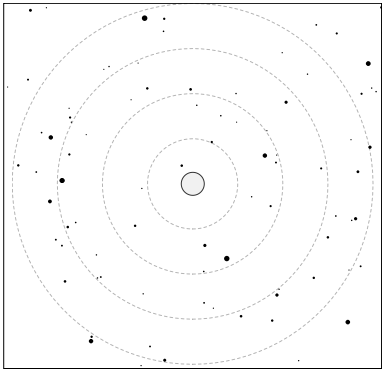


C86

BSC 67 = NGC 6397

RA : 17.7^h	PSA: 58	Type: GC
Dec: $-54°$	OSA: S18	Mag : 5.3
Con:	CSA:	Size : 31.0′

See: OSA DSC:TCO

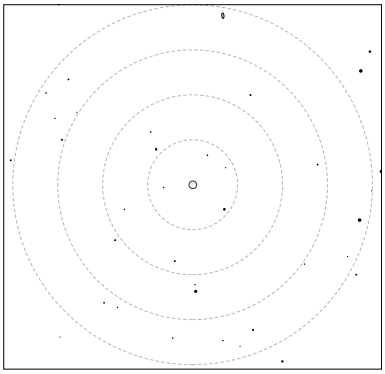


C87

BSC 115 = NGC 1261

RA : 03.2^h	PSA: 19	Type: GC
Dec: $-55°$	OSA:	Mag : 8.3
Con:	CSA:	Size : 6.8′

See: DSC:TCO

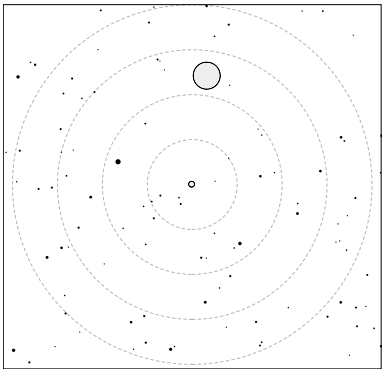


C88

BSC 103 = NGC 5823 = Mel 131

RA : 15.1^h	PSA: 59	Type: OC
Dec: $-56°$	OSA:	Mag : 7.9
Con:	CSA:	Size : 12.0′

See: DSC:TCO

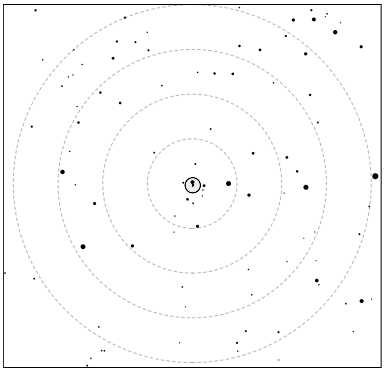


C89

BSC 33 = NGC 6087 = S Nor Cluster

RA : 16.3^h	PSA: 58/60	Type: OC
Dec: $-58°$	OSA: S18	Mag : 5.2
Con:	CSA:	Size : 15.0′

See: OSA DSC:TCO

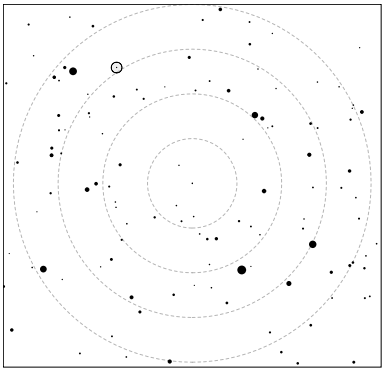


C90

NGC 2867

RA : 09.4^h	PSA: 39	Type: PN
Dec: $-58°$	OSA:	Mag : 9.7
Con:	CSA:	Size : 0.2′

See: DSC:TCO



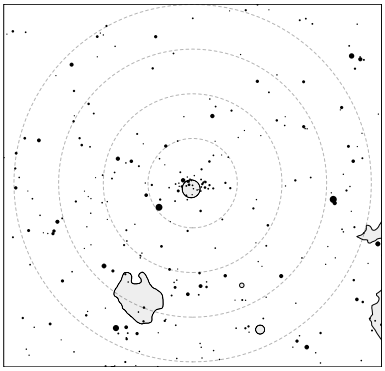
C91

BSC 9 = NGC 3532

RA : 11.1^h	PSA: 38/40	Type: OC
Dec: $-59°$	OSA: S9	Mag : 3.2
Con:	CSA:	Size : 50.0′

See: OSA DSC:TCO

Notes: C90 is 2.5° SW.



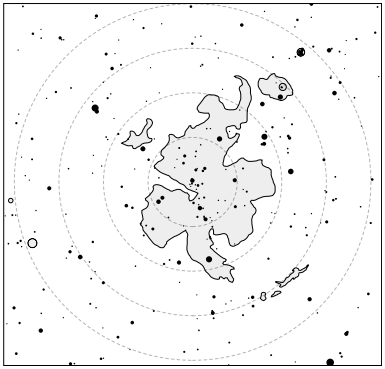
C92

BSC 3 = NGC 3372 = η Car Nebula

RA : 10.8^h	PSA: 38/40	Type: BN
Dec: $-60°$	OSA: S9	Mag : 4.5
Con:	CSA:	Size : 120.0′

See: OSA DSC:TCO

Notes: The nebula is variable in magnitude. C91 is 2.5° NE.

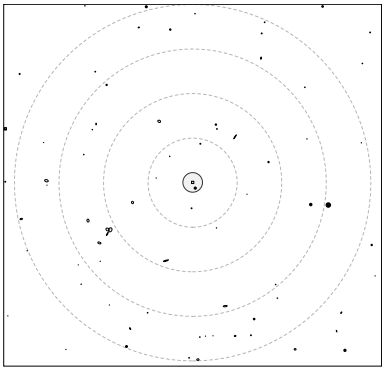


C93

BSC 34 = NGC 6752

RA : 19.2^h	PSA: 69/70	Type: GC
Dec: $-60°$	OSA: S24	Mag : 5.4
Con:	CSA:	Size : 29.0′

See: OSA DSC:TCO



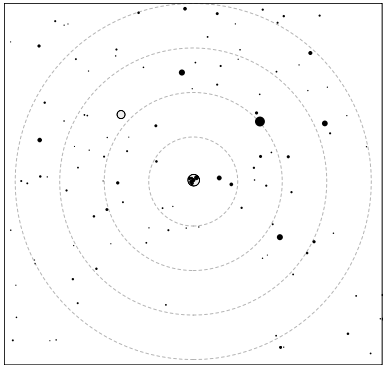
C94

BSC 20 = NGC 4755 = Mel 114 = Jewel Box

RA : 12.9^h	PSA: 49/50	Type: OC
Dec: $-60°$	OSA: S12	Mag : 4.2
Con:	CSA:	Size : 10.0′

See: OSA DSC:TCO

Notes: The northern edge of C99 is 1° S.

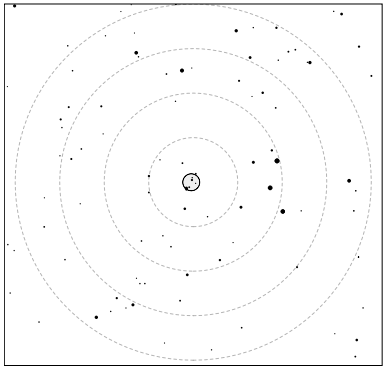


C95

BSC 29 = NGC 6025

RA : 16.1^h	PSA: 59/60	Type: OC
Dec: $-60°$	OSA:	Mag : 5.5
Con:	CSA:	Size : 15.0′

See: DSC:TCO

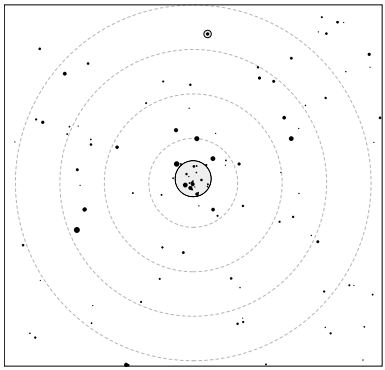


C96

BSC 13 = NGC 2516 = Southern Beehive Cluster

RA : 08.0^h	PSA: 28/30	Type: OC
Dec: $-61°$	OSA: S3	Mag : 3.5
Con:	CSA:	Size : 22.0′

See: OSA DSC:TCO



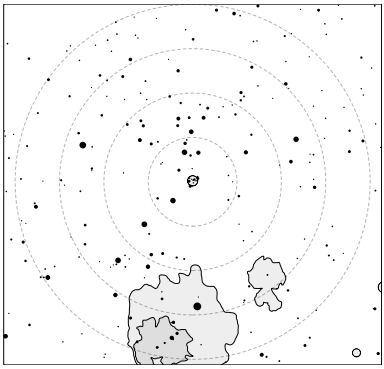
C97

BSC 32 = NGC 3766 = Cr 248

RA : 11.6^h	PSA: 38/40	Type: OC
Dec: $-62°$	OSA: S12	Mag : 5.3
Con:	CSA:	Size : 15.0′

See: OSA DSC:TCO

Notes: C100 is 1.8° S.



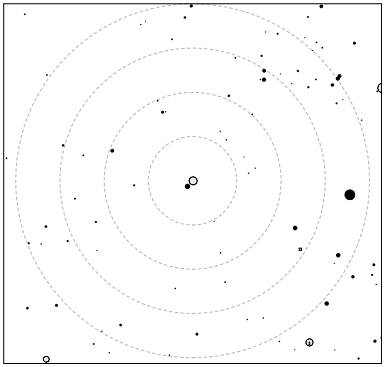
C98

BSC 70 = NGC 4609 = Coalsack Cluster

RA : 12.7^h	PSA: 49/50	Type: OC
Dec: $-63°$	OSA:	Mag : 6.9
Con:	CSA:	Size : 6.0′

See: DSC:TCO

Notes: Surrounded by C99.



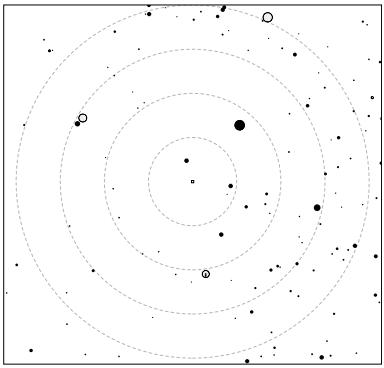
C99

Coalsack

RA : 12.5^h	PSA: 49/50	Type: DN
Dec: $-64°$	OSA: S12	Mag :
Con:	CSA:	Size : 7.0° × 5.0°

See: OSA DSC:TCO

Notes: Much bigger than the finder chart.
Surrounds C98. C94 is 1° N of the northern edge.



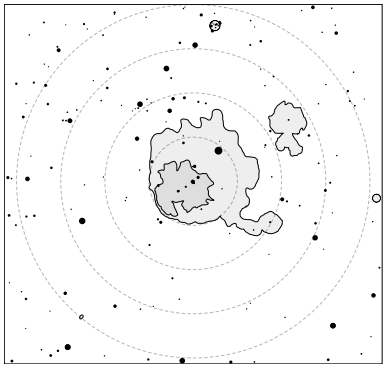
C100

BSC 22 = Cr 249 = λ Cen Cluster

RA : 11.6^h	PSA: 38/40	Type: OC	
Dec: $-63°$	OSA:	Mag : 2.9	
Con:	CSA:	Size : 65.0′	×
		40.0′	

See: DSC:TCO

Notes: C97 is 1.8° N.

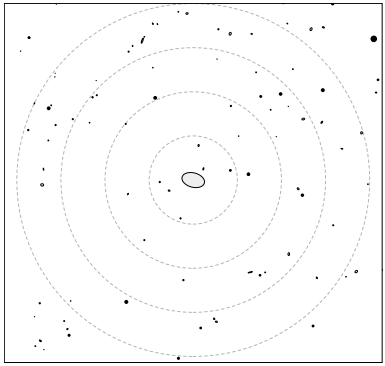


C101

BSC 143 = NGC 6744

RA : 19.2^h	PSA: 70	Type: Gal	
Dec: $-64°$	OSA:	Mag : 8.6	
Con:	CSA:	Size : 21.0′	×
		13.0′	

See: DSC:TCO

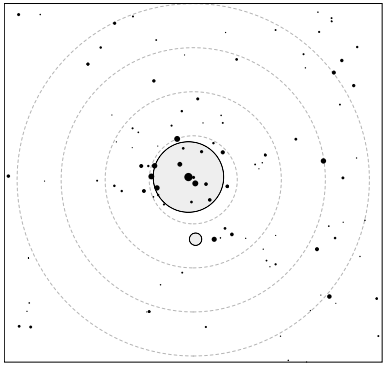


C102

BSC 5 = IC 2602 = Mel 102 = Cr 229 = θ Car Cluster

RA : 10.7^h	PSA: 38/40	Type: OC
Dec: $-64°$	OSA: S9	Mag : 1.9
Con:	CSA:	Size : 100.0′

See: OSA DSC:TCO



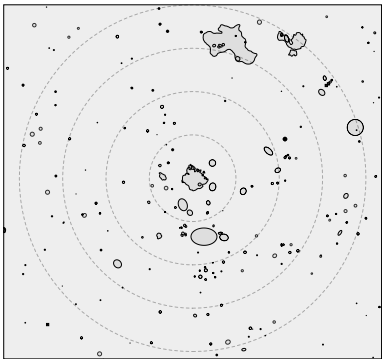
C103

BSC 109 = NGC 2070 = Tarantula Nebula

RA : 05.6^h	PSA: 20/30/D	Type: BN
Dec: $-69°$	OSA: S3	Mag : 4.0/8.3
Con:	CSA:	Size : 30.0′ × 20.0′

See: OSA DSC:TCO

Notes: The nebula is magnitude 4.0 (DSC:TCO). the cluster is magnitude 8.3 and has a diameter of 5′ (DSC:TCO).

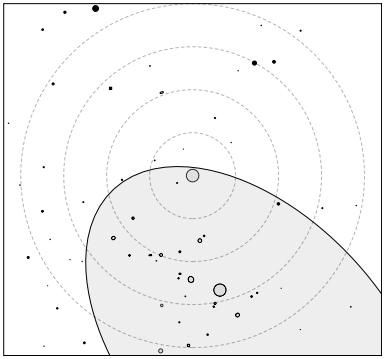


C104

BSC 64 = NGC 362 = Mel 4

RA : 01.1^h	PSA: 10/80	Type: GC
Dec: $-71°$	OSA: S0	Mag : 6.4
Con:	CSA:	Size : 14.0′

See: OSA DSC:TCO

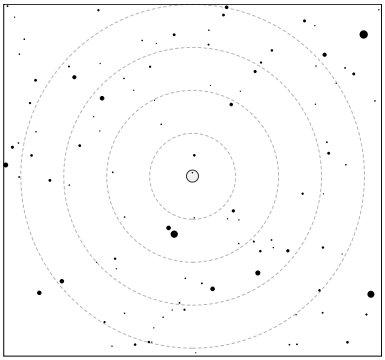


C105

BSC 97 = NGC 4833

RA : 13.0^h	PSA: 50	Type: GC
Dec: $-71°$	OSA:	Mag : 8.4
Con:	CSA:	Size : 14.0′

See: DSC:TCO

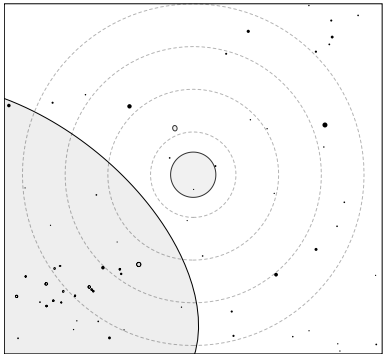


C106

BSC 26 = NGC 104 = Mel 1 = 47 Tuc

RA : 00.4^h	PSA: 10/80	Type: GC
Dec: $-72°$	OSA: S0	Mag : 4.0
Con:	CSA:	Size : 50.0′

See: OSA DSC:TCO

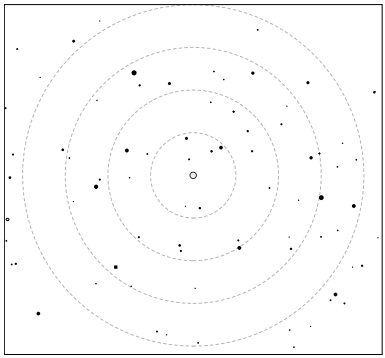


C107

BSC 139 = NGC 6101

RA : 16.4^h	PSA: 60	Type: GC
Dec: $-72°$	OSA:	Mag : 9.2
Con:	CSA:	Size : 11.0′

See: DSC:TCO

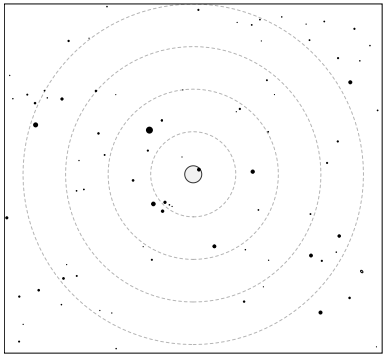


C108

BSC 99 = NGC 4372

RA : 12.4^h	PSA: 40/50	Type: GC
Dec: $-73°$	OSA:	Mag : 7.2
Con:	CSA:	Size : 5.0′

See: DSC:TCO

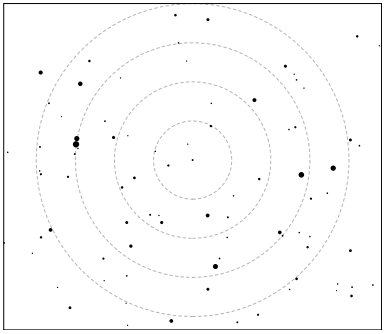


C109

NGC 3195

RA : 10.2^{h}	PSA: 40	Type: PN
Dec: -81°	OSA:	Mag : 11.6
Con:	CSA:	Size : $0.6'$

See: DSC:TCO



The Urban Observing Program Objects

The Urban Observing Program of the Astronomical League consists of bright objects selected by Terry Trees for northern observers under light-polluted skies. There are 87 deep-sky objects, 12 double stars, and 1 variable star. The deep-sky objects range in declination from −35 to +72 degrees, cover all seasons, and include 41 Messier objects and 14 Caldwell objects.

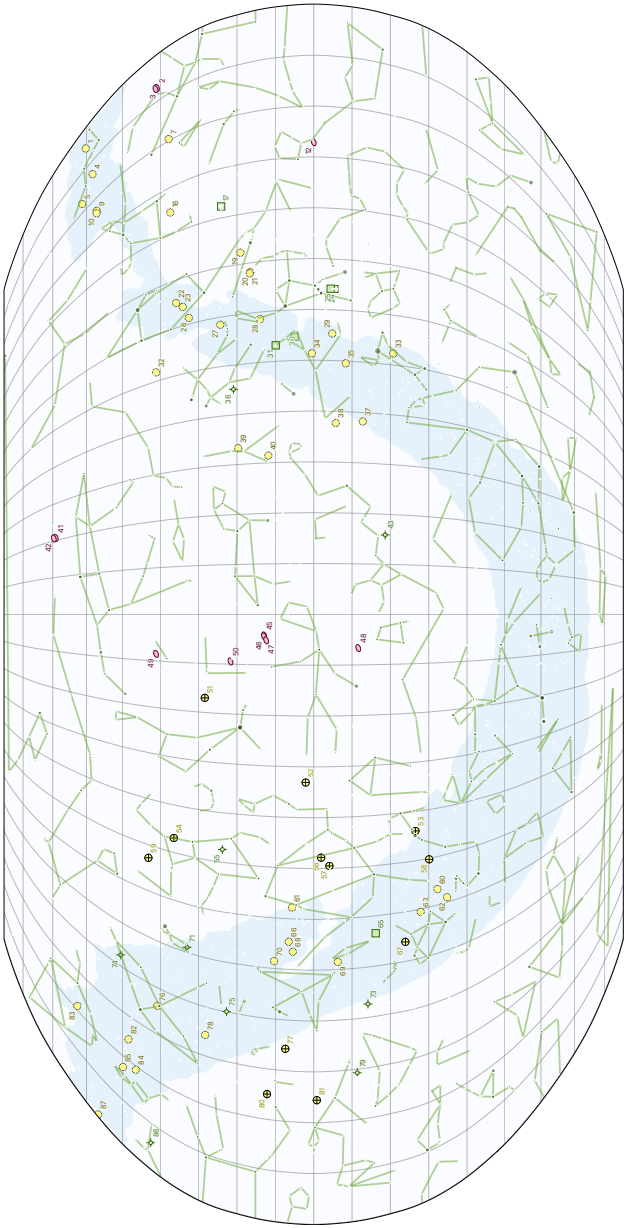
I give finder charts for the deep-sky objects only; the stars are all no fainter than magnitude 5.2 and have Bayer designations, and so a standard all-sky atlas is adequate. I use U1 to U87 to label the deep-sky objects, following the order in right ascension given by Trees, but this is not a standard designation. For completeness, I include all the deep-sky objects, even bright ones like the Hyades (U18).

The following table lists the deep-sky objects with their J2000 positions (decimal hours of right ascension and degrees of declination), the charts on which they appear in the *Pocket Sky Atlas*, their types, other names, and references to O'Meara's *Deep-Sky Companions* volumes (*The Messier Objects*, *The Caldwell Objects*, *The Secret Deep*, and *Hidden Treasures*).

I note that the Hyades (U18) and Coma Star Cluster (U44) are much bigger than the finder charts. They are really binocular objects and better located using a small-scale all-sky atlas. That said, be aware that the α Per Cluster (U15) and Coma Star Cluster (U44) are not labelled in the *Pocket Sky Atlas*.

Name	Position	Type	Other Names
U1	00.5 +60	OC	NGC 129
U2	00.7 +41	Gal	M32 = BSC 110 = NGC 221
U3	00.7 +41	Gal	M31 = BSC 11 = NGC 224 = Andromeda Galaxy
U4	01.3 +58	OC	C13 = BSC 62 = NGC 457 = E.T. Cluster = Owl Cluster
U5	01.8 +61	OC	C10 = BSC 76 = NGC 663 = Horseshoe Cluster
U6	01.8 +72	OC	Cr 463
U7	01.9 +38	OC	C28 = NGC 752 = Mel 12
U8	02.3 +60	OC	Stock 2
U9	02.3 +57	OC	NGC 869 = Mel 13 = Cr 24 = h Per Cluster = half of C14
U10	02.4 +57	OC	NGC 884 = Mel 14 = Cr 25 = χ Per Cluster = half of C14
U11	02.6 +56	OC	Tr 2
U12	02.7 −00	Gal	M77 = NGC 1068
U13	03.2 +63	OC	Tr 3
U14	03.3 +60	OC	Stock 23 = Pazmino's Cluster
U15	03.4 +49	OC	Mel 20 = Cr 39 = α Per Cluster
U16	03.5 +37	OC	NGC 1342
U17	03.8 +24	OC	M45 = BSC 4 = Mel 22 = Pleiades
U18	04.4 +16	OC	C41 = BSC 1 = Mel 25 = Cr 50 = Hyades
U19	04.8 +19	OC	NGC 1647
U20	05.2 +17	OC	NGC 1807 = Mel 29 = Cr 61
U21	05.2 +17	OC	NGC 1817 = Cr 60
U22	05.5 +36	OC	M38 = BSC 82 = NGC 1912
U23	05.6 +34	OC	M36 = BSC 58 = NGC 1960
U24	05.6 −05	BN	M42 = BSC 15 = NGC 1976 = Orion Nebula
U25	05.6 −04	OC	NGC 1981
U26	05.9 +33	OC	M37 = BSC 55 = NGC 2099
U27	06.2 +24	OC	M35 = BSC 31 = NGC 2168
U28	06.1 +14	OC	NGC 2169
U29	06.5 −05	OC	NGC 2232
U30	06.5 +05	OC	C50 = BSC 25 = NGC 2244
U31	06.7 +10	OC	NGC 2264
U32	06.8 +41	OC	NGC 2281
U33	06.8 −21	OC	M41 = BSC 21 = NGC 2287
U34	06.9 +00	OC	NGC 2301
U35	07.0 −08	OC	M50 = BSC 44 = NGC 2323
U36	07.5 +21	PN	C39 = NGC 2392 = Eskimo Nebula
U37	08.2 −13	OC	NGC 2539 = Mel 83
U38	08.2 −06	OC	M48 = BSC 37 = NGC 2548
U39	08.7 +20	OC	M44 = BSC 12 = NGC 2632 = Mel 88 = Beehive Cluster
U40	08.9 +12	OC	M67 = BSC 51 = NGC 2682
U41	09.9 +69	Gal	M81 = BSC 72 = NGC 3031
U42	09.9 +70	Gal	M82 = BSC 122 = NGC 3034
U43	10.4 −19	PN	C59 = BSC 125 = NGC 3242 = Ghost of Jupiter
U44	12.4 +26	OC	Mel 111 = Cr 256 = Coma Star Cluster
U45	12.4 +13	Gal	M84 = NGC 4374
U46	12.4 +13	Gal	M86 = NGC 4406
U47	12.5 +12	Gal	M87 = NGC 4486
U48	12.7 −12	Gal	M104 = BSC 133 = NGC 4594 = Sombrero Galaxy
U49	12.8 +41	Gal	M94 = BSC 134 = NGC 4736
U50	12.9 +22	Gal	M64 = BSC 149 = NGC 4826 = Black-Eye Galaxy

Name	Position	Type	Other Names
U51	13.7 +28	GC	M3 = BSC 53 = NGC 5272
U52	15.3 +02	GC	M5 = BSC 66 = NGC 5904
U53	16.4 −27	GC	M4 = BSC 43 = NGC 6121
U54	16.7 +36	GC	M13 = BSC 40 = NGC 6205 = Hercules Cluster
U55	16.7 +24	PN	NGC 6210
U56	16.8 −02	GC	M12 = BSC 94 = NGC 6218
U57	17.0 −04	GC	M10 = BSC 61 = NGC 6254
U58	17.0 −30	GC	M62 = BSC 81 = NGC 6266
U59	17.3 +43	GC	M92 = BSC 59 = NGC 6341
U60	17.7 −32	OC	M6 = BSC 18 = NGC 6405 = Butterfly Nebula
U61	17.8 +06	OC	IC 4665 = Mel 179 = Cr 349
U62	17.9 −35	OC	M7 = BSC 10 = NGC 6475
U63	18.1 −28	OC	NGC 6520
U64	18.1 −24	BN	M8 = BSC 45 = NGC 6523/6530 = Lagoon Nebula
U65	18.3 −16	BN	M17 = BSC 47 = NGC 6618 = Omega Nebula
U66	18.5 +07	OC	NGC 6633
U67	18.6 −24	GC	M22 = BSC 28 = NGC 6656
U68	18.6 +05	OC	IC 4756 = Mel 210
U69	18.9 −06	OC	M11 = NGC 6705 = Mel 213 = Cr 391 = Wild Duck Cluster
U70	18.9 +10	OC	NGC 6709
U71	18.9 +33	PN	M57 = BSC 127 = NGC 6720 = Ring Nebula
U72	19.4 +20	OC	Cr 399 = Brocchi's Cluster
U73	19.7 −14	PN	NGC 6818
U74	19.7 +51	PN	NGC 6826
U75	20.0 +23	PN	M27 = BSC 88 = NGC 6853 = Dumbell Nebula
U76	20.4 +41	OC	NGC 6910
U77	20.6 +07	GC	NGC 6934
U78	20.6 +28	OC	NGC 6940 = Mel 232
U79	21.1 −11	PN	NGC 7009
U80	21.5 +12	GC	M15 = BSC 54 = NGC 7078
U81	21.6 −01	GC	M2 = BSC 56 = NGC 7089
U82	21.5 +48	OC	M39 = BSC 36 = NGC 7092
U83	21.9 +63	OC	NGC 7160
U84	22.1 +47	OC	NGC 7209
U85	22.3 +50	OC	C16 = BSC 63 = NGC 7243
U86	23.4 +43	PN	NGC 7662
U87	23.9 +57	OC	NGC 7789 = Mel 245



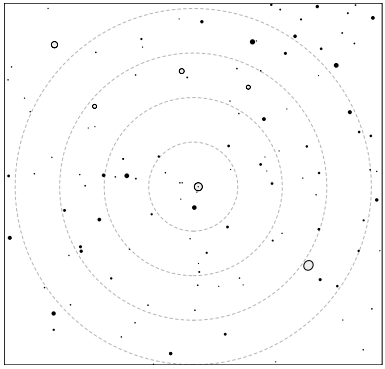
The distribution of Urban List objects over the whole sky

U1

NGC 129

RA : 00.5^h	PSA: 3	Type: OC
Dec: $+60°$	OSA:	Mag :
Con:	CSA:	Size : 21.0′

See: DSC:TSD



U2

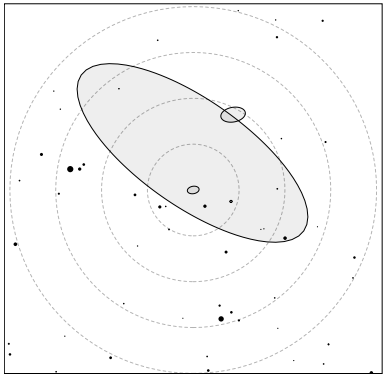
M32 = BSC 110 = NGC 221

RA : 00.7^h	PSA: 3	Type: Gal
Dec: $+41°$	OSA: N0	Mag : 8.2
Con: And	CSA: 2/7	Size : 11.0′ × 7.3′

See: OSA DSC:TMO

Notes: M31/U3 is 0.5° N. M110 is 1° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 2.5° N to M32/U2



U3

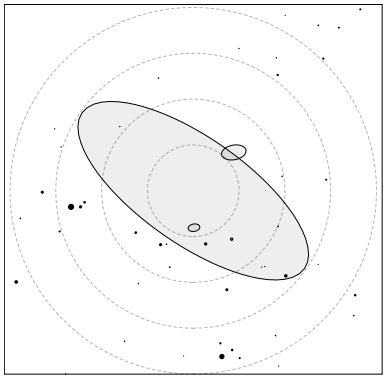
M31 = BSC 11 = NGC 224 = Andromeda Galaxy

RA : 00.7^h	PSA: 3	Type: Gal
Dec: $+41°$	OSA: N0	Mag : 3.4
Con: And	CSA: 2/7	Size : 3.0° × 1.0°

See: OSA DSC:TMO

Notes: M32/U2 is 0.5° S. M100 is 0.5° NW.

β And Mirach (mag 2.1)
→ 2.5° W
→ 3.0° N to μ And (mag 3.9)
→ 2.5° W
→ 3.0° N to M31/U3

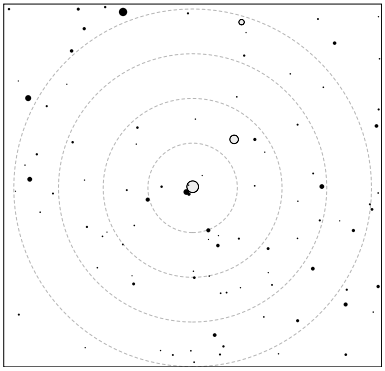


U4

C13 = BSC 62 = NGC 457 = E.T. Cluster = Owl Cluster

RA : 01.3^h	PSA: 1/3	Type: OC
Dec: +58°	OSA: N2	Mag : 6.4
Con:	CSA:	Size : 20.0′

See: OSA DSC:TCO



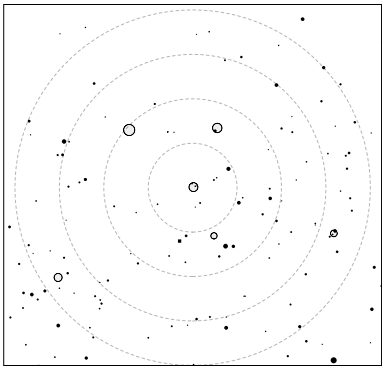
U5

C10 = BSC 76 = NGC 663 = Horseshoe Cluster

RA : 01.8^h	PSA: 1/2	Type: OC
Dec: +61°	OSA: N2	Mag : 6.7
Con: Cas	CSA: 1/2	Size : 15.0′

See: OSA DSC:TCO

Notes: M103 is 1.7° WSW. C8 is 3° NW.

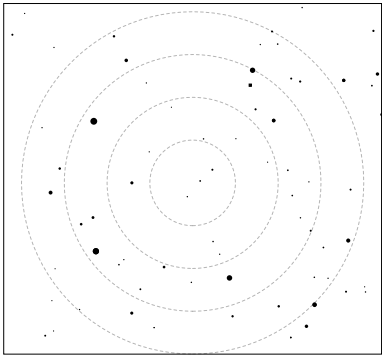


U6

Cr 463

RA : 01.8^h	PSA: 1	Type: OC
Dec: +72°	OSA:	Mag :
Con:	CSA:	Size : 36.0′

See: DSC:TSD

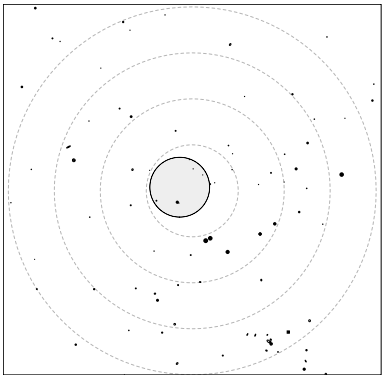


U7

C28 = NGC 752 = Mel 12

RA : 01.9^h	PSA: 2	Type: OC
Dec: +38°	OSA: N0	Mag : 5.7
Con:	CSA:	Size : 75.0′

See: OSA DSC:TCO



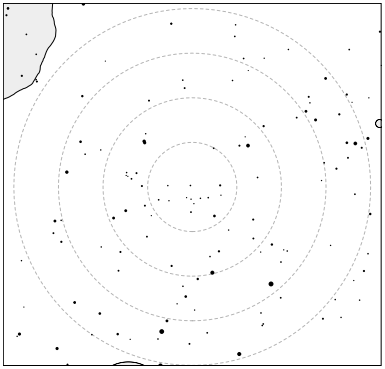
U8

Stock 2

RA : 02.3^h	PSA: 2	Type: OC
Dec: +60°	OSA:	Mag :
Con:	CSA:	Size : 60.0′

See: DSC:TSD

Notes: C14/U9/U10 is 2° S.



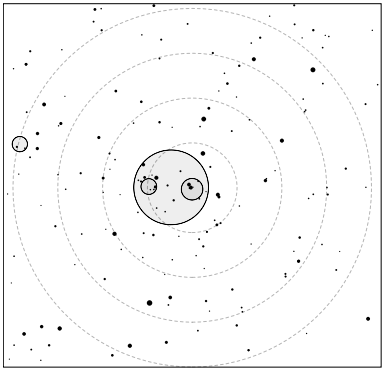
U9

NGC 869 = Mel 13 = Cr 24 = h Per Cluster = half of C14

RA : 02.3^h	PSA: 2	Type: OC
Dec: +57°	OSA: N2	Mag :
Con:	CSA:	Size : 18.0′

See: OSA DSC:TMO DSC:TCO

Notes: U9 and U10 form C14. U10 is 0.5° W. U8 is 2° N. U11 is 3.0° SE.



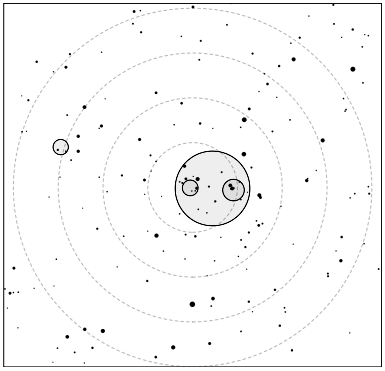
U10

NGC 884 = Mel 14 = Cr 25 = χ Per Cluster = half of C14

RA : 02.4^h	PSA: 2	Type: OC
Dec: $+57°$	OSA: N2	Mag :
Con:	CSA:	Size : $18.0'$

See: OSA DSC:TMO DSC:TCO

Notes: U9 and U10 form C14. U9 is 0.5° E. U8 is 2° N. U11 is 2.5° SE.



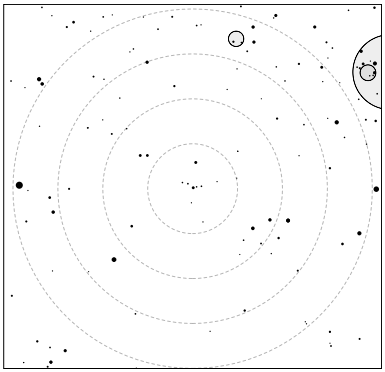
U11

Tr 2

RA : 02.6^h	PSA: 2	Type: OC
Dec: $+56°$	OSA:	Mag :
Con:	CSA:	Size : $20.0'$

See:

Notes: C14/U9/10 are 2.5° NW.

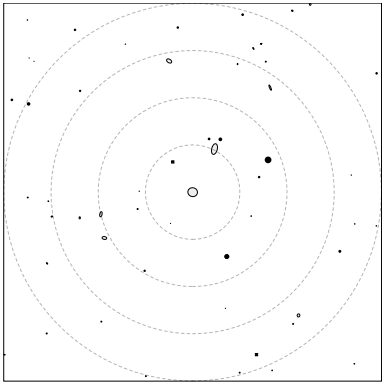


U12

M77 = NGC 1068

RA : 02.7^h	PSA: 4	Type: Gal
Dec: $-00°$	OSA: E0	Mag : 8.9
Con: Cet	CSA: 8	Size : $8.2' \times 7.3'$

See: OSA DSC:TMO

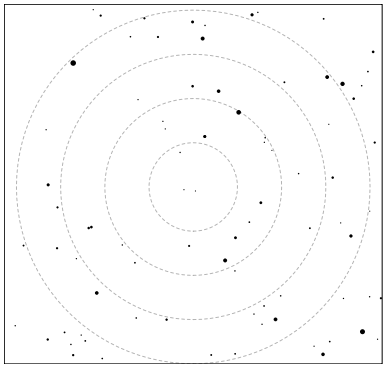


U13

Tr 3

RA : 03.2^h	PSA: 11/13	Type: OC
Dec: +63°	OSA:	Mag :
Con:	CSA:	Size : 23.0′

See:

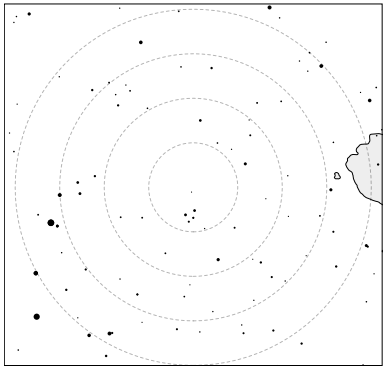


U14

Stock 23 = Pazmino's Cluster

RA : 03.3^h	PSA: 11/13	Type: OC
Dec: +60°	OSA:	Mag :
Con:	CSA:	Size : 15.0′

See:



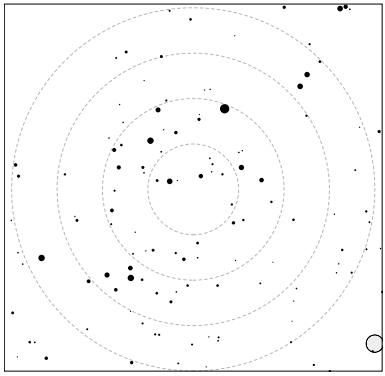
U15

Mel 20 = Cr 39 = α Per Cluster

RA : 03.4^h	PSA: 3	Type: OC
Dec: +49°	OSA:	Mag :
Con:	CSA:	Size : 3.1°

See:

Notes: Not shown in the PSA.

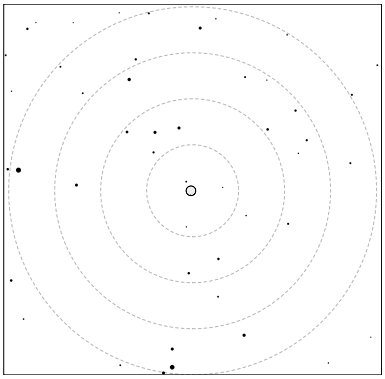


U16

NGC 1342

RA : 03.5^h	PSA: 13	Type: OC
Dec: +37°	OSA:	Mag :
Con:	CSA:	Size : 14.0′

See: DSC:TSD

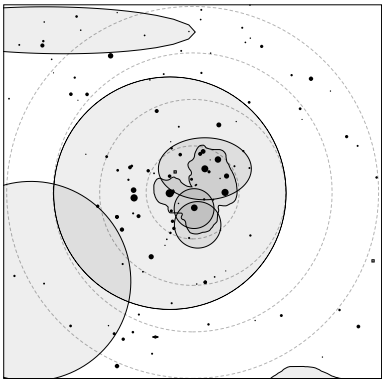


U17

M45 = BSC 4 = Mel 22 = Pleiades

RA : 03.8^h	PSA: 15/A	Type: OC
Dec: +24°	OSA: E3	Mag : 1.5
Con: Tau	CSA: 2/3/8/9	Size : 120.0′

See: OSA DSC:TMO



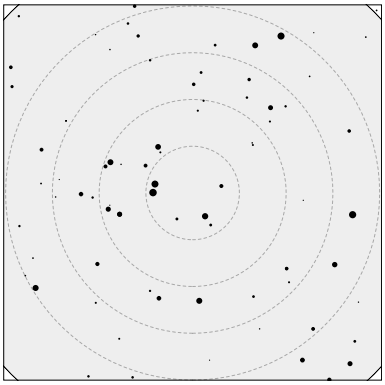
U18

C41 = BSC 1 = Mel 25 = Cr 50 = Hyades

RA : 04.4^h	PSA: 15	Type: OC
Dec: +16°	OSA: E3	Mag : 0.5
Con:	CSA:	Size : 5.5°

See: OSA DSC:TCO

Notes: Much bigger than the finder chart.



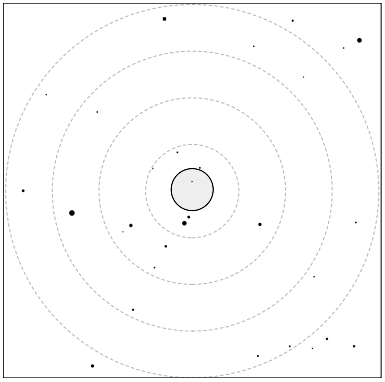
U19

NGC 1647

RA : 04.8^h	PSA: 15	Type: OC
Dec: $+19°$	OSA: E3	Mag :
Con:	CSA:	Size : 45.0′

See: OSA DSC:TCO DSC:HT

α Tau Betelgeuse (mag 0.3–0.9)
→ 2.5° N
→ 2.5° E to U19



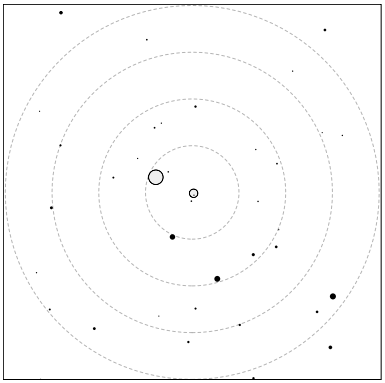
U20

NGC 1807 = Mel 29 = Cr 61

RA : 05.2^h	PSA: 14	Type: OC
Dec: $+17°$	OSA:	Mag :
Con:	CSA:	Size : 17.0′

See: DSC:TSD

Notes: U21 is 0.5° WSW.



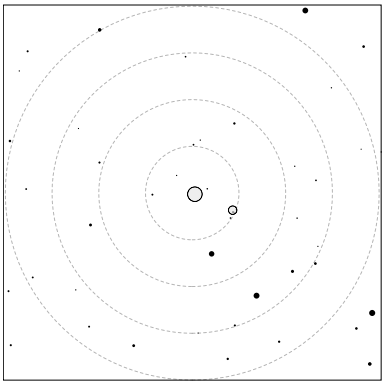
U21

NGC 1817 = Cr 60

RA : 05.2^h	PSA: 14	Type: OC
Dec: $+17°$	OSA:	Mag :
Con:	CSA:	Size : 15.0′

See: DSC:TSD

Notes: U20 is 0.5° ENE.



U22

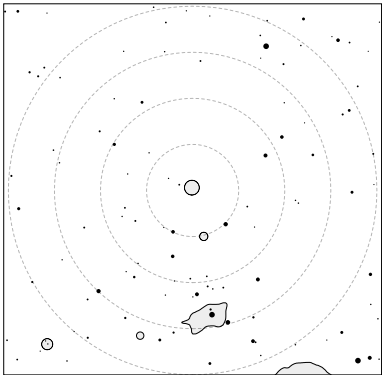
M38 = BSC 82 = NGC 1912

RA : 05.5^h	PSA: 12	Type: OC
Dec: $+36°$	OSA: N6	Mag : 6.4
Con: Aur	CSA: 3	Size : 15.0′

See: OSA DSC:TMO

Notes: M36/U23 is 2.5° SE.

β Tau Elnath (mag 1.7)
→ 7° N to M38/U23

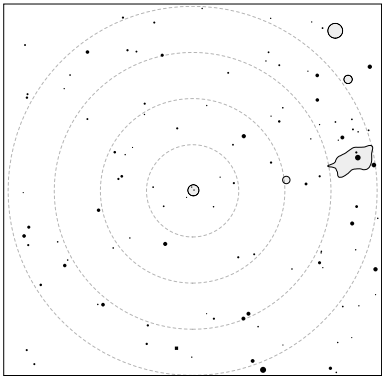


U23

M36 = BSC 58 = NGC 1960

RA : 05.6^h	PSA: 12	Type: OC
Dec: $+34°$	OSA: N6	Mag : 6.0
Con: Aur	CSA: 3	Size : 10.0′

See: OSA DSC:TMO



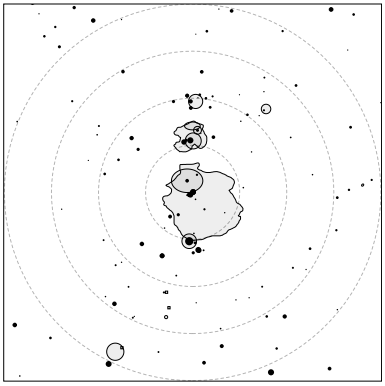
U24

M42 = BSC 15 = NGC 1976 = Orion Nebula

RA : 05.6^h	PSA: 16/B	Type: BN
Dec: $-05°$	OSA: E4	Mag : 3.7
Con: Ori	CSA: 9	Size : 90.0′ × 60.0′

See: OSA DSC:TMO

Notes: M42/U24 surrounds M43. U25 is 1° N.



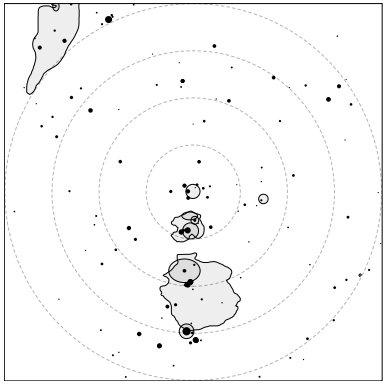
U25

NGC 1981

RA : 05.6^h	PSA: 16/B	Type: OC
Dec: $-04°$	OSA: E4	Mag :
Con:	CSA:	Size : 25.0′

See: OSA DSC:HT

Notes: M42/U24 and M43 are 1° S.



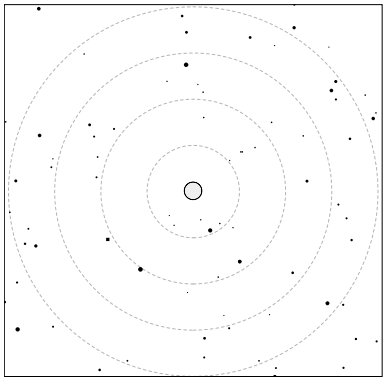
U26

M37 = BSC 55 = NGC 2099

RA : 05.9^h	PSA: 12	Type: OC
Dec: $+33°$	OSA: N6	Mag : 5.6
Con: Aur	CSA: 3	Size : 15.0′

See: OSA DSC:TMO

β Tau Elnath (mag 1.7)
→ 6° W to point 1° N of 136 Tau
→ 4° N to M37/U26

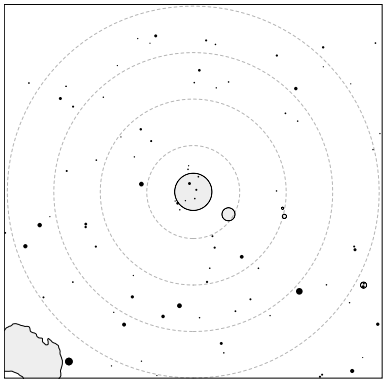


U27

M35 = BSC 31 = NGC 2168

RA : 06.2^h	PSA: 14	Type: OC
Dec: $+24°$	OSA: E7	Mag : 5.1
Con: Gem	CSA: 3/9	Size : 25.0′

See: OSA DSC:TMO

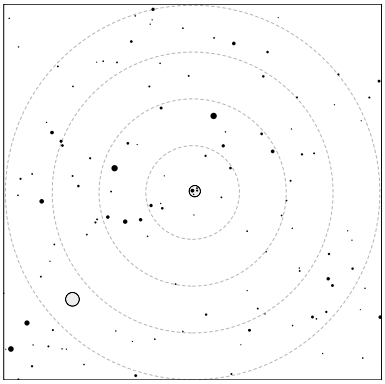


U28

NGC 2169

RA : 06.1^h	PSA: 14/25	Type: OC
Dec: $+14°$	OSA:	Mag :
Con:	CSA:	Size : $6.0'$

See: DSC:HT



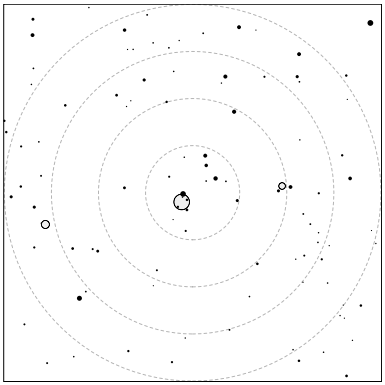
U29

NGC 2232

RA : 06.5^h	PSA: 25/27	Type: OC
Dec: $-05°$	OSA:	Mag :
Con:	CSA:	Size : $29.0'$

See: DSC:TSD

κ Ori Saiph (mag 2.1)
→ 3° N
→ 11° E passing γ Mon to β Mon
→ 2.5° N to U29



U30

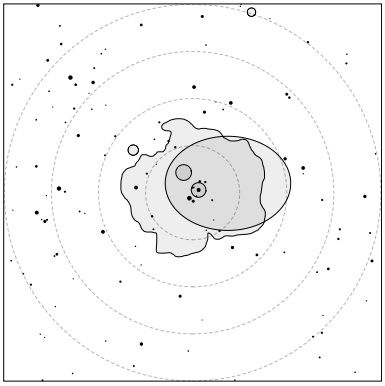
C50 = BSC 25 = NGC 2244

RA : 06.5^h	PSA: 25/E	Type: OC
Dec: $+05°$	OSA: E9	Mag : 4.4
Con:	CSA:	Size : $30.0'$

See: OSA DSC:TCO

Notes: Surrounded by C49.

α Ori Betelgeuse (mag 0.3–0.9)
→ 3° S
→ 7° E to 8 Mon
→ 2° E to C50/U30



U31

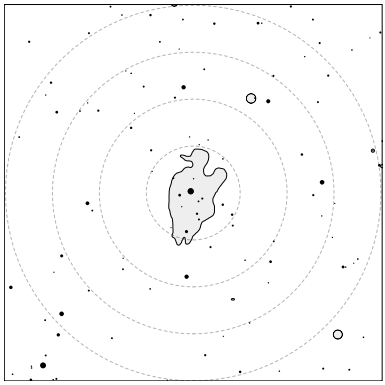
NGC 2264

RA : 06.7^h	PSA: 25/E	Type: OC
Dec: +10°	OSA: E7	Mag :
Con:	CSA:	Size : 20.0′

See: OSA DSC:HT

Notes: C46 is 1.25° SSW.

γ Gem Alhena (mag 1.9)
→ 7° S passing ξ Gem and 30 Gem to U31

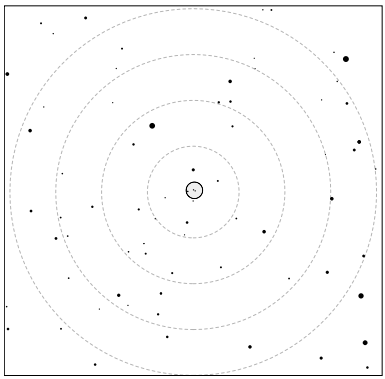


U32

NGC 2281

RA : 06.8^h	PSA: 23	Type: OC
Dec: +41°	OSA: N6	Mag :
Con:	CSA:	Size : 14.0′

See: OSA DSC:TSD



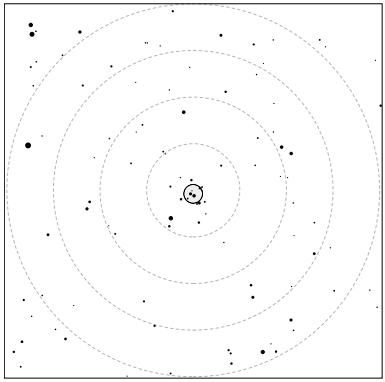
U33

M41 = BSC 21 = NGC 2287

RA : 06.8^h	PSA: 27	Type: OC
Dec: −21°	OSA: E6	Mag : 4.5
Con: CMa	CSA: 9/15	Size : 39.0′

See: OSA DSC:TMO

α CMa Sirius (mag −1.5)
→ 4° S to M41/U33

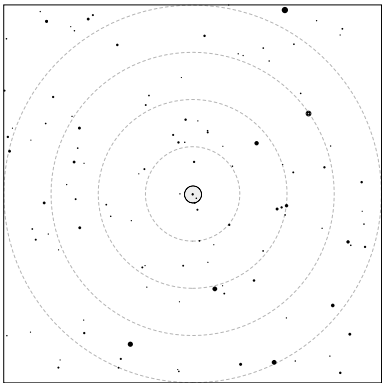


U34

NGC 2301

RA : 06.9^h	PSA: 25/27	Type: OC
Dec: $+00°$	OSA: E6	Mag :
Con:	CSA:	Size : 12.0′

See: OSA DSC:HT



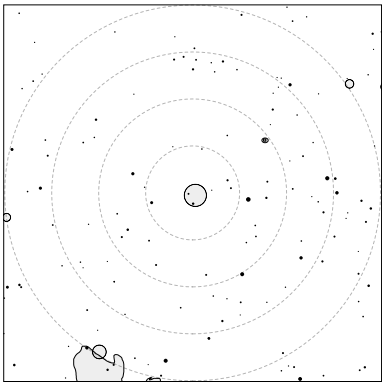
U35

M50 = BSC 44 = NGC 2323

RA : 07.0^h	PSA: 27	Type: OC
Dec: $-08°$	OSA: E8	Mag : 5.9
Con: Mon	CSA: 9	Size : 15.0′

See: OSA DSC:TMO

α CMa Sirius (mag -1.5)
→ 3° E to ι CMa
→ 5° N passing μ CMa to θ CMa
→ 4° N to asterism of three stars
→ 2° E to M50/U35



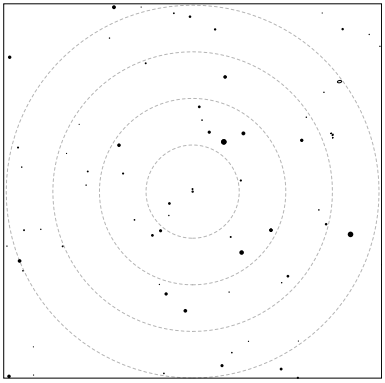
U36

C39 = NGC 2392 = Eskimo Nebula

RA : 07.5^h	PSA: 25	Type: PN
Dec: $+21°$	OSA: E7	Mag : 9.2
Con:	CSA:	Size : 0.8′×0.7′

See: OSA DSC:TMO DSC:TCO

Notes: The open cluster NGC 2420 is 2° N (see DSC:TCO).

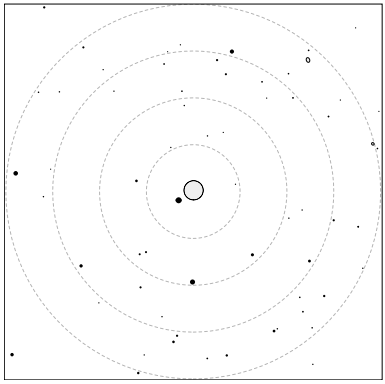


U37

NGC 2539 = Mel 83

RA : 08.2^h	PSA: 26	Type: OC
Dec: $-13°$	OSA: E8	Mag :
Con:	CSA:	Size : 21.0′

See: OSA DSC:HT

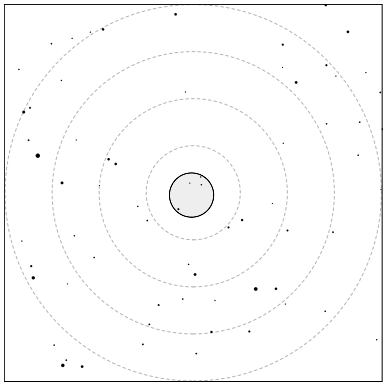


U38

M48 = BSC 37 = NGC 2548

RA : 08.2^h	PSA: 26	Type: OC
Dec: $-06°$	OSA: E10	Mag : 5.8
Con: Hya	CSA: 9/10	Size : 30.0′

See: OSA DSC:TMO

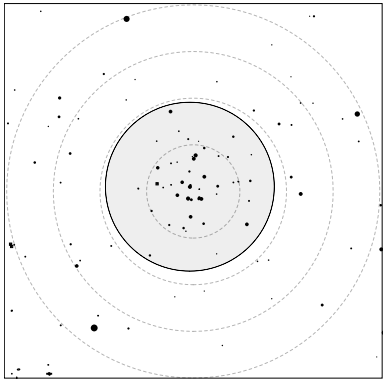


U39

M44 = BSC 12 = NGC 2632 = Mel 88 = Beehive Cluster

RA : 08.7^h	PSA: 24	Type: OC
Dec: $+20°$	OSA: E9	Mag : 3.1
Con: Cnc	CSA: 3/4/9/10	Size : 70.0′

See: OSA DSC:TMO

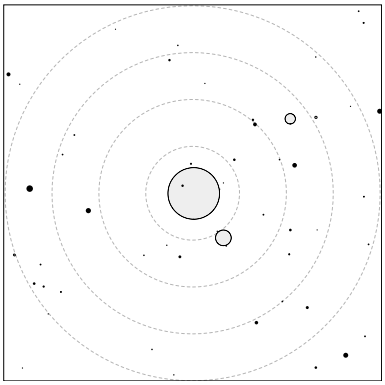


U40

M67 = BSC 51 = NGC 2682

RA : 08.9^h PSA: 24 Type: OC
Dec: +12° OSA: E9 Mag : 6.0
Con: Cnc CSA: 10 Size : 25.0′

See: OSA DSC:TMO



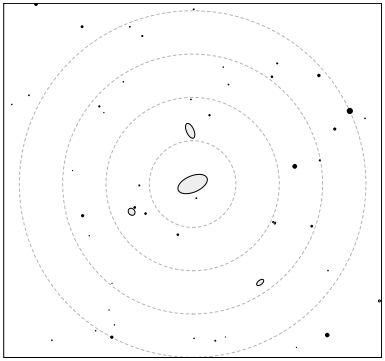
U41

M81 = BSC 72 = NGC 3031

RA : 09.9^h PSA: 31 Type: Gal
Dec: +69° OSA: N8 Mag : 6.9
Con: UMa CSA: 1/4 Size : 24.0′ × 13.0′

See: OSA DSC:TMO

Notes: M82/U42 is 0.6° N.



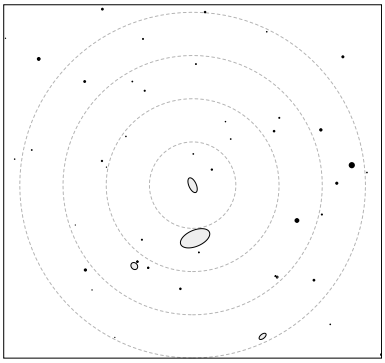
U42

M82 = BSC 122 = NGC 3034

RA : 09.9^h PSA: 31 Type: Gal
Dec: +70° OSA: N8 Mag : 8.4
Con: UMa CSA: 1/4 Size : 12.0′ × 5.6′

See: OSA DSC:TMO

Notes: M81/U41 is 0.6° S.

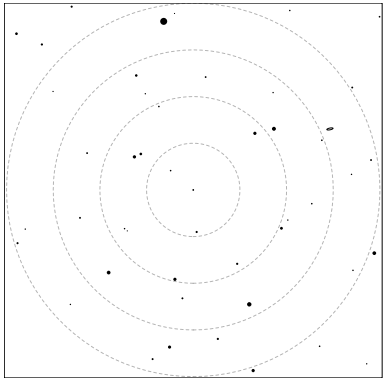


U43

C59 = BSC 125 = NGC 3242 = Ghost of Jupiter

RA : 10.4^h	PSA: 36/37	Type: PN
Dec: $-19°$	OSA: E10	Mag : 7.3
Con:	CSA:	Size : $0.8' \times 0.6'$

See: OSA DSC:TMO DSC:TCO



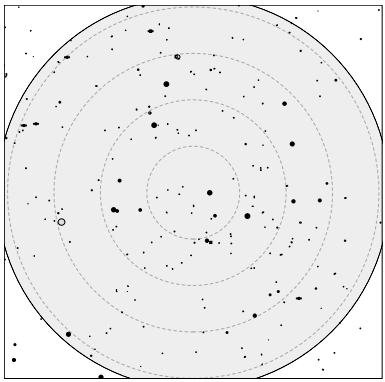
U44

Mel 111 = Cr 256 = Coma Star Cluster

RA : 12.4^h	PSA: 45	Type: OC
Dec: $+26°$	OSA: E13	Mag :
Con:	CSA:	Size : $4.6°$

See: OSA DSC:HT

Notes: Not shown in the PSA.



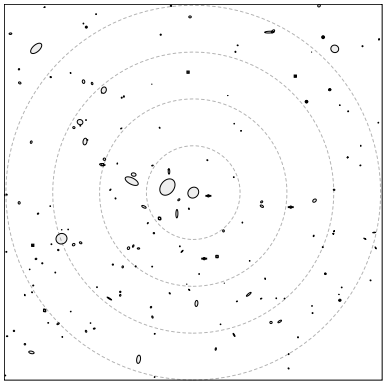
U45

M84 = NGC 4374

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: $+13°$	OSA: E14	Mag : 9.1
Con: Vir	CSA: 4/10/11	Size : $5.1' \times 4.1'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



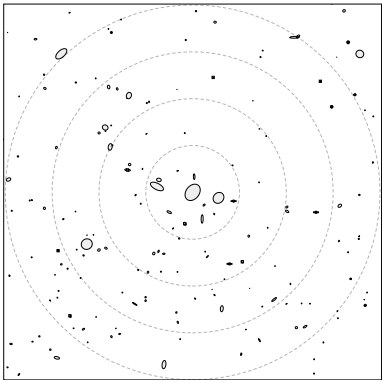
U46

M86 = NGC 4406

RA : 12.4^h	PSA: 45/C	Type: Gal
Dec: $+13°$	OSA: E14	Mag : 8.9
Con: Vir	CSA: 4/10/11	Size : $12.0' \times 9.3'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.



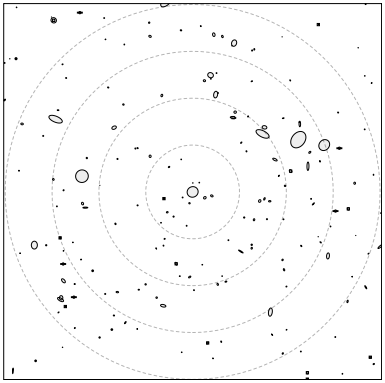
U47

M87 = NGC 4486

RA : 12.5^h	PSA: 45/C	Type: Gal
Dec: $+12°$	OSA: E14	Mag : 8.6
Con: Vir	CSA: 10/11	Size : $7.1'$

See: OSA DSC:TMO

Notes: In the Virgo Cluster.

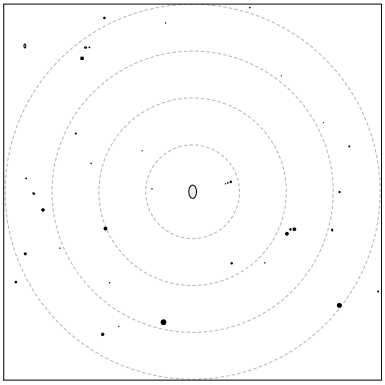


U48

M104 = BSC 133 = NGC 4594 = Sombrero Galaxy

RA : 12.7^h	PSA: 47	Type: Gal
Dec: $-12°$	OSA: E12	Mag : 8.0
Con: Vir	CSA: 10/11	Size : $7.1' \times 4.4'$

See: OSA DSC:TMO

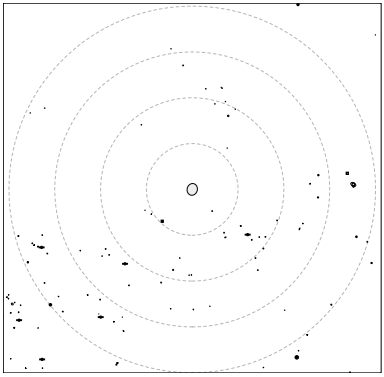


U49

M94 = BSC 134 = NGC 4736

RA : 12.8^h PSA: 43 Type: Gal
Dec: +41° OSA: N12 Mag : 8.2
Con: CVn CSA: 4/5 Size : 11.5′ × 10.0′

See: OSA DSC:TMO

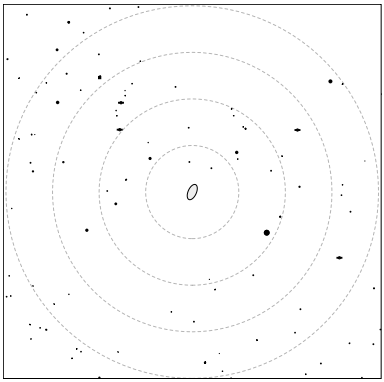


U50

M64 = BSC 149 = NGC 4826 = Black-Eye Galaxy

RA : 12.9^h PSA: 45 Type: Gal
Dec: +22° OSA: E13 Mag : 8.5
Con: Com CSA: 5/11 Size : 9.2′×4.6′

See: OSA DSC:TMO

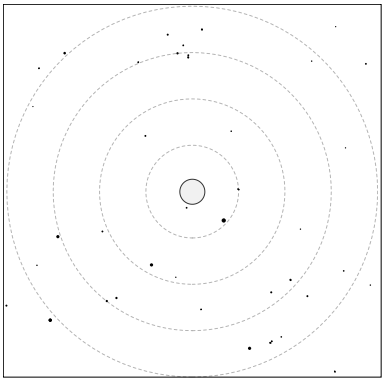


U51

M3 = BSC 53 = NGC 5272

RA : 13.7^h PSA: 43/44 Type: GC
Dec: +28° OSA: E15 Mag : 5.9
Con: CVn CSA: 5 Size : 19.0′

See: OSA DSC:TMO

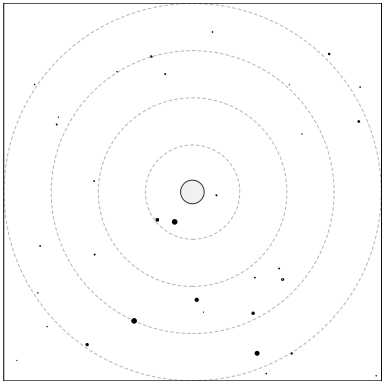


U52

M5 = BSC 66 = NGC 5904

RA : 15.3^h	PSA: 55/57	Type: GC
Dec: $+02°$	OSA: E15	Mag : 5.7
Con: Ser	CSA: 11/12	Size : 23.0′

See: OSA DSC:TMO

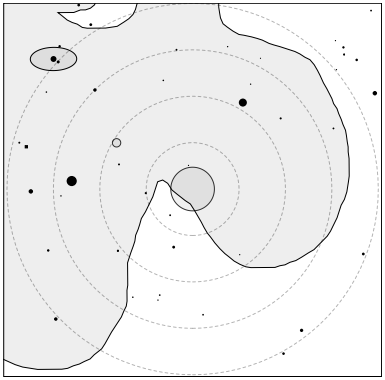


U53

M4 = BSC 43 = NGC 6121

RA : 16.4^h	PSA: 56/58	Type: GC
Dec: $-27°$	OSA: E18	Mag : 5.4
Con: Sco	CSA: 11/12/17/18	Size : 36.0′

See: OSA DSC:TMO



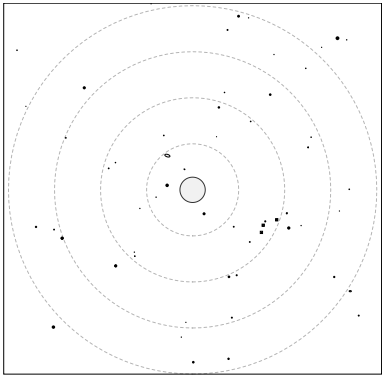
U54

M13 = BSC 40 = NGC 6205 = Hercules Cluster

RA : 16.7^h	PSA: 52	Type: GC
Dec: $+36°$	OSA: N14	Mag : 5.3
Con: Her	CSA: 5/6	Size : 20.0′

See: OSA DSC:TMO

ζ Her (mag 2.8)
$\rightarrow$ 5° N (2/3 distance to η Her) to M13/U54

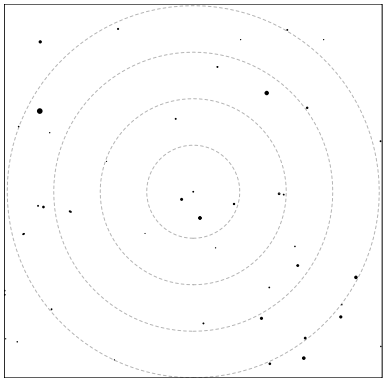


U55

NGC 6210

RA : 16.7^h	PSA: 54	Type: PN
Dec: $+24°$	OSA: E19	Mag :
Con:	CSA:	Size : 0.8′

See: OSA DSC:HT

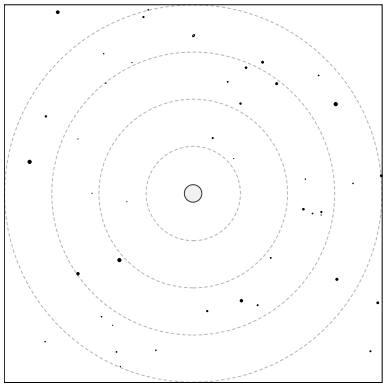


U56

M12 = BSC 94 = NGC 6218

RA : 16.8^h	PSA: 54/56	Type: GC
Dec: $-02°$	OSA: E17	Mag : 6.1
Con: Oph	CSA: 12	Size : 16.0′

See: OSA DSC:TMO

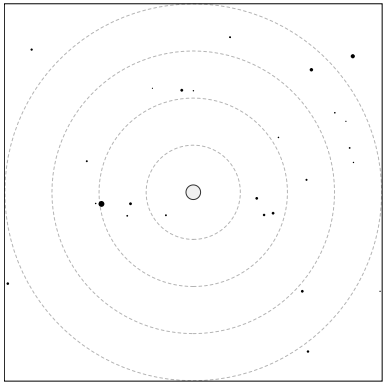


U57

M10 = BSC 61 = NGC 6254

RA : 17.0^h	PSA: 54/56	Type: GC
Dec: $-04°$	OSA: E17	Mag : 6.6
Con: Oph	CSA: 12	Size : 20.0′

See: OSA DSC:TMO



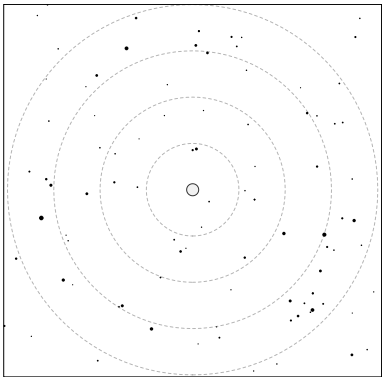
U58

M62 = BSC 81 = NGC 6266

RA : 17.0^h	PSA: 58	Type: GC
Dec: $-30°$	OSA: E18	Mag : 6.7
Con: Oph	CSA: 17/18	Size : 14.0′

See: OSA DSC:TMO

ε Sco (mag 2.3)
→ 2° E
→ 4° N to M63



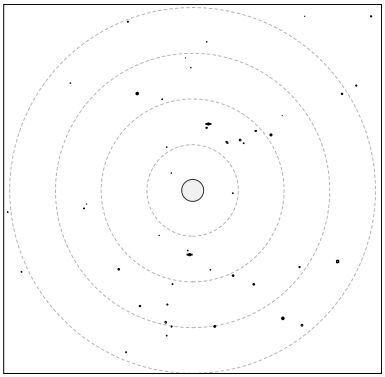
U59

M92 = BSC 59 = NGC 6341

RA : 17.3^h	PSA: 52	Type: GC
Dec: $+43°$	OSA: N14	Mag : 6.5
Con: Her	CSA: 5/6	Size : 14.0′

See: OSA DSC:TMO

π Her (mag 3.2)
→ 6.5° N to M92/U59



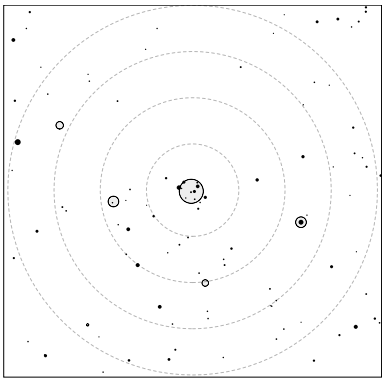
U60

M6 = BSC 18 = NGC 6405 = Butterfly Nebula

RA : 17.7^h	PSA: 58/69	Type: OC
Dec: $-32°$	OSA: E18	Mag : 4.2
Con: Sco	CSA: 18	Size : 33.0′

See: OSA DSC:TMO

λ Sco Shaula (mag 1.6)
→ 5° N
→ 1° E to M6/U60

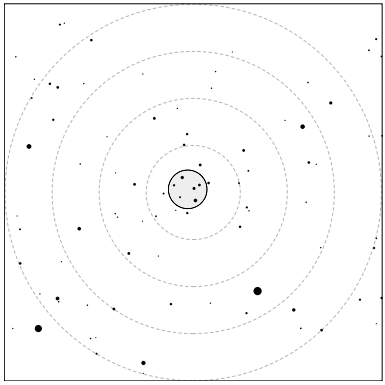


U61

IC 4665 = Mel 179 = Cr 349

RA : 17.8^h	PSA: 54	Type: OC
Dec: $+06°$	OSA: E17	Mag :
Con:	CSA:	Size : 70.0′

See: OSA DSC:HT



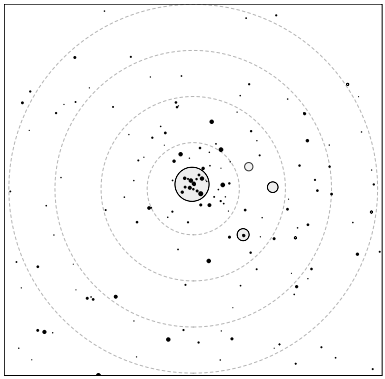
U62

M7 = BSC 10 = NGC 6475

RA : 17.9^h	PSA: 58/69	Type: OC
Dec: $-35°$	OSA: E18	Mag : 2.8
Con: Sco	CSA: 18	Size : 75.0′

See: OSA DSC:TMO

ε Sgr Kaus Australis (mag 1.8)
→ 6° W to M7/U62

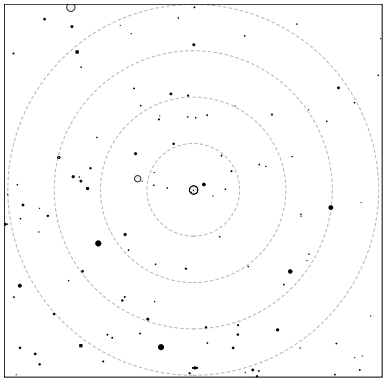


U63

NGC 6520

RA : 18.1^h	PSA: 67	Type: OC
Dec: $-28°$	OSA:	Mag :
Con:	CSA:	Size : 6.0′

See: DSC:HT



U64

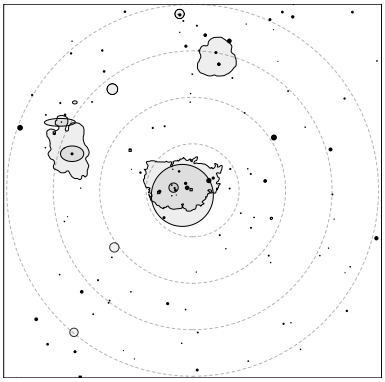
M8 = BSC 45 = NGC 6523/6530 = Lagoon Nebula

RA : 18.1^h	PSA: 67	Type: BN	
Dec: $-24°$	OSA: E20	Mag : 3.0	
Con: Sgr	CSA: 12/18	Size : 45.0′	×
		30.0′	

See: OSA DSC:TMO

Notes: M20 is 1.5° N. M21 is 2.0° N.

γ Sgr Alnasl (mag 3.0)
→ 6° N to M8/U64



U65

M17 = BSC 47 = NGC 6618 = Omega Nebula

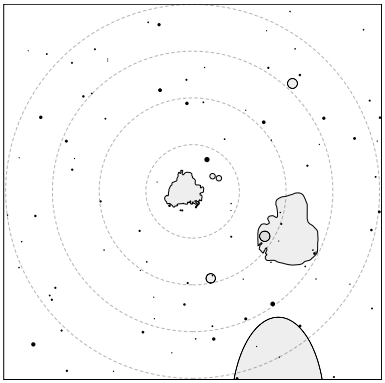
RA : 18.3^h	PSA: 67	Type: BN	
Dec: $-16°$	OSA: E20	Mag : 6.0	
Con: Sgr	CSA: 12	Size : 20.0′	×
		15.0′	

See: OSA DSC:TMO

Notes: M18 is 1° S. M24 is 3° SW.

ν Oph (mag 3.3)
→ 5° E
→ 6° S to M17/U65

μ Sgr (mag 3.8)
→ 5° N passing M24 and M18
→ 2° E to M17

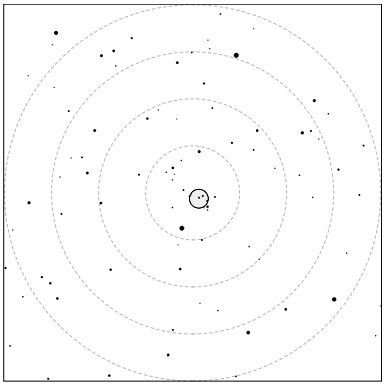


U66

NGC 6633

RA : 18.5^h	PSA: 65	Type: OC	
Dec: $+07°$	OSA: E19	Mag :	
Con:	CSA:	Size : 20.0′	

See: OSA DSC:HT

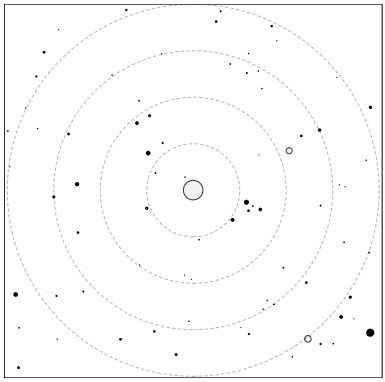


U67

M22 = BSC 28 = NGC 6656

RA : 18.6^h	PSA: 67	Type: GC
Dec: $-24°$	OSA: E20	Mag : 5.2
Con: Sgr	CSA: 12/18	Size : 32.0′

See: OSA DSC:TMO

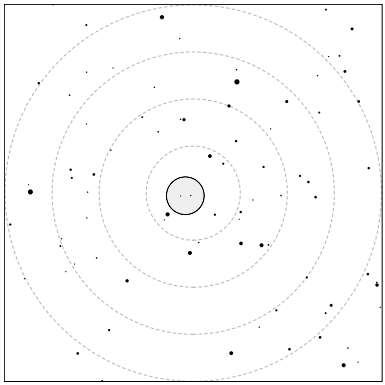


U68

IC 4756 = Mel 210

RA : 18.6^h	PSA: 65	Type: OC
Dec: $+05°$	OSA: E19	Mag :
Con:	CSA:	Size : 40.0′

See: OSA DSC:HT

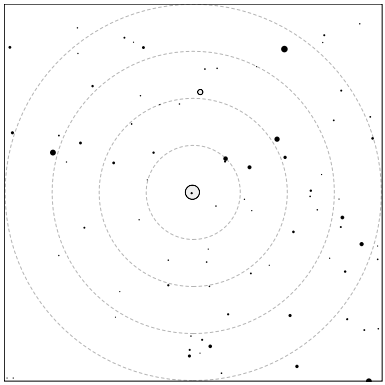


U69

M11 = NGC 6705 = Mel 213 = Cr 391 = Wild Duck Cluster

RA : 18.9^h	PSA: 65/67	Type: OC
Dec: $-06°$	OSA: E19	Mag : 5.3
Con: Sct	CSA: 12	Size : 13.0′

See: OSA DSC:TMO

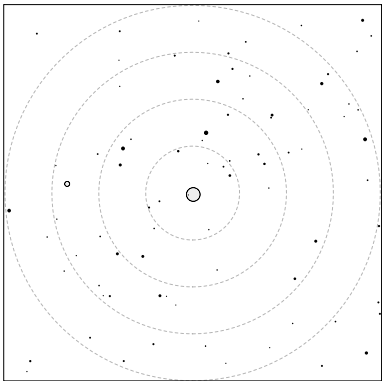


U70

NGC 6709

RA : 18.9^h	PSA: 65	Type: OC
Dec: +10°	OSA:	Mag :
Con:	CSA:	Size : 13.0′

See: DSC:HT



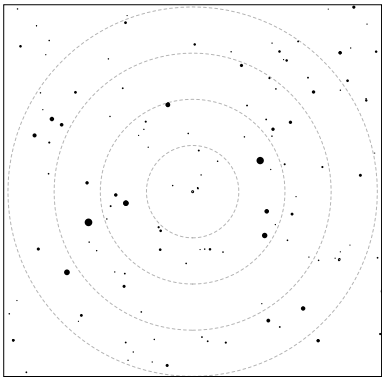
U71

M57 = BSC 127 = NGC 6720 = Ring Nebula

RA : 18.9^h	PSA: 63	Type: PN
Dec: +33°	OSA: N18	Mag : 8.8
Con: Lyr	CSA: 6/7	Size : 1.1′

See: OSA DSC:TMO

γ Lyr Sulaphat (mag 3.2)
→ 1.2° W
→ 0.4° N to M57/U71

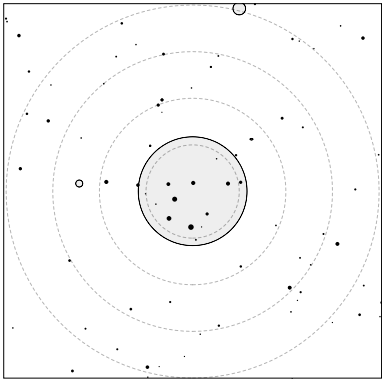


U72

Cr 399 = Brocchi's Cluster

RA : 19.4^h	PSA: 64/65	Type: OC
Dec: +20°	OSA:	Mag :
Con:	CSA:	Size : 60.0′

See: DSC:TMO DSC:HT



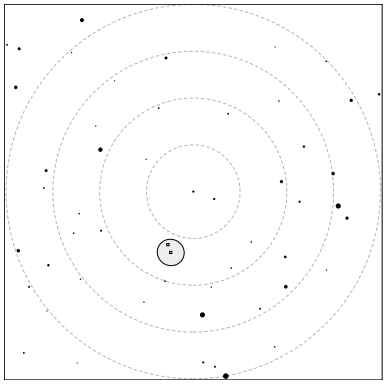
U73

NGC 6818

RA : 19.7^h	PSA: 66	Type: PN
Dec: $-14°$	OSA: E22	Mag :
Con:	CSA:	Size : $0.4' \times 0.3'$

See: OSA DSC:HT

Notes: C57 is 0.7° SSE.

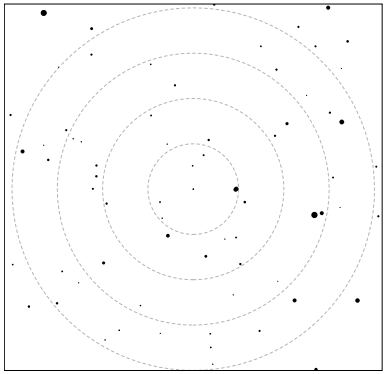


U74

NGC 6826

RA : 19.7^h	PSA: 62	Type: PN
Dec: $+51°$	OSA: N18	Mag :
Con:	CSA:	Size : $0.5' \times 0.4'$

See: OSA

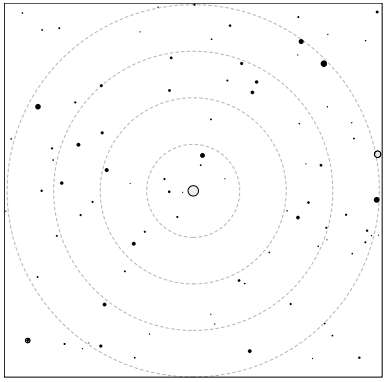


U75

M27 = BSC 88 = NGC 6853 = Dumbell Nebula

RA : 20.0^h	PSA: 64	Type: PN
Dec: $+23°$	OSA: E21	Mag : 7.3
Con: Vul	CSA: 6/7/12/13	Size : $8.0' \times 6.0'$

See: OSA DSC:TMO

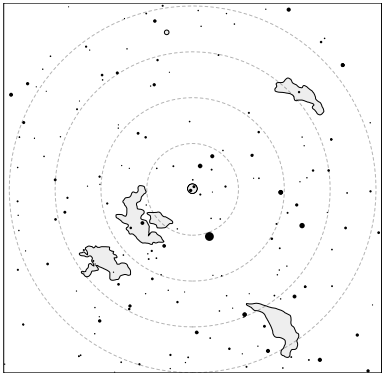


U76

NGC 6910

RA : 20.4^h	PSA: 62	Type: OC
Dec: +41°	OSA:	Mag :
Con:	CSA:	Size : 7.0′

See: DSC:TSD

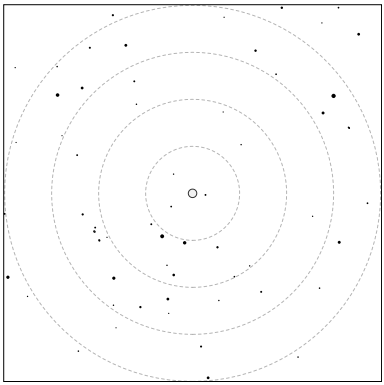


U77

NGC 6934

RA : 20.6^h	PSA: 64	Type: GC
Dec: +07°	OSA:	Mag :
Con:	CSA:	Size : 5.9′

See:

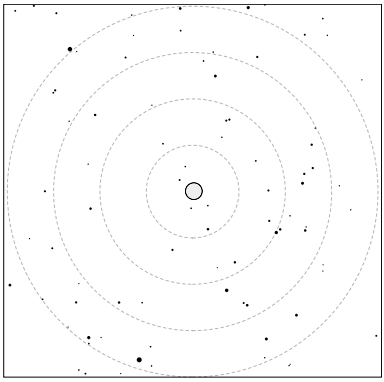


U78

NGC 6940 = Mel 232

RA : 20.6^h	PSA: 62	Type: OC
Dec: +28°	OSA: N20	Mag :
Con:	CSA:	Size : 31.0′

See: OSA DSC:HT

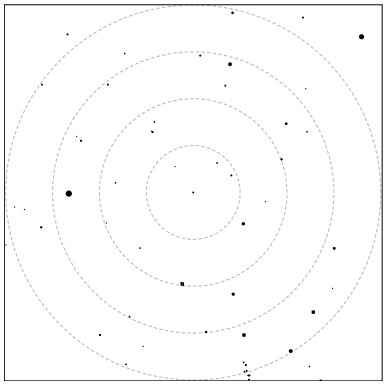


U79

NGC 7009

RA : 21.1^h	PSA: 77	Type: PN
Dec: $-11°$	OSA: E24	Mag :
Con:	CSA:	Size : $0.7' \times 0.4'$

See: OSA

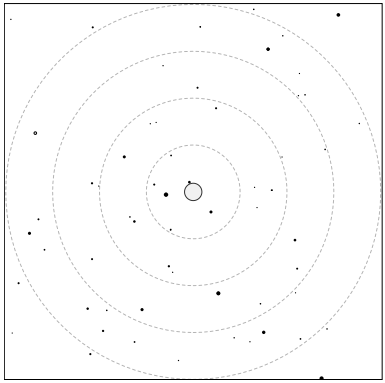


U80

M15 = BSC 54 = NGC 7078

RA : 21.5^h	PSA: 75	Type: GC
Dec: $+12°$	OSA: E23	Mag : 6.0
Con: Peg	CSA: 13	Size : 18.0′

See: OSA DSC:TMO

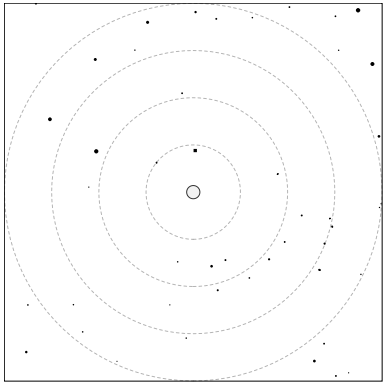


U81

M2 = BSC 56 = NGC 7089

RA : 21.6^h	PSA: 75/77	Type: GC
Dec: $-01°$	OSA: E24	Mag : 6.3
Con: Aqr	CSA: 13	Size : 16.0′

See: OSA DSC:TMO

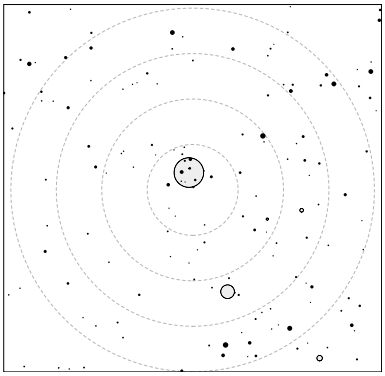


U82

M39 = BSC 36 = NGC 7092

RA : 21.5^h	PSA: 73	Type: OC
Dec: +48°	OSA: N24	Mag : 4.6
Con: Cyg	CSA: 1/6/7	Size : 31.0′

See: OSA DSC:TMO

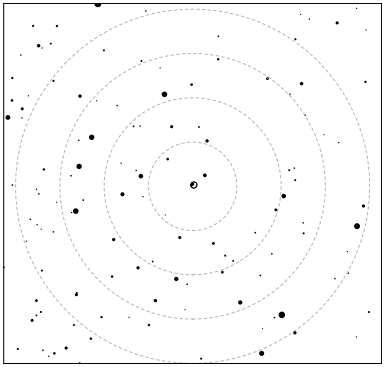


U83

NGC 7160

RA : 21.9^h	PSA: 71	Type: OC
Dec: +63°	OSA:	Mag :
Con:	CSA:	Size : 7.0′

See: DSC:TSD



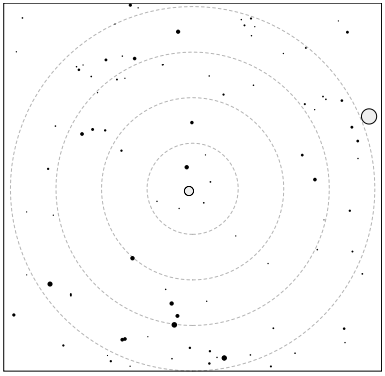
U84

NGC 7209

RA : 22.1^h	PSA: 73	Type: OC
Dec: +47°	OSA: N24	Mag :
Con:	CSA:	Size : 25.0′

See: OSA DSC:TSD

Notes: C19 is 2° WNW.

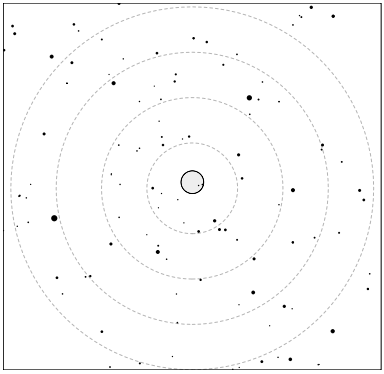


U85

C16 = BSC 63 = NGC 7243

RA : 22.3^h	PSA: 73	Type: OC
Dec: +50°	OSA: N24	Mag : 6.4
Con:	CSA:	Size : 30.0′

See: OSA DSC:TCO

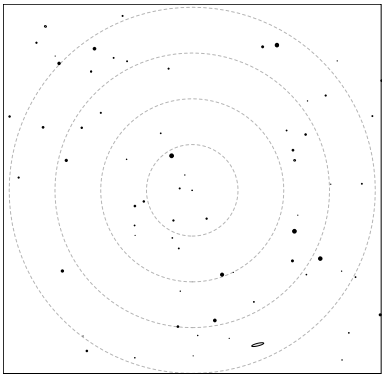


U86

NGC 7662

RA : 23.4^h	PSA: 72	Type: PN
Dec: +43°	OSA: N24	Mag :
Con:	CSA:	Size : 0.5′

See: OSA



U87

NGC 7789 = Mel 245

RA : 23.9^h	PSA: 72	Type: OC
Dec: +57°	OSA: N22	Mag :
Con:	CSA:	Size : 15.0′

See: OSA DSC:TCO DSC:HT

